# AFM exam-day pack — priority, what ICAI asked, exam steps

Order = marks set in Sep 2025, Jan 2026, May 2026. Paper: 15 MCQs (30, no negative marking) + Q1 compulsory + any 4 of Q2–Q6. Every Q2–Q6 has a 4-mark theory part (c).

| # | Chapter | Marks set (3 papers) | Share | Plan |
|---|---|---|---|---|
| 1 | Ch 9 Derivatives Analysis and Valuation | 45 | 12.7% | Done |
| 2 | Ch 10 Foreign Exchange Exposure and Risk Management | 41 | 11.6% | Done |
| 3 | Ch 6 Portfolio Management | 38 | 10.7% | Must |
| 4 | Ch 8 Mutual Funds | 37 | 10.5% | Must (easy) |
| 5 | Ch 5 Security Valuation | 34 | 9.6% | Must |
| 6 | Ch 14 Mergers, Acquisitions and Corporate Restructuring | 29 | 8.2% | Must |
| 7 | Ch 11 International Financial Management | 25 | 7.1% | Steps only |
| 8 | Ch 12 Interest Rate Risk Management | 21 | 5.9% | FRA + cap/floor only |
| 9 | Ch 2 Risk Management | 14 | 4.0% | Theory |
| 10 | Ch 1 Financial Policy and Corporate Strategy | 12 | 3.4% | EFR/SGR + theory |
| 11 | Ch 15 Startup Finance | 12 | 3.4% | Theory |
| 12 | Ch 3 Advanced Capital Budgeting Decisions | 12 | 3.4% | Sensitivity/probability MCQ |
| 13 | Ch 4 Security Analysis | 12 | 3.4% | Theory |
| 14 | Ch 7 Securitization | 12 | 3.4% | Theory |
| 15 | Ch 13 Business Valuation | 10 | 2.8% | EV/EVA only |

**Exam-hall rules:** attempt all 15 MCQs → Q1 compulsory → read Q2–Q6 for 10 min, pick 4 → every part: formula → numbers → decision (step marks) → never skip a theory (c): 5–6 SM points.

12.7% of marks

# ⑨ Ch 09 – Derivatives Analysis and Valuation

| Attempt | Q | M | Asked (MCQ / numerical / theory) |
|---|---|---|---|
| Sep 2025 | MCQ CS-III 7-9 | 6 | Long strangle (buy 3-month call at 750 and put at 600 on the same stock): net gain/loss at expiry for unchanged, lower and higher price (MCQ) |
| Sep 2025 | Q4(b) | 4 | Hedging a long and a short stock position with Nifty futures using betas: number of contracts and overall profit/loss on close-out (Numerical) |
| Jan 2026 | Q1(c) | 4 | Long strangle (buy call at 620 and put at 480): profit/loss at three expiry prices and upper and lower break-even points (Numerical) |
| Jan 2026 | Q2(c) | 4 | Real options versus financial options on four points: underlying asset, pay-off, exercise period, approach (Theory) |
| Jan 2026 | Q3(a) | 6 | Long futures plus short call: profit/loss on futures and on option net of brokerage, overall profit and total brokerage (Numerical) |
| Jan 2026 | Q6(b) | 4 | Real option: value of abandonment option by the risk-neutral method (up/down project values, abandonment price, risk-free rate) (Numerical) |
| May 2026 | MCQ CS-II 4-6 | 6 | Real option to wait: NPV today and after one year in high/low scenarios, risk-neutral returns and probability, PV of the option to wait (MCQ) |
| May 2026 | Q2(b) | 4 | Six options (calls and puts, three strikes): ITM/ATM/OTM, intrinsic and time value, exercise or decline and net profit/loss per contract at +150 and -150 expiry prices (Numerical) |
| May 2026 | Q6(b) | 7 | Black-Scholes: hedge ratio (N(d1)), probability of expiry above exercise price, value of call and put (Numerical) |

## 5.2.5 Uses / Advantages of Stock Index Futures

Asked: Sep 2025 Q4(b) 4m

- → **Change weightings / risk exposure** (e.g. 60% US / 40% Japan: buy US index futures, sell Nikkei).
- → **Separate market timing from selection**: buy index futures + PV in risk-free = market exposure; sell unwanted stocks and hold futures until new picks are decided.
- → **Index arbitrage (program trading)**: futures above fair value → buy the basket, sell futures.
- → **Hedge portfolio value** against systematic risk: **short hedge / sell-hedge** when long the portfolio. Cash and futures must move similarly; amounts invested should be the same. Perfect hedges rare due to **basis risk**.
- → Flexibility; leverage; keep the portfolio through corrections without selling; sell as readily as buy (same margin); **transfer risk quickly and efficiently** (vs once-a-day MF pricing).

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**SM hedge example:** portfolio $1.2 million, β 1.00, α 6%; market −15% + 1% dividend = −14%, expected −8%. To halve β: S&P 240 × $500 = $120,000 per contract → sell **5 contracts** ($600,000). Cheaper than selling shares (capital gains tax, commissions).

| Formula (from TYK answers) | Terms / logic |
|---|---|
| No. of contracts = (Value of portfolio × β) ÷ (Futures price × lot size) | β converts rupee value into index-equivalent exposure. Long shares → **sell**; short shares → **buy**. Cash has β 0 and is not hedged. |
| Contracts to change β = P(β − β′) ÷ F | P portfolio value, β current, β′ target, F value of one contract; reduce β → sell, increase → buy. |
| Target β via risk-free: β′ = β × (1 − p) | p = share of portfolio moved to risk-free (β 0). |
| Portfolio β (mixed) = Σ wβ; or = % change in portfolio ÷ % change in index | Weights on market values. |

12.7% of marks

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**TYK P5:** futures 5,000 + 5,000(0.09 − 0.06) × 4/12 = **5,050**; contract ₹ 2,52,500; 10,10,000 × 1.5 ÷ 2,52,500 = **6 contracts**; at 4,500 (1 month left) futures 4,511.25 → gain 6 × (5,050 − 4,511.25) × 50 = **₹ 1,61,625**. **TYK P7 (MF):** 1.1 × 90 crore ÷ (4,300 × 50) = **4,605**; 10% fall: loss 9.90 crore = gain on futures 9.90 crore; cash not hedged. **TYK P8:** β = 0.85 × 1.6 = **1.36**; shares −3.2% → index −2%; long ₹ 100 lakh futures → MTM paid 2 lakh; value 85 × 0.968 + 13 = 95.28 lakh (−4.72%) → β = 0.0472/0.02 = **2.36**. **TYK P9:** Right long 50 × 1.25 → short 62.5 lakh; Wrong short 25 × 0.90 → long 22.5; Fair short 20 × 0.75 → long 15 → net **short ₹ 25 lakh** index futures. **TYK P10 (Ram):** (10,000 × 22 × 1.5 − 5,000 × 40 × 2) ÷ 1,000 = **70 contracts long**; outlay ₹ 90,000, close-out ₹ 78,550 → **loss ₹ 11,450**. **TYK P11:** β **0.849**; F = 5,900 $e^{(0.0998)(58/365)}$ = **5,994.28** (alt. $e^{0.105\times58/365}$ → 5,999.24); hedge 1,88,54,860 × 0.849 ÷ (5,994.28 × 200) = **13.35** (13 or 14); to β 0.6: **3.92** ≈ **4**. **TYK P12:** β **1.30**; 0.91 = 1.30(1 − p) → p 30% → risk-free **₹ 1,500 lakh**; dispose A 0.90, B 1.20, C 0.60 lakh shares; (1.30 − 0.91) × 5,000 lakh ÷ (8,125 × 200) = **120 contracts sold**; Nifty +2% → shares 5,130 − MTM 39 = 5,091 → 1.82% ÷ 2% = β **0.91**. **TYK P13:** β = 10,95,832.30 ÷ 9,94,450 = **1.102**; $F_{May}$ = 8,500 $e^{0.0333}$ (interpolated 1.03387) = **8,788**; $F_{June}$ 8,500 $e^{0.05}$ = **8,935.80**; hedge to June at 8,850 × 25: 100% **4.953** ≈ **5**, 50% **2.47** ≈ **3**, 120% **5.94** ≈ **6**.

## 6.6 Pay-off scenarios (6.6.1–6.6.4)

Asked: Sep 2025 MCQ 6m; Jan 2026 Q1(c) 4m; May 2026 Q2(b) 4m

| Position | Net pay-off at expiry (per share) | Break-even / SM example |
|---|---|---|
| Long call (buyer) | Max(0, Spot − X) − premium; exercise only if Spot > X | X + premium. X 40, premium 5 → loss 5 below 40, BE **45** |
| Short call (writer) | Mirror image of long call | Max gain = premium |
| Long put (buyer) | Max(0, X − Spot) − premium; exercise only if Spot < X | X − premium. X 40, premium 5 → BE **35** |
| Short put (writer) | Mirror image of long put | Max gain = premium |

Note (app-written): SM 6.6.3 (p 9.21) prints the put buyer's gain as "(Spot Price − Exercise Price − Premium)"; for a put the gain is (Exercise Price − Spot Price − Premium), which is what gives ICAI's own break-even of ₹ 35.

- **Buy call + buy put on the same stock** (asked as "long strangle" when strikes differ): total premium paid upfront; at expiry only one leg can be in the money. Loss = total premium when price is between the strikes. (App-written from the TYK answers:) upper break-even = call strike + total premium; lower break-even = put strike − total premium.
- App-written labels (asked May 2026): call is in the money if Spot > X, at the money if Spot = X, out of the money if Spot < X (put the reverse); intrinsic value = value if exercised now (never negative); time value = premium − intrinsic value (SM 7: an option with zero theoretical value can still have positive market value while time remains).

12.7% of marks

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**TYK P15 (Mr A):** call 550 @ 30, put 450 @ 5, 100 shares; price 350 → put gain (450 − 350) × 100 = 10,000 − premium 3,500 = **net gain ₹ 6,500**. **TYK P19 (Delta Corp, same data):** at 500 neither exercised → **loss ₹ 3,500**; at 350 → **gain ₹ 6,500**; at 600 → call (600 − 550) × 100 − 3,500 = **gain ₹ 1,500**. **TYK P16 (ABC):** call 42 @ 2 + put 40 @ 1, cost ₹ 300; at 43 → −300 + 100 = **loss ₹ 200**; at 36 → −300 + 400 = **gain ₹ 100**. **TYK P20 (VCC, X 220):** call net pay-off at 200...240 = −6, −6, −6, 4, 14; put = 15, 5, −5, −5, −5; call gainful above **₹ 226** (220 + 6), put below **₹ 215** (220 − 5). **TYK P17 (written put, X 300):** expected value = 120 × 0.1 + 40 × 0.2 + 20 × 0.5 = **₹ 30** (not exercised at 320, 400). **TYK P18 (call X 150):** expected price **₹ 160.50**; value at that "exercise price prevails" = **nil**; expected value held to maturity = 10 × 0.5 + 30 × 0.10 + 40 × 0.15 = **₹ 14**.

*Note (app-written): in TYK P18's alternative table (p 9.83) ICAI prints probabilities 0.1 and 0.2 for prices 120 and 140 (the question gives 0.05 and 0.20); those rows are "not exercised", so the ₹ 14 answer is unaffected.*

## 6.7 Comparison with Single Stock Futures

Asked: Jan 2026 Q3(a) 6m

| Point | Options | Stock futures |
|---|---|---|
| Nature | Buyer has right, not obligation | Long is obliged to square off at or before expiry |
| Movement | Premium moves by **delta**; may fluctuate independently of the stock | Closely follows the underlying |
| Price of investing | Premium paid = sunk cost; lost if not in the money | Initial margin + maintenance (MTM) margin |

*Futures + option combined (Jan 2026): profit on futures = (close − entry) × lot; profit on written call = premium − Max(0, Spot − X) per unit; deduct brokerage on each leg.*

## 7.3 Black-Scholes Model

Asked: May 2026 Q6(b) 7m TYK T3

- → Theoretical price = what a writer needs as compensation for writing a call and **completely hedging** by buying stock. Hedger's view of future prices is irrelevant → **risk-neutral valuation** (underlying assumed to earn the risk-free rate). Based on a normal distribution of returns.
- → **Assumptions**: **European** options; **no transaction costs**; short-term interest rates **known and constant**; stocks pay **no dividend**; price follows a **random walk**; returns **normally distributed**; **variance constant** over the option's life.
- → **Advantage**: speed. **Limitation**: cannot accurately price **American** options — values only at expiry, ignores early exercise.

$$\text{Call} = S \cdot N(d_1) - X \cdot e^{-rt} \cdot N(d_2); \quad d_1 = [\log_e(S/X) + (r + v^2/2)t] \div v\sqrt{t}; \quad d_2 = d_1 - v\sqrt{t}$$

12.7% of marks

| Piece | What it is / logic | Illus 3(i) figures |
|---|---|---|
| S, X, t, r, v | Current price; strike; time as fraction of a year; **continuously compounded** risk-free rate; annual volatility (SD of returns) | 415, 400, 0.25, 5%, 22% |
| $\log_e(S/X)$ | How far the price is above (+) or below (−) the strike, in log terms (app-written gloss) | $\log_e(1.0375) = 0.03681$ |
| $(r + v^2/2)t$ | Growth at r plus half the variance over the option's life | $(0.05 + 0.0242) \times 0.25 = 0.01855$ |
| $\div v\sqrt{t}$ | SD of return over the life → $d_1$ is a number of SDs (z) | $0.22 \times 0.5 = 0.11$ → $d_1$ = **0.5033** |
| $d_2 = d_1 - v\sqrt{t}$ | Same as using $(r - v^2/2)t$ in the numerator | $(0.03681 + 0.00645)/0.11$ = **0.3933** |
| $N(d_1)$ | **Hedge ratio (delta)**: shares held per call for a fully hedged position; $S \cdot N(d_1)$ = value of those shares | 0.6928 → $415 \times 0.6928 = 287.512$ |
| $N(d_2)$ | **Probability** that spot at expiry > strike (call exercised) | 0.6529 |
| $X \cdot e^{-rt} \cdot N(d_2)$ | Holder seen as having **borrowed** PV of strike × $N(d_2)$ | $400/1.012578 \times 0.6529 = 257.916$ |

- → **Reading N() from ICAI tables**: **one-tail** table (area to the left or right): N(+z) = 1 − tail area; N(−z) = tail area. **Cumulative** table: read directly. **Two-tail** table: N(+z) = 1 − two-tail ÷ 2. Not on the table → **interpolate** between the two nearest z.
- → Interpolation (Illus 3): N(0.39) = 0.6517; difference to N(0.40) is 0.0037 per 0.01 → $0.0037 \times 0.00327/0.01 = 0.0012$ → N(0.39327) = **0.6529**.
- → **Put-call parity** (as printed in Illus 3): $\mathbf{V_p = -V_s + V_c + PV(E)}$ — put = call − share price + PV of exercise price. Illus 3 writes the call as $V_0 = V_s\,N(d_1) - E\,e^{-rt}\,N(d_2)$: $V_s = S$, $E = X$.
- → **Dividend** before expiry: adjusted S = S − PV(dividend), $PV = D \times e^{-rt}$; then normal B-S.

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**Illus 3 (TIC Ltd, X 400, 3 months, 5%, v 22%):** (i) S 415: $N(d_1) = N(0.50327) = 1 - 0.3072$ = **0.6928**; $N(d_2) = N(0.39327) = 1 - 0.3471$ = **0.6529**; value 287.512 − 257.916 = ₹ **29.60** > ₹ 25 → underpriced, **worth buying**. (ii) S 380: $\log_e(0.95) = -0.05129$; $d_1$ = **−0.297636**, $d_2$ = **−0.407636**; N = 0.3830 / 0.3418; 145.54 − 135.02 = ₹ **10.52**. (iii) Put = −380 + 10.52 + 395.06 = ₹ **25.58** (or with 395.03 → ₹ **25.55**). (iv) S 408, dividend ₹ 10 in 2 months: PV = $10 \times e^{-0.05 \times 0.16666}$ = **9.917** → S 398.083; $\log_e(0.9952) = -0.00481$; $d_1 = 0.01374/0.11$ = **0.125**, $d_2$ = **0.015**; N = 0.5498 / 0.5060; 218.866 − 199.8858 = ₹ **18.98**.

*Note (app-written) on Illus 3(iii) (p 9.31): ICAI lists "Current Market Value = ₹ 415" although the put uses 380, and gives two answers because PV of strike is taken as 400/1.0125 = 395.06 in one line and 400/1.012578 = 395.03 in another. Keep both of ICAI's figures.*

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**TYK P25 (S 80, X 75, v 0.40, 6 months, 12%; one-tail table; ICAI's answer prints $C = SN(d_1) - Ke^{-rt}N(d_2)$ with σ: K = X, σ = v):** $d_1 = [0.0646 + (0.12 + 0.08)0.5] \div (0.40 \times 0.7071) = 0.1646/0.2828$ = **0.5820**; $d_2 = 0.5820 - 0.2828$ = **0.2992**; interpolating tails (0.55: 0.2912, 0.60: 0.2743; 0.25: 0.4013, 0.30: 0.3821) → $N(d_1)$ = **0.7197**, $N(d_2)$ = **0.6176**; call = $80 \times 0.7197 - (75/1.062) \times 0.6176 = 57.57 - 43.61$ = ₹ **13.96**. ICAI's teaching note gives the same z as cumulative (0.25: 0.5987, 0.30: 0.6179, 0.55: 0.7088, 0.60: 0.7257) and two-tail (0.8026, 0.7642, 0.5823, 0.5485) areas.

12.7% of marks

## 9.1 How Real Option is different from Financial Option

Asked: Jan 2026 Q2(c) 4m

| Basis | Financial options | Real options |
|---|---|---|
| Underlying | Traded assets — shares, bonds, commodities | Projects, not traded |
| Pay-off | Specified in contract, fixed | Estimated from project cash flows, can vary |
| Exercise period | Short, maximum up to 1 year | Mostly from end of year 1, longer |
| Approach | Traded → "**Priced**" | Used for decisions → "**Valued**" |

→ **Long call**: right to invest / enter / expand. **Long put**: right to sell or abandon; flexibility to exit. **Short call**: promise to sell if counterparty buys (commitment to disinvest). **Short put**: promise to buy if counterparty sells (commitment to invest).

## 9.3.2 Abandonment Option

Asked: Jan 2026 Q6(b) 4m

→ Like an **American put**: exercise if the value from the project's assets (disposal) exceeds PV of continuing. Pay-off = Max(0, abandonment value − project value).

→ Risk-neutral p from returns: **risk-free rate = p × up return + (1 − p) × down return** (as Illus 5: 8% = p × 30% + (1 − p) × (−40%)); value = expected pay-off ÷ (1 + risk-free rate).

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**Illus 5 (IPL)**: PV 100 crore → 130 (+30%) / 60 (−40%); disposal 80 → pay-off 0 / **20**. 0.08 = 0.30p − 0.40(1 − p) → p = 0.48/0.70 = **0.686**; expected 0.314 × 20 = **6.28 crore**; ÷ 1.08 = **₹ 5.81 crore** (value of put).

## 9.3.3 Timing Option

Asked: May 2026 MCQ 6m

→ Third choice besides accept/reject: **delay** (better information later) — like an **American call**.

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**Illus 6 (MIS Ltd, solar plant ₹ 2.50 crore, saving ₹ 21 lakh p.a. forever, WACC 10%, risk-free 8%)**: today 21 lakh/0.10 = 2.1 crore → NPV **−0.4 crore**. After 1 year: 35 lakh → 3.50 crore, NPV **+1 crore**, return 40%; 12 lakh → 1.20 crore, NPV **−1.3 crore**, return −52%. 0.40p − 0.52(1 − p) = 0.08 → 0.92p = 0.60 → p = **0.652**. Expected pay-off = 0.652 × 1 + 0.348 × (−1.30) = **₹ 19.96 lakh**; PV 19.96/1.08 = **₹ 18.48 lakh** (or /1.10 = 18.15 lakh) → **wait and see**.

*Note (app-written) on Illus 6 (p 9.47): ICAI includes the down-state NPV of −1.3 crore in the expected pay-off. By SM 7's own rule that an option's value cannot be negative, a firm that waits would not invest in the down state (pay-off 0), giving 0.652 × 1 crore before discounting. Keep ICAI's printed ₹ 19.96 / 18.48 lakh; follow the question's wording in the exam.*

## Exam steps

**Exam steps — futures price and arbitrage**: 1 → pick formula by what is given (simple S + S·r·t − div; compound $(1 + r/n)^{nt}$; continuous $Se^{rt}$, $(S - PV\ div)e^{rt}$, $Se^{(r-y)t}$) → 2 → compare with quoted futures → 3 → quoted > fair: borrow, buy spot, sell futures; quoted < fair: sell spot, invest, buy futures → 4 → profit = difference at expiry (TYK P2, P4, P6).

**Exam steps — beta hedge with index futures**: 1 → portfolio β = Σ wβ (cash β 0) → 2 → contract value = futures price × lot → 3 → contracts = value × β ÷ contract value (long cash → sell; short cash → buy; net long and short positions) → 4 → change β: P(β − β′) ÷ F → 5 → close-out P/L = change in shares + change in futures (TYK P5, P7–P13).

12.7% of marks

**Exam steps — margin account:** 1 → initial = $\mu + 3\sigma$; maintenance = % of initial → 2 → daily MTM = lot × change in closing price → 3 → balance < maintenance → call to restore initial (TYK P14).

**Exam steps — option pay-offs / call + put combinations:** 1 → total premium paid → 2 → at each expiry price, exercise only the leg in the money (call $S > X$, put $S < X$) → 3 → net = exercise gain × lot − total premium → 4 → break-evens: call strike + total premium, put strike − total premium (TYK P15, P16, P19, P20).

**Exam steps — binomial / risk-neutral:** 1 → pay-offs at expiry → 2 → u, d, r (as given: compound, simple, $e^{rt}$) → 3 → $p = (r - d)/(u - d)$ → 4 → value each node backwards: $[p \cdot C_u + (1 - p) \cdot C_d] \div r$ → 5 → hedge $\Delta = (C_u - C_d)/(S_u - S_d)$ shares per call written (Illus 1, 2, TYK P21, P23, P24).

**Exam steps — riskless hedge:** 1 → $\Delta = (C_u - C_d)/(S_u - S_d)$ → 2 → end value of $\Delta$ shares − call written (same both states) → 3 → $(\Delta S_0 - C_0) \times r$ = end value → $C_0$ → 4 → expected return uses real probabilities (TYK P22).

**Exam steps — Black-Scholes:** 1 → adjust S for PV of dividend ($D\,e^{-rt}$) → 2 → $\log_e(S/X)$, $(r \pm v^2/2)t$, $v\sqrt{t}$ → $d_1$, $d_2 = d_1 - v\sqrt{t}$ → 3 → N() from table type given (one-tail: 1 − area for +z, area for −z), interpolate → 4 → $C = S \cdot N(d_1) - X\,e^{-rt}\,N(d_2)$ → 5 → put = −S + C + PV(X) → 6 → compare with market premium; state $N(d_1)$ = hedge ratio, $N(d_2)$ = probability of exercise (Illus 3, TYK P25).

**Exam steps — real options:** 1 → identify type (growth = European call; abandonment = American put; timing = American call) → 2 → up/down values and returns → 3 → p from risk-free rate = p × up return + (1 − p) × down return → 4 → option pay-off in each state → 5 → PV of expected pay-off at $r_f$; patent → B-S with $e^{-Dt}$ (Illus 4, 5, 6).

11.6% of marks

# 10 Ch 10 — Foreign Exchange Exposure and Risk Management

| Attempt | Q | M | Asked (MCQ / numerical / theory) |
|---|---|---|---|
| Sep 2025 | MCQ CS-IV 10-12 | 6 | Forward quotes from swap points: interpolating swap points for a broken date (2 months 15 days), forward bid/ask, annualised premium/discount (MCQ) |
| Sep 2025 | Q1(b) | 4 | Importer's yen payable: expected loss if left unhedged (expected spot after a 10% decline) versus forward cover; recommendation (Numerical) |
| Sep 2025 | Q6(b) | 7 | Receivable hedge for a UK company: money-market hedge (borrow now) versus 12-month forward; net GBP proceeds and advice (Numerical) |
| Jan 2026 | MCQ CS-II 4-6 | 6 | Early delivery of a forward sale contract: swap loss, interest on outlay of funds, net conversion rate realised (MCQ) |
| Jan 2026 | Q2(a) | 6 | Import payment choices: pay now with 1% discount (cash plus loan) versus 60-day or 90-day credit with interest, using forward and spot rates; cheapest option (Numerical) |
| Jan 2026 | Q3(b) | 4 | Cross rates (bid and ask) from two direct quotes and arbitrage test through direct versus cross-rate route (Numerical) |
| May 2026 | MCQ CS-V 13-15 | 6 | Forward purchase contract: maturity settlement cost, closing out and rebooking, early settlement outflow, mark-to-market gain/loss on cancellation (MCQ) |
| May 2026 | Q5(c) | 2 | Why transaction exposure is generally more critical than translation exposure (Theory) |

## 5.4 Cross Rates

Asked: Jan 2026 Q3(b) 4m

- **Cross rate**: rate between a pair of currencies neither of which is the currency of the country where it is quoted (CAD/GBP quoted in India). In the Bloomberg table, rows are direct quotes for the first-column currency; columns are indirect quotes for the first-row currency.
- **Chain rule** (SM TYK P6, P3): **₹/$ x $/¥ = ₹/¥**; two-way: $(£/₹)_{ask} = (£/\$)_{ask} \times (\$/₹)_{ask}$. *App-written generalisation:* bid leg × bid leg gives the cross bid, ask leg × ask leg gives the cross ask; if one leg is quoted the other way, divide: ₹/HK$ = ₹ 42.85 / 7.5880 (TYK P7).
- Pick each leg from the side the bank actually deals on: when the bank buys the third currency, it buys $ at ask locally and sells $ for the third currency at the bid abroad.
- **Arbitrage test**: compute the cross rate; if a quoted rate differs, go round the markets (buy where cheap, sell where dear) and compare the end amount with the start.

| SM item | Working | ICAI answer |
|---|---|---|
| TYK P3 (buy ₹ 25 m with £) | (£/₹) = (1/1.5260) × (1/61.3625) = £ 0.01068 ≈ 0.0107 | **£ 2,67,500** |
| TYK P7 (HK$ 1 crore sold at ₹ 5.70) | Cover: buy $ at 42.85, sell $ for HK$ at 7.5880 → ₹ 42.85 ÷ 7.5880 = ₹ 5.6471 | Profit **₹ 5,29,000** |
| TYK P8 (DKK 10 lakh sold at ₹ 6.5150 = ₹ 65,15,000) | London: 10,00,000 ÷ 11.4200 = £ 87,565.67 × 74.3200 = ₹ 65,07,881. New York: ÷ 7.5670 = $ 1,32,152.77 × 49.2625 = ₹ 65,10,176 | Cover in **London**, profit **₹ 7,119** (NY ₹ 4,824; saving ₹ 2,295) |
| TYK P9 (SGD 25 lakh remitted late) | 28 Jan: 45.90 × 1.7850 ÷ 3.1575 = 25.9482 + 0.125% (0.0324) = 25.9806. 4 Feb: 45.97 × 1.7775 ÷ 3.1380 = 26.0394 + 0.0325 = 26.0719 | Loss to importer **₹ 2,28,250** |
| TYK P13 (3-market arbitrage, USD 1 crore) | $ → ₹ at 48.30 = ₹ 48,30,00,000 → £ at 77.52 = £ 62,30,650.155 → $ at 1.6231 = $ 1,01,12,968.26 | Gain **USD 1,12,968.26** |
| TYK P46 (sold $ 10 lakh at € 1.4400, squared up at 1.4450) | € 14,40,000 − € 14,45,000 = loss € 5,000; €/₹ = 31.4500 ÷ 1.4400 = ₹ 21.8403 | Loss **₹ 1,09,201.50** |
| TYK P47 (i) (buy £ 1 m with CHF) | Buy $ from Bank A at CHF 1.4655; buy £ from Bank B at $ 1.7650 → 1.7650 × 1.4655 = CHF 2.5866 | **CHF 25,86,600** |

11.6% of marks

*Note (app-written): TYK P3 — ICAI labels the rate $(£/₹)_{ask}$ but multiplies the reciprocals of $/£ 1.5260 and ₹/$ 61.3625; rounding 0.01068 up to 0.0107 gives £ 2,67,500, whereas 25,000,000 × 0.010679 ≈ £ 2,66,980. TYK P8 — the London rupee cost is printed "₹ 65,07,88"; the profit line shows ₹ 65,07,881.*

## 5.8 Broken period forward rate

Asked: Sep 2025 MCQ CS-IV 6m

- Interbank forwards are for standard amounts and **standard month-end dates**; customer deals are odd amounts and odd dates (export bill realised 23 Feb). Points for the odd date are **interpolated** — assumed to grow **uniformly (pro rata)** over the period.
- SM: premium for the odd date = **10*53/59** paise, "53 & 59 are broken & full periods", i.e. full-period points × broken period ÷ full period. Between two standard dates (TYK P5): (c) = (b) − (a) for the month between, (d) = (c)/2 for 15 days, (e) = (a) + (d).
- Then: **merchant forward rate = spot ± broken points ± bank's margin** (margin deducted for exporter, added for importer).

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**SM example:** 1 Jan spot ₹ 62, 2-month (28 Feb) premium 10 paise → 62.10; margin 3 paise → importer 62.13, exporter 62.07. For 23 Feb: 10 × 53/59 = **8.98 paise** → 62.0898 ≈ 62.09 → exporter **62.06**, importer **62.12**.

**TYK P5 (Sep 2025 MCQ type):** 2m points 70/90, 3m 160/186 → 30 days 90/96 → 15 days 45/48 → 2m 15d = **115/138** (1 point = 0.0001) → rate for 20 June = 66.2525 + 0.0115 = **66.2640** / 67.5945 + 0.0138 = **67.6083**. Annual premium on average rate: bid 0.0115 ÷ 66.2583 × 12/2.5 × 100 = **0.0833%**; ask 0.0138 ÷ 67.6014 × 12/2.5 × 100 = **0.0980%**.

## 10. Foreign Exchange Exposure

Asked: May 2026 Q5(c) 2m

- **Exposure**: "a contracted, projected or contingent cash flow whose magnitude is not certain at the moment; depends on FX rates and interest rates". FX exposure = parts of business affected if exchange rates change.
- Examples: traveller (destination currency appreciates); **exporter** invoicing in FC (FC falls → lower home revenue); **importer** (FC appreciates → higher cost); fund managers owning foreign assets (currency falls); trade in local currency where rates move the negotiated price.
- Timing view: **translation** and **transaction** exposure look at the moment of the rate change (accounting / settled obligations); **operating (economic)** exposure looks at expected future cash flows after an unexpected change.

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**TYK T1** ("operations in foreign exchange are exposed to a number of risks") → this para 10 (three exposures). **TYK P1:** bond return (5,250 − 5,000 + 350)/5,000 = 12% for a US investor; non-US investor with $ up 2%: 1.12 × 1.02 − 1 = **14.24%**.

## 11. (B) External techniques — (i) Money Market Hedge (MMH)

Asked: Sep 2025 Q6(b) 7m

- Fixes today the home-currency value of a future FC flow by **borrowing and investing in the money markets** at today's spot (SM: owner expecting $ 1 m in six months locks the INR value now).
- **Advantages:** (a) fixes future rate — no downside risk; (b) **flexible amount**; (c) usable where **forwards are not available**. **Disadvantages:** (a) **more complicated** than a forward; (b) no benefit from favourable movements.

11.6% of marks

| Step (SM TYK P29 wording) | FC receivable (exporter) | FC payable (importer) |
|---|---|---|
| 1 Identify / Create | FC is an asset → create an FC **liability** | FC is a liability → create an FC **asset** |
| 2 Borrow / Deposit FC | **Borrow FC** = receivable ÷ (1 + FC borrowing rate for period) | **Deposit FC** = payable ÷ (1 + FC deposit rate for period) |
| 3 Convert at spot | Sell FC at the rate worse for you (bid in direct quote) | Buy FC with home currency (ask in direct quote), borrowed at home |
| 4 Invest / Borrow home | Invest home currency at home **deposit** rate | Home loan repaid at home **borrowing** rate |
| 5 Settle | Customer's FC repays the FC loan; keep home deposit maturity | FC deposit matures = payable; pay home loan + interest |
| 6 Compare | With forward receipt (higher is better) | With forward cost (lower is better) |

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**TYK P28 (Rohit, £ 5,00,000 receivable):** borrow £ 5,00,000 ÷ 1.0125 = £ 4,93,827 × ₹ 56 = ₹ 2,76,54,312; invest at 12% for 3 months = **₹ 2,84,83,941**; gain ₹ 4,83,941 over ₹ 56 × 5,00,000.

**TYK P29 (UK exporter, $ 3,50,000):** borrow at 9% (2.25%/quarter) $ 3,42,298.29; convert at ask 1.5905 (indirect quote) = £ 2,15,214.27; invest at 5% → **£ 2,17,904.45**; forward 3,50,000 ÷ 1.6140 = £ 2,16,852.54 → MMH better by **£ 1,051.91**.

**TYK P38 (Columbus, payable £ 3,42,000):** forward 3,42,000 ÷ 0.9520 = $ 3,59,244; MMH deposit £ 3,42,000 ÷ 1.025 = £ 3,33,658 ÷ 0.9830 = $ 3,39,429 loan × 1.0325 = **$ 3,50,460** → MMH cheaper. **Receipt € 5,90,000 (4 months):** forward × 1.9510 = **$ 11,51,090**; MMH borrow € ÷ 1.0533 = € 5,60,144 × 1.8890 = $ 10,58,113 × 1.0383 = $ 10,98,639 → forward better.

**TYK P39 (b):** £ 3,00,000 ÷ 1.045 = £ 2,87,081 × $ 2 = $ 5,74,162 × 1.055 = **$ 6,05,741**. **TYK P40 (ii):** $ 3,64,897 ÷ 1.0225 = $ 3,56,867 × 0.64033 = £ 2,28,512 + 3.5% interest £ 7,998 = **£ 2,36,510**.

## 12. Forward Contract

Asked: Sep 2025 Q1(b) 4m

Asked: Jan 2026 Q2(a) 6m

- → Simplest derivative: obliges one party to buy and the other to sell a specified quantity at a **specific price on a specified future date**. FX forwards, FRAs, commodities. Importers can cover all liabilities by forwards.
- → **Hedge choice** (all SM answers): exporter **sells FC forward** (bank's bid); importer **buys FC forward** (bank's ask). Compare: forward outcome vs unhedged at expected spot.
- → **Expected spot** with probabilities = Σ (rate × probability) (TYK P15, P39).
- → **Pay now vs credit:** pay now = spot cost + borrowing interest for the period; pay later = (FC + supplier's interest) × forward rate (+ bank loan interest for any gap). Choose the lower outflow.

11.6% of marks

| SM item | Working | ICAI answer |
|---|---|---|
| TYK P17 (importer, 108 lakh yen) — Sep 2025 Q1(b) type | Spot 108/30 = 3.6 yen/₹; −10% → 3.24 → cost 33.33 lakh; forward 3.3 → 32.73 lakh | Loss unhedged **₹ 3.33 lakh**; with forward **₹ 2.73 lakh** → take forward |
| TYK P6 (JKL, export ¥ 1 crore) | Cross ₹/¥: 62.22/102.34 = 0.6080; expected 65/124 = 0.5242; forward 66.50/110.35 = 0.6026; actual 66.25/110.85 = 0.5977 | Unhedged loss **₹ 8,38,000**; forward loss ₹ 54,000; actual unhedged loss ₹ 1,03,000 → forward justified |
| TYK P4 (ABC UK, Can$ 5 lakh) | Forward Can$ 2.535 → £ 1,97,239. Spot 2.55 → £ 1,96,078; 2.40 → £ 2,08,333; 2.50 → £ 2,00,000 | Gain £ 1,161; loss £ 11,094; loss £ 2,761 |
| TYK P11 (iii) ($ 69,000) | Now 69,000 × 46 = 31,74,000 + 10% for 2 months 52,900 = 32,26,900; forward 69,000 × 47 = 32,43,000 | Encash **after 2 months** |
| TYK P15 (Multinational) | EV = 0.24 + 0.34 + 0.45 + 0.38 + 0.40 = $ 1.81 vs forward $ 1.80 | **Retain** receivable (don't sell forward) |
| TYK P12 (Z Ltd, USD 2 m) — Jan 2026 Q2(a) type | (i) 2 m × 57.10 = ₹ 11,42,00,000 + loan 10% 30 days ₹ 9,51,667; (ii) (2 m + 8% 30 days $ 13,333) × 57.50 | (i) **₹ 11,51,51,667** vs (ii) ₹ 11,57,66,648 → pay in 60 days |
| TYK P22 (DEF, US$ 1 crore) | (i) 63.15 cr + 9.5% 30 days 0.50 cr; (ii) 1.00646 cr × 63.45 | (i) **₹ 63.65 cr** vs (ii) ₹ 63.86 cr → (i) |
| TYK P21 ($ 1,30,000) | Now: 1,30,000 × 48.36 = 62,86,800 + OD 15% 3 months 2,35,755 = 65,22,555. Later: 1,31,625 × 48.83 | Later **₹ 64,27,249** → pay after 3 months |
| TYK P35 (Gibralter, $ 1,00,000) | (i) $ 1,02,500 × 60.30 = 61,80,750; (ii) 1,00,000 × 60.55 = 60,55,000 + 14% 3 months 2,11,925 | (i) **₹ 61,80,750** vs ₹ 62,66,925 → (i) |
| TYK P42 (Sun, 3,400 lakh yen) | Loan 18% quarterly: 1,000 × 1.045² = 1,092.03. LC: commission 10.00 + interest 0.90 = 10.90; 3,433.53 lakh yen ÷ 345 × 100 = 995.23 | **₹ 1,006.13 lakh**; cheaper by ₹ 85.90 lakh → accept |
| TYK P19 (Excel, $ 1 lakh) | (45.20 − 45.60)/45.60 × 365/60 × 100; loss (45.20 − 45.50) × 1,00,000 | Discount **5.33%**; loss **₹ 30,000** |
| TYK P23 ($ 60,000, forward 64, 2% upfront) | Premium 60,000 × 64 × 2% = 76,800 + 12% for 6 months 4,608 = hedging cost 81,408 | At 68: gain 1,58,592; 62: loss 2,01,408; 70: gain 2,78,592; 65: loss 21,408 |
| TYK P24 (AKC, indirect quotes) | Hedged at 3m forward ask (2.427, 0.0216, 0.0178) = ₹ 1,33,38,719, contribution 26,08,719; unhedged at 3m spot ask = ₹ 1,32,75,578, contribution 25,45,578 | C/S **19.56%** vs 19.17% → hedge |

## 12.2 Fate of Forward Contracts

Asked: Jan 2026 MCQ CS-II 6m

Asked: May 2026 MCQ CS-V 6m

→ Three fates, each on due date / early / late: **(A) Delivery**, **(B) Cancellation**, **(C) Extension**.

→ **Core rule — cancel by the reverse deal**: bank's **sale** contract (importer) is cancelled at the bank's **buying** rate; bank's **purchase** contract (exporter) at the bank's **selling** rate. Margin deducted from buying, added to selling. Difference vs contract rate is charged to / paid to the customer (+ flat charges).

11.6% of marks

| Fate | Rate applied (SM) | SM figures |
|---|---|---|
| **Delivery on due date** | Contract rate, whatever the spot. | Illus 3: $ 1,00,000 at 65.5000 → debit **₹ 65,50,000** |
| **Early delivery** | Contract rate + charges: (a) **swap difference** (bank undoes its cover: sells/buys spot now, reverses at forward for original date); (b) **interest on outlay of funds** at not less than **PLR** from early date to due date (if inflow, bank may pass on term-deposit interest). | Illus 4 below |
| **Late delivery** | Current (spot) rate on delivery date, after applying **automatic cancellation**. | — |
| **Cancellation on due date** | Reverse at **spot** (TT) rate ± margin; difference + flat charges. | Illus 5 below |
| **Early cancellation** | Reverse at the **forward rate for the original due date** (opposite side). | Illus 6; TYK P44, P45 |
| **Late cancellation** | Automatic cancellation provisions. | — |
| **Extension on due date** | Cancel at spot (as cancellation on due date) + **rebook** at forward for the new period. | Illus 7; TYK P32 |
| **Extension before due date** | Cancel at forward for original date + rebook at current forward for new date. | Illus 8 |
| **Late extension** | Current rates after automatic cancellation. | Illus 9 |

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**Illus 4 (Mr. X exporter, $ 1,00,000 at 65.40, delivered 28/30 Nov instead of 31 Dec; spot 65.22/65.27, 1-month 10/15, PLR 18%):** (a) **swap loss**: bank sells spot 65.22, buys forward 65.27 + 0.15 = 65.42 → 0.20 × 1,00,000 = **₹ 20,000**; (b) **outlay**: bank pays customer 65.40, sells spot 65.22 → 0.18 × 1,00,000 × 31/365 × 18% = **₹ 275**; charges ₹ 20,275; net inflow 65,40,000 − 20,275 = **₹ 65,19,725**.

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**Illus 5 (import $ 2,50,000 at 65.3450 cancelled on due date):** spot buying 65.2900 − 0.10% (0.0653) = 65.2247 ≈ **65.2250**; 65.3450 − 65.2250 = 0.12 × 2,50,000 = ₹ 30,000 + flat ₹ 100 = **₹ 30,100**.

**Illus 6 (purchase AUD 1,00,000 at 47.2500, cancelled after 2 of 3 months):** at 1-month forward sale 47.5200 → 0.27 → **₹ 27,000**. **TYK P44:** CHF 10,000 at 27.25, cancelled at 1-month sale 27.52 → **₹ 2,700**. **TYK P45:** USD/₹ 49.4455 + April 0.4200 = 49.8655 + 0.10% = 49.91537; ÷ 1-month CHF 1.5150 = 32.9474 ≈ 32.9475 − 32.4000 = 0.5475 → **₹ 54,750**.

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**Illus 7 (export $ 1,00,000 at 62.5200, extended on due date by 1 month):** cancel at spot selling 62.7200 + 0.10% = 62.78272 ≈ **62.7825** → 0.2625 → **₹ 26,250**; rebook at 1-month buying 62.6400 − 0.06264 = 62.57736 ≈ **₹ 62.5775**. **TYK P32 (import $ 20,000 at 42.32, extended 3 months):** cancel at spot buying 41.5000 − 0.075% = 41.4689 ≈ 41.47 → 8,46,400 − 8,29,400 = **₹ 17,000**; new rate 41.5200 + 0.93% (0.3861) = 41.9061 + 0.20% (0.0838) = 41.9899 ≈ **₹ 41.99**.

**Illus 8 (purchase $ 50,000 at 59.6000, on 5 May extended to 5 July):** cancel at June forward selling 59.2425 + 0.0592 = 59.3017 ≈ 59.3025 → 0.2975 × 50,000 = **₹ 14,875 payable to the customer**; rebook at July buying 59.6300 − 0.0596 = 59.5704 ≈ **₹ 59.5700**.

*Note (app-written): Illus 8 says "rebooked with the delivery 15th July"; the request and rate used are for 5th July.*

11.6% of marks

**Automatic cancellation (FEDAI Rule 6):** an overdue forward contract with no instructions on or before due date is **automatically cancelled within 3 working days** after maturity. Customer pays any exchange difference but is **not entitled to any profit**. Late delivery → fresh contract at spot; extension after due date → forward rate for the desired period. Charges:

- → (i) **Exchange difference**: spot rate of offsetting position on the cancellation date (or 3 working days, **whichever is earlier**) vs contract rate.
- → (ii) **Swap loss**: bank offsets at spot on the due date and takes the opposite position at the earliest forward.
- → (iii) **Interest on outlay of funds**: (bank's interbank cover rate − spot of opposite position on due date) × FC amount, for the period from due date to actual cancellation or 3 working days, **whichever is later**.
- → Any profit to the bank is **not passed on** to the customer.

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**Illus 9 (importer $ 2,00,000 at 64.4000, cover 64.2800, due 10 June, extension asked 14 June to 10 Aug; margin 0.10%, interest 12%):** (i) cancellation rate = spot buying 13 June 63.6800 − 0.0636 = 63.6163 ≈ **63.6175**; (ii) 1,28,80,000 − 1,27,23,500 = **₹ 1,56,500**; (iii) swap on 10 June: sell spot 63.8000, buy June forward 63.9500 → 0.15 → **₹ 30,000**; (iv) outlay 64.2800 − 63.8000 = 0.48 × 2,00,000 = ₹ 96,000 × 12% for 3 days = **₹ 96**; (v) new rate August selling 64.2500 + 0.0643 = 64.3143 ≈ **₹ 64.3150**; (vi) total cost **₹ 1,86,596**.

Note (app-written): Illus 9 — ₹ 96,000 × 12% × 3/365 = ₹ 94.68; ICAI's ₹ 96 equals a 360-day year. Also 63.6800 − 0.0636 is 63.6164 (ICAI prints 63.6163 using the unrounded margin 0.06368); the rounded rate 63.6175 is unaffected.

## Exam steps

**Exam steps — which rate:** write each quote as "home per foreign" (direct) → customer sells FC = bid, buys FC = ask (bank's view: buys at bid, sells at ask) → indirect quote: pick the side that gives the customer less → margin: deduct from buying, add to selling → round as told (0.0025 / 0.0001 / whole rupee).

**Exam steps — forward rate from points / broken date:** ascending points = premium (add), descending = discount (deduct) → points × 0.0001 → for an odd date, interpolate pro rata (points for full period × days ÷ full days, or add the share of the next month's increment) → add margin → annualise [(Forward Rate − Spot rate)/(Spot rate)] x 12/months x 100.

**Exam steps — cross rate / arbitrage:** chain ₹/$ x $/¥ = ₹/¥ (bid × bid, ask × ask; divide where a leg is inverted) → compare with the quoted rate → route: buy in the cheaper market, sell in the dearer → end amount − start = gain.

**Exam steps — IRP arbitrage:** $(1 + r_D) = {}^{F}/_{S}\,(1 + r_F)$ with period rates → annualised premium vs interest differential → borrow the currency that is "cheap" after the forward, convert spot, invest, convert back at forward, repay loan + interest → gain.

11.6% of marks

**Exam steps — hedge comparison (exporter / importer):** forward (amount × forward) → MMH (PV at FC rate → spot → compound at home rate) → futures (contracts = exposure ÷ size; futures P/L + spot conversion − margin interest) → options (whole contracts, exercise if favourable, premium at spot + interest, balance at forward) → unhedged (expected spot = Σ rate × probability) → receivable: pick highest; payable: pick lowest.

**Exam steps — fate of forward contract:** identify bank's side (sale to importer / purchase from exporter) → reverse at opposite rate: spot (on due date), forward for original date (early), spot on cancellation date (after due date, auto-cancel in 3 working days) ± margin → difference × amount + flat charges → early delivery / late: add swap loss (spot vs forward reversal of cover) + interest on outlay (PLR / given rate × days) → extension: rebook at forward for new date ± margin → late: profit to bank not passed on.

**Exam steps — import payment options:** pay now = FC × spot (ask) + borrowing interest to due date; pay later = (FC + supplier interest) × forward (ask); gap periods add bank-loan interest → lowest rupee outflow wins.

**Exam steps — exchange and cash position:** list purchases vs sales (all deals, incl. forwards, bills, cancellations) → overbought / oversold → Nostro: only ready deliveries (TT remitted / received) → spot deal brings Nostro to target (also moves position) → forward deal sets the required overbought/oversold position.

10.7% of marks

# ⑥ Ch 06 — Portfolio Management

| Attempt | Q | M | Asked (MCQ / numerical / theory) |
|---|---|---|---|
| Sep 2025 | Q4(a) | 6 | Single-index risk split: systematic risk (beta squared x market variance) and unsystematic risk; two-stock portfolio risk; effect of a 60:40 reallocation (Numerical) |
| Jan 2026 | MCQ CS-III 7-9 | 6 | CAPM/SML from portfolio data: market return, risk-free rate from average portfolio return and betas, expected return of one share (MCQ) |
| Jan 2026 | Q5(a) | 8 | Ranked table by Treynor ratio, cut-off point and securities in the optimal portfolio (single index model) (Numerical) |
| May 2026 | MCQ CS-I 1-3 | 6 | Sovereign Gold Bond (SGB): post-tax annual interest and real post-tax return; indexed cost and capital gain; net amount after capital gains tax (MCQ) |
| May 2026 | Q2(a) | 6 | Two-factor APT: required return = Rf + sum of risk premium x beta, tested against actual return; long and short amounts from two simultaneous equations for target sensitivities (Numerical) |
| May 2026 | Q4(a) | 6 | Sharpe single-index model optimal portfolio with given cut-off rate: securities selected, proportions, CAPM expected return and value, consistency with CAPM (Numerical) |

## 4.4.2 Portfolio Risk

**Asked: Sep 2025 Q4(a) 6m (two-stock risk, 60:40 reallocation)**

→ Portfolio variance = weighted variances of constituents **+** weighted covariances among pairs. **Not** a simple weighted average of variances.

→ **Covariance**: how returns of two securities move relative to each other — same direction consistently (+), opposite (−), independent (close to 0).

| Formula | Terms / logic | SM example figures |
|---|---|---|
| $Cov_{AB} = \Sigma[(R_A - \bar{R}_A)(R_B - \bar{R}_B)] \div N$ (with probabilities: $\Sigma\ p \times dev_A \times dev_B$) | Product of each period's deviations from the means, averaged over N observations. | 4 years, $\bar{R}_x$ 15, $\bar{R}_y$ 13: Σ products −42 → Cov = −42/4 = **−10.5**. |
| $r_{AB} = Cov_{AB} \div (\sigma_A\ \sigma_B)$; $Cov_{AB} = \sigma_A\ \sigma_B\ r_{AB}$ | Correlation ranges −1 to +1: −1 perfect negative, +1 perfect positive, 0 independent. | — |
| $\sigma_p^2 = x_1^2\sigma_1^2 + x_2^2\sigma_2^2 + 2x_1x_2\ r_{12}\ \sigma_1\sigma_2$ | $x_1 + x_2 = 1$. Last term = $2x_1x_2Cov_{12}$. $\sigma_p = \sqrt{\sigma_p^2}$. | A: r 20%, σ 50%; B: r 25%, σ 30%; $r_{AB}$ −0.60; 40:60 → $R_p$ **23%**; $\sigma_p^2$ = 400 + 324 − 432 = **292**; $\sigma_p$ **17.09%**. |

→ Return and risk depend on (a) returns, risks and covariances of securities — **parametric**, no control; (b) **proportions** — the investor's **choice** variable.

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**TYK P3 (B σ 10%, ER 12%; D σ 18%, ER 20%; r 0.15):** (i) 100% B: 12%, σ 10%; (ii) 50:50: 16%, $\sigma^2$ 25 + 81 + 13.5 = 119.50, σ **10.93%**; (iii) 75:25: 14%, $\sigma^2$ 56.25 + 20.25 + 10.125 = 86.625, σ **9.31%**; (iv) 100% D: 20%, 18%. Return → (iv) best; risk → (iii) best. **TYK P4:** E(A) 13%, E(B) 15%, 40:60 → **14.2%**; $\sigma_A = \sigma_B$ = 3%; Cov 9; r = 9/(3 × 3) = **1**; $\sigma_p = \sqrt{(1.44 + 3.24 + 4.32)}$ = **3%**. **TYK P6 (10 years):** $\bar{R}_1$ 14.8, $\bar{R}_2$ 21; Σ products −8 → Cov **−0.8**; $\sigma_1 = \sqrt{20.76}$ = 4.56, $\sigma_2 = \sqrt{8.40}$ = 2.90 → r = −0.8/13.22 = **−0.0605**.

## 7. Single Index Model (Sharpe Index Model)

**Asked: Sep 2025 Q4(a) 6m**

→ Co-movement between stocks is due only to movement in the **market index**; no effects beyond the market (no fundamental factor effects) explain co-movement. β and α estimated by regression.

10.7% of marks

| Formula | Terms / logic | SM / TYK figures |
|---|---|---|
| $R_i = \alpha_i + \beta_i R_m + \in_i$ | $\alpha_i$ intercept; $\beta_i$ slope (sensitivity; $\beta$ 1.5 → +1.5% per +1% market); $R_m$ index return; $\in_i$ error term (unsystematic). Return = market part + part independent of market. $\alpha + \beta R_m$ = **characteristic line**. | TYK P11: $\beta = (508 - 6 \times 5.75 \times 6.33)/(439.25 - 6 \times 5.75^2) = 289.615/240.875 =$ **1.202**; $\alpha = 6.33 - 1.202 \times 5.75 =$ **−0.58** → line $-0.58 + 1.202\ R_m$. |
| $\sigma_i^2 = \beta_i^2\sigma_m^2 + \in_i^2$ | Total variance = **systematic** (explained by index) + **residual/unsystematic** (unexplained). | TYK P11: $\sigma_m^2$ 240.86/6 = 40.14; $\sigma_A^2$ 497.34/6 = 82.89; systematic $1.202^2 \times 40.14 =$ **57.99**; unsystematic **24.90**. |
| Systematic = $\beta_i^2 \times \sigma_m^2$; Unsystematic $\in_i^2 = \sigma_i^2 -$ Systematic risk = Variance of Security Return $(1 - r^2)$ (Illus 1) | $r^2$ (coefficient of determination) = % of variation explained by the index. | Illus 1 below. |
| $\sigma_{ij} = \beta_i\beta_j\sigma_m^2$ | Covariance between two stocks comes only through the market. | TYK P14: Cov A,B = 0.80 × 1.14 × 225 = **205.2**; A,C **108**; B,C **153.9**. |
| $\sigma_p^2 = [(\Sigma X_i\beta_i)^2\sigma_m^2] + [\Sigma X_i^2\in_i^2]$ | First part = $\beta_p^2\sigma_m^2$ (portfolio systematic); second = weighted residual variances (weights **squared**). $X_i$ = portion of stock i; $\in_i^2$ = variation not related to the index. | Illus 1: **3.5876**. |
| $R_P = \Sigma\ x_i(\alpha_i + \beta_i R_m)$; $\alpha_P = \Sigma\ x_i\alpha_i$; $\beta_P = \Sigma\ x_i\beta_i$ | Portfolio return, alpha and beta are weighted averages using investment proportions. | — |

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**Illus 1 (X, Y, Sensex):** $\sigma_m^2$ 2.25. X: $\beta$ 0.71, variance 6.30, $r^2$ 0.18 → systematic $(0.71)^2 \times 2.25 =$ **1.134**; unsystematic 6.30 − 1.134 = **5.166** (= 6.30 × 0.82). Y: $\beta$ 0.685, variance 5.86 → systematic **1.056**, unsystematic **4.804**. Equal weights: $[(0.355 + 0.3425)^2 \times 2.25] + [(0.5)^2(5.166) + (0.5)^2(4.804)] = 1.0946 + 2.493 =$ **3.5876**.

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**TYK P14 (equal weights):** matrix diagonals 400, 324, 144 ($\sigma^2$) → total variance **200.244**; $\beta_p = (0.80 + 1.14 + 0.60)/3 =$ **0.8467**; systematic $(0.8467)^2(15)^2 =$ **161.302**; unsystematic **38.942**. **TYK P34:** $\beta_p$ **0.66**; $\sigma_m^2$ 0.01 → residual A 0.0134, B 0.0225, C 0.0879; SIM variance 0.004356 + 0.014072 = **0.018428**; Markowitz (9-cell matrix with covariances) **0.0363**. **TYK P37:** $\beta_p$ **1.295**; $\sigma_p = [543.36 + 3.0625 + 10.89 + 0.5625 + 3.24]^{½} = \sqrt{561.115} =$ **23.69%** (ICAI alternative: 25.03%).

*Note (app-written): TYK P37's "alternative answer" (SM p. 125) weights each security's total variance by w (not $w^2$) and adds no covariance, so its 25.03% is not the single-index portfolio risk. The formula above gives ICAI's main answer, 23.69%.*

## 8. Capital Asset Pricing Model (CAPM)

Asked: Jan 2026 MCQ CS-III 7-9 6m

→ Portfolio risk = unsystematic (averted by diversification) + systematic (market-related, measured by β). Only **non-diversifiable** risk is priced; market β = 1.0; portfolio β = Σ proportion × β.

→ Three aspects: market not concerned with diversifiable risk; not concerned with investor holding a diversified portfolio; **compensation only for non-diversifiable risk**.

**Assumptions (TYK T3):**

10.7% of marks

→ **Efficient market** (full information, prices reflect intrinsic value; Hampton — CAPM usable for capital budgeting); **rational investment goals**; **risk aversion**.

→ Assets **divisible and liquid**; **borrow freely at risk-less rate**; **no brokerage, taxes or transaction cost**; **no bankruptcy**; capital market in **equilibrium**.

→ Required return = minimum expected return for investors to buy and hold the asset.

| Formula | Terms / logic | SM figures |
|---|---|---|
| $R_j = R_f + \beta(R_m - R_f)$ | $R_f$ risk-free return; $(R_m - R_f)$ **market risk premium** (reward for market risk); β × premium = stock's risk premium (β tells how much riskier than market). | $R_f$ 5%, $R_m$ 15% → premium 10%; 30% riskier → 10% + 3% = 13% → 5% + 1.3 × 10% = **18%**. Premium 6%, β 1.2 → **7.2%**. |
| **SML**: required return (Y-axis) vs β (X-axis); intercept $R_f$, slope $(R_m - R_f)$ | Relationship between **systematic risk** and return; higher β → higher premium. | — |
| $\beta_p = \Sigma\, w_i\beta_i$ (risk-free β = 0, market β = 1) | Mixing T-bills and market changes β linearly. | Illus 4. |

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**Illus 2:** β 1.40, $R_m$ 14%, $R_f$ 10% → 10% + 5.6% = **15.6%**; premium +2.50 points → $R_m$ 16.5% → 10% + 1.40 × 6.5% = **19.10%**. **Illus 3:** premium 10%, β 0 / 0.25 / 0.42 / 1.00 / 1.67 → **0% / 2.50% / 4.20% / 10.00% / 16.70%**. **Illus 4:** $R_f$ 5%, $R_m$ 13% → premium **8%**; T-bill:market 100:0 β 0 → 5%; 70:30 β 0.3 → **7.40%**; 30:70 β 0.7 → **10.60%**; 0:100 β 1 → 13%.

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**TYK P13:** β aggressive (40 − 4)/(25 − 7) = **2**; defensive (18 − 9)/(25 − 7) = **0.50**; expected returns **22%** and **13.5%**; $R_m$ 16% → SML: **7.5% + $\beta_i$ 8.5%**; alphas ($R_s = \alpha + \beta R_m$): $\alpha_A$ **−10%**, $\alpha_D$ **5.5%**. **TYK P16:** premium 7% → returns 10.15/9.45/15.05/19.95% → ₹ 1,45,933 on ₹ 9,05,000 = **16.13%**; β **1.30** (weighted 1.3035 → 16.1245%). **TYK P19:** $R_f$ = 7/140 = 5%; 15% = 5% + β(10%) → β 1; 1 = 225/$\sigma_m^2$ → $\sigma_m$ **15%**; 1 = $\sigma_X$ × 0.75/15 → $\sigma_X$ **20%**. **TYK P20 (two correctly priced shares):** 19.8 = $R_f$ + 1.2P, 17.1 = $R_f$ + 0.9P → P = 2.7/0.3 = 9 → $R_m$ **18%**, $R_f$ **9%**; SML = 9% + β × 9%. **TYK P21:** B better (higher return, lower risk); 70:30 → 22.6%, $\sigma^2$ 0.1374, σ **37%** (37.06% also correct); $R_m - R_f$ = −2/−0.38 = 5.26% → $R_f$ **17.48%**, $R_m$ **22.74%**; $\beta_p$ **0.974**.

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**TYK P23:** 20% each → β **1.22**; 15% A, 30% B, 15% C, 30% D, 10% E → β **1.335**; at 12% for β 1 → **14.64%** and **16.02%**. **TYK P25 (₹ 520 lakh):** CAPM returns 9 / 10 / 9.60 / 10.40 / 11% → portfolio **10.208%** (avg β 1.104); "Better" has β 1 = Nifty's 10% → replacement makes no difference. **TYK P36 (beta adjustment):** market value ₹ 12,00,000, $\beta_p$ **1.108**; to 0.8 → 0.8/1.108 = 72.2% → total ₹ 16,62,050 → buy risk-free **₹ 4,62,050** (alternative: 13,30,000/(12,00,000 + x) = 0.8 → **₹ 4,62,500**); to 1.2 → 108.30% → ₹ 11,08,030 → divest **₹ 91,970** (alternative **₹ 91,667**). **TYK P39:** $\beta_{asset}$ = 1.5 × 60/100 = **0.9** (debt riskless); cost of capital 8% + 0.9 × 10 = **17%** (or WACC 23% × 3/5 + 8% × 2/5 = 17%).

*Note (app-written): TYK P36's two methods differ only by rounding of 72.2% and 108.30% (SM p. 123–125); the direct equation gives ₹ 4,62,500 and ₹ 91,667. TYK P23(iii) takes return = 12% × β with no risk-free rate given (SM p. 111).*

10.7% of marks

## 8.1 Risk free Rate of Return

**Asked: Jan 2026 MCQ ($R_f$ from average portfolio return)**

→ CAPM has **only one** risk-free rate. Returns include both **dividend and capital appreciation** — factor both into β and required return.

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**Illus 5 (Pearl Ltd):** T-bill lowest bid 4.3% (conservative), 4.6% (aggressive) → moderate average **4.45%**; 15.50% = 4.45% + β(12% − 4.45%) → β = 11.05/7.55 = **1.464**. **Illus 6 (Jaykay):** returns = $[D_1 + (P_1 - P_0)]/P_0$: 25.62%, 20.07%, 17.05%; market = index appreciation + dividend yield: 12.62%, 21.79%, 5.32%; β = (851.36 − 830.55)/(662.36 − 525.89) = 20.81/136.47 = **0.15** → less affected by market. **TYK P8 (Krishna):** returns 12.24/32.41/16.13% (avg 20.26); market 10.81/16.33/16.79% (avg 14.64); β **1.897**; $R_f$ 6% → expected 15.12% (Sell), 25.60% (Buy), 26.47% (Sell).

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**TYK P26:** dividends 7,025 + capital gain 5,500 on cost 75,000 → $R_m$ **16.7%**; average β (0.6 + 0.8 + 0.6 + 0.01)/4 = 0.50 → 15.7 = $R_f$ + 0.50(16.7 − $R_f$) → $R_f$ **14.7%**; Gold 15.90%, Silver 16.30%, Bronze 15.90%, GOI 14.72%. **TYK P27:** $R_m$ = (146 + 145)/1,105 = **26.33%**; Epsilon 23.86%, Sigma 22.63%, Omega 20.17%, GOI 14.12% → average **20.20%**. **TYK P28:** $R_m$ = (5,800 + 5,000)/68,000 = **15.88%**; X 15.70%, Y 15.62%, Z 15.44%, PSU 15.18% → **15.49%**.

*Note (app-written): TYK P26's alternative (SM p. 113–114) computes β 0.387 and $R_f$ 5.83% from the GOI bond, then stops at 15.7% = 5.83% + 0.387($R_m$ − 5.83%) without solving. Use the main answer.*

## 9. Arbitrage Pricing Theory Model (APT)

**Asked: May 2026 Q2(a) 6m**

→ Alternative to CAPM: uses the asset's expected return and the **risk premia of several macro-economic factors**. Steps: (a) risk premium for each factor × its beta (sensitivity); (b) add $R_f$.

→ CAPM = one factor (market); APT needs no particular factor. **Stephen Ross** factors: **inflation, level of industrial production, risk premium, term structure of interest rates**. No surprise in factors → actual = expected.

| Formula | Terms / logic | SM figures |
|---|---|---|
| $E(R_i) = R_f + \lambda_1\beta_1 + \lambda_2\beta_2 + \ldots + \lambda_n\beta_n$ | $\lambda_n$ = factor price / risk premium of factor n; $\beta_n$ = sensitivity to factor n. Expected return in equilibrium. | Illus 9 below. |
| With surprises: $E(R) = R_f + \beta_1(EV_1 - AV_1) + \ldots + \beta_n(EV_n - AV_n)$ | (EV − AV) = surprise in factor value; β = sensitivity to that macro factor. (ICAI's TYK P38 uses actual − expected.) | TYK P38. |
| b prop. 1 = Σ (position in security ÷ own funds) × its Factor 1 sensitivity (same for b prop. 2) — as in TYK P15 | Position ÷ own funds; short sale and borrowing are negative weights; risk-free asset has sensitivity 0. | TYK P15. |

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**Illus 9:** 8% + 1.3 × 4% + 0.3 × 1% + 0.2 × (−4%) = 8 + 5.2 + 0.3 − 0.8 = **12.7%**. **TYK P38 (Tamarind):** (actual − expected) × β: 0 + 2.63 + 1.63 + 3.40 + 0.50 = 8.16 → 9.25 + 8.16 = **17.41%**. **TYK P15:** (i) +1.50 in A, −0.5 in B → $b_1$ 1.50 × 0.80 − 0.50 × 1.50 = **0.45**, $b_2$ **0.30**; (ii) borrow ₹ 1,00,000: A 3, B −1, $R_f$ −1 → **0.90** and **0.60**; (iii) 15 = 10 + 0.80$\lambda_1$ + 0.60$\lambda_2$, 20 = 10 + 1.50$\lambda_1$ + 1.20$\lambda_2$ → $\lambda_1$ 0, $\lambda_2$ = 5/0.60 = **8.33%**.

10.7% of marks

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**TYK P40 (Nirmal Kumar):** (a) APT: small cap growth 5.9925%, small cap value 8.8525%, large cap growth 8.478%, large cap value 7.535% → market index **7.7526%** (Method II 4.5 + 6.85 − 7.3675 + 3.77 = 7.7525%); (b) CAPM: 9.98, 10.665, 12.48, 10.3225 → **11.33%**; (c) target β 1 with small cap value (0.90) and large cap growth (1.165): $X_1$ + $X_2$ = 1, $0.90X_1 + 1.165X_2 = 1$ → $X_1 = 0.165/0.265$ = **62.3%**, $X_2$ **37.7%**.

*Note (app-written): in TYK P40(b) ICAI takes 12.45 for large cap growth in the total, while its own line above gives 12.48 (SM p. 127); with 12.48 the total is about 11.35%. In TYK P38 the rounded products add to 8.16; the unrounded total is 8.15 (17.40%).*

## II. Sharpe's Optimal Portfolio

Asked: Jan 2026 Q5(a) 8m · Asked: May 2026 Q4(a) 6m

→ Simplified variant of Markowitz — far less data and computation. Based on **excess return over $R_f$ to beta**; ranking plus a unique **cut-off point $C^*$**.

| Step / formula | Terms / logic |
|---|---|
| (a) $(R_i - R_f) \div \beta_i$ for each stock (Treynor's ratio); (b) rank highest to lowest | Excess return per unit of systematic risk. |
| (c) $C_i = \sigma_m^2 \Sigma_{i=1}^{N} [(R_i - R_f)\beta_i \div \sigma_{ei}^2] \div \{1 + \sigma_m^2 \Sigma_{i=1}^{N} [\beta_i^2 \div \sigma_{ei}^2]\}$ | $\sigma_m^2$ = market index variance; $\sigma_{ei}^2$ = stock's unsystematic (residual) variance. Sums run from the top-ranked stock down to stock i (cumulative in ranked order, as in TYK P35). |
| (d) $C^*$ = highest $C_i$ | Select stocks whose $(R_i - R_f)/\beta_i$ is **above** $C^*$; reject those below. If a stock belongs to the optimal portfolio, all higher-ranked ones do too. |
| (e) $Z_i^o = (\beta_i \div \sigma_{ei}^2) \times [(R_i - R_f)/\beta_i - C^*]$; $X_i^o = Z_i^o \div \Sigma_{j=1}^{N} Z_i$ | $Z_i$ gives relative investment; dividing by $\Sigma Z$ makes weights total 1 (all funds invested). |

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**TYK P35 (Ramesh, $\sigma_m^2$ 10, $R_f$ 7%):** excess return/β: A 5.33, B 2.5, C 1.2, D 2, E 0.83, F 4.67 → rank A, F, B, D, C, E. $C_i$: $C_A = 10 \times 0.30/[1 + 10 \times 0.056]$ = **1.923**; $C_F = 10 \times 0.65/[1 + 10 \times 0.131]$ = **2.814**; $C_B$ 2.668; $C_D$ 2.542; $C_C$ 2.165; $C_E$ 2.047 → **$C^*$ = 2.814** → select A and F. $Z_A = (1.5/40)(5.33 - 2.814)$ = **0.09435**; $Z_F = (1.5/30)(4.67 - 2.814)$ = **0.0928** → A **50.41%**, F **49.59%**.

*Note (app-written): the opening line of section II says ranking is "based on the Sharpe Ratio", but step (a) and TYK P35 rank by excess return to beta, i.e. Treynor's ratio (SM p. 53).*

## 16.2 Gold

Asked: May 2026 MCQ CS-1 1-3 6m (SGB)

→ Jewellery suffers **making charges** on buying and selling back. Alternatives:

→ **Gold bars/coins** — but cost of physical storage.

→ **Sovereign Gold Bonds** — Government securities in **grams of gold**, issued by **RBI on behalf of GoI**; pay issue price in cash, redeemed in cash at market price of gold; periodical interest; no storage risk/cost, no making charges or purity issue; held in RBI books or demat.

→ **Gold ETFs** — traded on exchanges like stock; invest in gold bullion; transparent holdings; much lower expenses.

→ **E-gold** — from 2010, National Spot Exchange Ltd (NSEL); 1 unit = 1 gram, held in demat; backed by gold with custodian.

*The May 2026 SGB MCQs also computed post-tax interest, real return, indexed cost and capital gains tax; the SM's 16.2 gives only the features above.*

10.7% of marks

## Exam steps

**Exam steps — expected return and SD:** returns = $[D_1 + (P_1 - P_0)] \div P_0$ (net cost if cum-benefit) → $E(R) = \Sigma\, p \times R$ (or mean) → deviations → $\sigma^2 = \Sigma\, p \times dev^2$ (or ÷ n) → $\sigma = \sqrt{}$ (TYK P1, P2, P32).

**Exam steps — two-asset portfolio and minimum risk:** means → deviations → Cov = $\Sigma\, dev_A \times dev_B$ (× p or ÷ n) → $r_{AB} = Cov_{AB} \div \sigma_A\sigma_B$ → $\sigma_p^2 = x_1^2\sigma_1^2 + x_2^2\sigma_2^2 + 2x_1x_2r_{12}\sigma_1\sigma_2$ → min-risk $X_{ABC} = (\sigma_X^2 - Cov._{AX}) \div (\sigma_A^2 + \sigma_X^2 - 2Cov._{AX})$ (TYK P3, P4, P6, P7).

**Exam steps — three-asset portfolio:** write the 3 × 3 matrix (diagonal $\sigma^2$, off-diagonal $r\sigma\sigma$ or Cov) → multiply each cell by row weight × column weight → add all 9 → √ (4.4.5 example, TYK P34(iv)).

**Exam steps — beta:** regression $\beta = [n\Sigma XY - (\Sigma X)(\Sigma Y)] \div [n\Sigma X^2 - (\Sigma X)^2]$ with X = market, Y = security; or $\beta = Cov \div \sigma_m^2$; or $\beta = r_{im}\sigma_i \div \sigma_m$ → $\alpha = \bar{Y} - \beta\bar{X}$ → characteristic line $\alpha + \beta R_m$ (Illus 6, TYK P8–P12).

**Exam steps — single index risk split:** systematic = $\beta^2\sigma_m^2$ → unsystematic $\in_i^2 = \sigma_i^2 -$ systematic (or $\sigma_i^2(1 - r^2)$) → $\beta_P = \Sigma x_i\beta_i$ → $\sigma_p^2 = [(\Sigma X_i\beta_i)^2\sigma_m^2] + [\Sigma X_i^2\in_i^2]$; for a new mix (e.g. 60:40) only change w (Illus 1, TYK P14, P34, P37).

**Exam steps — CAPM valuation:** $R_f$ (T-bill / bond yield = coupon ÷ price) and $R_m$ (dividend + capital gain ÷ cost) → β → required $R_j = R_f + \beta(R_m - R_f)$ → compare with expected: required < expected → undervalued, buy; > → sell; = → hold; for price use $P_0 = D_1 \div (K_e - g)$ vs CMP (Illus 7, 8, TYK P18, P29, P30, P33).

**Exam steps — $R_f$ and $R_m$ from data:** two correctly priced securities → subtract the two CAPM equations → $(R_m - R_f)$ → back-substitute; portfolio average return with average β → solve $R_f$ (TYK P20, P21, P26).

**Exam steps — beta adjustment / target beta:** $\beta_p = \Sigma$ (value ÷ total) × β → required total = (Σ value × β) ÷ target β → risk-free to add (or divest) = required total − current total; target β from two stocks: $w_1 + w_2 = 1$ and $w_1\beta_1 + w_2\beta_2$ = target (TYK P36, P40(c)).

10.7% of marks

**Exam steps — APT:** $R_f + \Sigma\ \lambda_k\beta_k$ (or $\Sigma\ \beta \times$ surprise) → compare with actual return (mispricing) → portfolio sensitivity = Σ (position ÷ own funds) × β, short and borrowing negative → for target sensitivities solve two simultaneous equations in the weights (Illus 9, TYK P15, P38, P40).

**Exam steps — Sharpe / Treynor / Jensen ranking:** $\beta = r\sigma_p \div \sigma_m$ if not given → Sharpe $(R_p - R_f) \div \sigma_p$ → Treynor $(R_p - R_f) \div \beta$ → Jensen $R_p - [R_f + \beta(R_m - R_f)]$ → rank highest first (TYK P17, P41, P42).

**Exam steps — Sharpe optimal portfolio:** $(R_i - R_f) \div \beta$ → rank descending → for each row $(R_i - R_f)\beta_i \div \sigma_{ei}^2$ and $\beta_i^2 \div \sigma_{ei}^2$, keep running totals of both → $C_i = \sigma_m^2 \times$ first total ÷ (1 + $\sigma_m^2 \times$ second total) → $C^*$ = highest $C_i$ (or given) → select ratios above $C^*$ → $Z_i = (\beta_i \div \sigma_{ei}^2)(\text{ratio} - C^*)$ → $X_i = Z_i \div \Sigma Z$ (TYK P35).

**Exam steps — rebalancing:** CPPI: cushion = portfolio − floor → target shares = multiplier × cushion → buy/sell the gap against bonds. Constant mix: revalue shares at new price → total → restore target % → switch the difference. Buy and hold: value = bonds (floor) + shares × new price (13 examples).

10.5% of marks

# ⑧ Ch 08 – Mutual Funds

| Attempt | Q | M | Asked (MCQ / numerical / theory) |
|---|---|---|---|
| Sep 2025 | MCQ CS-II 4-6 | 6 | Mutual fund accounts: closing net cash balance, total NAV, and NAV per unit after a dividend per unit is distributed (MCQ) |
| Sep 2025 | Q3(b) | 4 | Mutual fund holding period return: payout of dividend and capital gain versus reinvestment at average NAV; choose the better option (Numerical) |
| Sep 2025 | Q5(b) | 7 | Mutual fund: units bought after entry load, NAV after appreciation, distribution reinvested at year-end NAV, return and real return after exit load, tax and inflation (Numerical) |
| Jan 2026 | Q1(a) | 6 | Dividend (bonus) scheme: back-solve opening NAV per unit from average annual yield, two bonus issues (1:4 and 2:5) and closing NAV (Numerical) |
| Jan 2026 | Q6(a) | 6 | New fund offer: closing NAV per unit after underwriting commission, marketing cost, gains, regulated expenses, income distribution policy (Numerical) |
| May 2026 | Q3(b) | 4 | Dividend plan versus bonus plan of a mutual fund: total return and holding period return after entry/exit loads; which plan is better (Numerical) |
| May 2026 | Q4(b) | 4 | Closed-ended fund: market price from NAV and premium/discount, total, average and geometric return per period for the investor, geometric growth of NAV, interpretation (Numerical) |

## 2, 2.1 Classification; Functional classification

Asked: May 2026 Q4(b) 4m (closed-ended)

Three classifications: **Functional**, **Portfolio**, **Ownership**.

| Point | Open-ended | Close-ended | Interval |
|---|---|---|---|
| Entry/exit | Any time | Invest at NFO or buy on stock exchange after listing; exit by selling on exchange, at expiry, or in repurchase period | Purchase and redemption only in pre-specified intervals (monthly, quarterly, annually) at NAV-based prices |
| Capital / life | Unlimited capital, indefinite redemption period | Limited life; corpus liquidated at end | Maturity not defined |
| Listing | — | Listed | Not required (in-built redemption window) |
| Fresh units | Continuously | Only at NFO | During interval period at NAV-based prices |

→ Interval = combination of open and close-ended; differs from close-ended on: **not listed**, **fresh issue** in interval, **maturity not defined**.

→ A close-ended fund trades through the day at prices that may be **more or less than its NAV** (SM 3.9).

Note (app-written): for May 2026 Q4(b) type — market price of a close-ended unit = NAV × (1 + premium %) or NAV × (1 − discount %); investor's return is on market price paid/received, the fund's growth is on NAV.

## 6, 6.1 Terms — Net Asset Value (NAV)

Asked: Sep 2025 MCQ CS-II 4-6 6m

Asked: Jan 2026 Q6(a) 6m

→ NAV = value per unit of portfolio holding = what a unit holder would get per unit if the fund were redeemed. Returns depend on **NAV** and **costs**.

→ Three aspects: (i) **net** value — market value of assets less liabilities; (ii) **changes daily**; (iii) **per unit** — ÷ outstanding units.

→ Investors' subscription = **unit capital** (liability side); investments = assets.

10.5% of marks

| Formula | Terms / logic | SM figures |
|---|---|---|
| NAV per unit = Net assets of the scheme ÷ Number of units outstanding | Per-unit claim of a holder. | TYK P7 (Cinderella, Scheme Rudolf): shares 4,20,20,000 − accrued exp 2,50,000 − other liab 2,00,000 = 4,15,70,000 ÷ 10,00,000 = **₹ 41.57** |
| Net assets = MV of investments + Receivables + Other accrued income + Other assets − Accrued expenses − Other payables − Other liabilities | Everything at **market value**; accrued (unpaid) items count. | |
| New units issued = Amount invested ÷ NAV at that time | New money enters at current NAV, so existing NAV is undisturbed; cash not invested stays as a bank asset. | TYK P8, P9 |

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**TYK P8 (L, M, N, P Ltd; 6,00,000 units):** (i) 4,00,000 + 93,72,000 + 72,24,000 + 3,03,06,000 = 4,73,02,000 ÷ 6,00,000 = ₹ 78.8366 → **₹ 78.84**. (ii) Mr. A ₹ 30,00,000; 8,000 M Ltd bought (M now 38,000 = 1,18,71,200); cash 5,00,800; total 5,03,02,000; units = 6,00,000 + 30,00,000 ÷ 78.8366 = **6,38,053**. (iii) 2 Feb: 4,10,000 + 1,36,80,000 + 76,62,000 + 3,02,34,000 + 5,00,800 = 5,24,86,800 ÷ 6,38,053 = **₹ 82.26**.

**TYK P9 (Fair Return MF; 8,00,000 units):** (a) 9,52,38,000 ÷ 8,00,000 = ₹ 119.0475 → **₹ 119.05**. (b) Mr. X ₹ 50,00,000; 18,000 C Ltd bought (C now 28,000 = 74,03,200); cash 2,40,800; total 10,02,38,000; units = 8,00,000 + 50,00,000 ÷ 119.0475 = **8,42,000**. (c) 2 April: 10,27,87,200 ÷ 8,42,000 = **₹ 122.075**.

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**TYK P18 (fund flow → NAV; 10,00,000 units of ₹ 10, investments 98,00,000):** opening cash 100 − 98 lakh = 2,00,000 + dividend 12,00,000 + interest 56,000 (7% G-sec) + 45,000 (9% deb) + 50,000 (10% deb) = 15,51,000 − opex 5,00,000 = cash **10,51,000**. Assets: cash 10,51,000 + G-sec 8,00,000 (par) + 50,000 shares @ 175 = 87,50,000 + 9% unlisted deb 5,00,000 (cost) + 10% deb @ 90% = 4,50,000 = 1,15,51,000 → NAV **₹ 11.55**. If ₹ 0.80/unit dividend paid: 1,15,51,000 − 8,00,000 = 1,07,51,000 → **₹ 10.75**.

**TYK P19 (regular income scheme; 20 lakh units):** listed shares 20 × 2,300 ÷ 1,000 = 46.00; cash 1.23; unlisted bonds 1 × 80% = 0.80; listed bonds at NAV-date value 8.00; dividend accrued 0.80; other fixed-interest at cost 4.50 → assets 61.33 cr − (payable on shares 6.32 + exp accrued 0.75 = 7.07) = 54.26 cr ÷ 20,00,000 = **₹ 271.30**.

**TYK P15 (ABC MF, 20 lakh units @ ₹ 10, initial exp 12 lakh, portfolio 185 lakh):** opening bank 200 − 185 − 12 = 3.00 + sale 63 + dividend 2 = 68 − purchases 56 − expenses paid 90% × 8 = 7.20 − capital gain distributed 80% × (63 − 60) = 2.40 − dividend distributed 80% × 2 = 1.60 → closing bank 0.80. Net assets 198 + 0.80 − arrears 0.80 = 198 → NAV **₹ 9.90**. Per unit: income received (2.40 + 1.60) ÷ 20 = 0.20, less fall in value (200 − 198) ÷ 20 = 0.10 → 0.10 on ₹ 10 = **1% monthly, 12% annual**.

10.5% of marks

## 6.1 (contd.) Return on MF units — formulas used in ICAI answers

Asked: Sep 2025 Q3(b) 4m

Asked: Sep 2025 Q5(b) 7m | Asked: Jan 2026 Q1(a) 6m | Asked: May 2026 Q3(b) 4m

| Formula | Terms / logic | SM figures |
|---|---|---|
| Return = $[(NAV_1 - NAV_0) + D_1 + CG_1] \div NAV_0 \times 100$ | Distributions taken in **cash**: change in NAV + dividend (income) + capital gain distribution, per unit, over opening NAV. | TYK P4: (13 − 12.25) + 1.25 + 1.00 = 3.00 ÷ 12.25 = **24.49%**. P5: 0.60 + 0.90 + 0.75 = 2.25 ÷ 8.50 = **26.47%**. P14: 0.70 + 0.75 + 0.60 = 2.05 ÷ 8.75 = **23.43%**. |
| $r = [(NAV_t - NAV_{t-1}) + I_t + G_t] \div NAV_{t-1}$ | Same, for one period (month); $I_t$ income, $G_t$ capital gain distribution. Annual = monthly × 12. | TYK P3: (0.06 + 0.0375 + 0.03) ÷ 20 = 0.1275 ÷ 20 = **0.6375% p.m. = 7.65% p.a.** TYK P6: 15% p.a. → 1.25% p.m.; 0.0125 = ($NAV_t$ − 65.78 + 0.50 + 0.32) ÷ 65.78 → $NAV_t$ = **₹ 65.78** (no change). |
| Additional units = Total amount reinvested ÷ reinvestment price; Return = (Value of all units at year end − Price paid for original units) ÷ Price paid for original units | Distributions **reinvested**: compare closing value of all units with original cost. (app-written general form of ICAI's working) | P4: 2.25 × 300 = 675 ÷ 12.50 = 54; 354 units × 13 = 4,602 vs 3,675 → **25.22%**. P5: 1.65 × 200 = 330 ÷ 8.75 = 37.71; 237.71 units = 2,163.16 vs 1,700 → **27.24%**. P14: 1.35 × 300 = 405 ÷ 8.65 = 46.82; 346.82 × 9.45 = 3,277.45 vs 2,625 → **24.85%** → reinvestment preferable. |
| Effective yield p.a. = h ÷ a × 365 ÷ i × 100 | ICAI table: a = investment; c = units (a ÷ b, b = opening NAV); e = total NAV at end (c × d); f = e − a; g = dividend received; h = total yield (f + g); i = number of days. Units = investment ÷ entry NAV. Days counted including entry date (1.12 to 31.3 = 121). | TYK P10 (Mr. Sinha): X 47,619.05 units × 10.40 = 4,95,238.12; yield (−4,761.88 + 9,500) = 4,738.12 over 121 days → **2.859%**; Y 2,500 over 90 days → **10.139%**; Z −1,000 over 31 days → **(−) 23.55%**. |
| Period of holding (days) = Yield (%) × 365 ÷ Effective yield p.a. (%) | Reverse of the above (ICAI working, TYK P11). | TYK P11 (Mr. Y): units 19,417.48 / 39,603.96 / 20,000; total NAV 7,99,068.77; yields 5,029.17 / −3,960.40 / 9,000; total yield 10,068.77 ÷ 8,00,000 = 1.2586%; yield % 2.5146 / −0.9901 / 4.5 → **95 / 31 / 68 days** → invested **27.12.17 / 01.03.18 / 23.01.18** (alternative 26.12.17 / 28.02.18 / 22.01.18). |

- → **Dividend reinvestment plan:** dividend = units held × **face value ₹ 10** × dividend %; new units = dividend ÷ NAV on that date (ex-dividend). Keep a running unit balance.
- → **Bonus plan:** ratio a : b → a bonus units for every b held (5 : 4 on 10,000 = 12,500). No cash; value = final units × final NAV.
- → **Annual (average) return** = Net gain ÷ Investment × 12 ÷ months held × 100 (simple). Alternative **IRR** = LR + [NPV at LR ÷ (NPV at LR − NPV at HR)] × (HR − LR).
- → Annualised yield → yield for the period = annualised yield × months ÷ 12 (ICAI: 153.33 × 9 ÷ 12 = 115%; used to back-solve NAVs).

10.5% of marks

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**TYK P17 (Mr. K, ₹ 2,00,000 each, 1.4.2005 to 31.3.2010 = 5 years): Plan D**: 2,00,000 ÷ 38.20 = 5,235.60 units; dividends reinvested: 10% → 5,235.60 ÷ 39.10 = 133.90 (5,369.50); 15% → 8,054.25 ÷ 44.20 = 182.22 (5,551.72); 13% → 7,217.24 ÷ 45.05 = 160.20 (5,711.92); 16% → 9,139.07 ÷ 44.80 = 204 (5,915.92); maturity 40.40 × 5,915.92 = 2,39,003.17; gain 39,003.17 ÷ 2,00,000 ÷ 5 = **3.90%** (IRR: NPV 4% −3,557, 2% 16,473 → **3.645%**). **Plan B**: 2,00,000 ÷ 35.60 = 5,617.98; 1:5 → 1,123.60 (6,741.58); 1:8 → 842.70 (7,584.28); 1:10 → 758.43 (8,342.71); × 39.70 = 3,31,205.59; gain 1,31,205.59 → **13.12%** (IRR: NPV 13% −20,235, 8% 25,413 → **10.78%**).

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**TYK P16 (Chanakya Opportunity Fund; ₹ 1,00,000 each at ₹ 10 on 1.4.1999; redeemed 31.07.2009 = 124 months; STT 0.2% on redemption; STCG 10% on units held under a year, LTCG exempt):**

- **Mrs. Charu, Plan A (reinvestment)**: 10,000 → 20% 20,000 ÷ 30.70 = 651.47 (10,651.47) → 70% 74,560.29 ÷ 58.42 = 1,276.28 (11,927.75) → 40% 47,711.00 ÷ 42.18 = 1,131.13 (13,058.88) → 25% 32,647.20 ÷ 46.45 = 702.85 (13,761.73) → 40% 55,046.92 ÷ 48.10 = 1,144.43 (**14,906.16**). × 53.75 = 8,01,206.10 − STT 1,602.41 = 7,99,603.69 − STCG 10% on 1,144.43 × (53.64 − 48.10) = 6,340 → 634 → 7,98,969.69; gain 6,98,969.69 × 12 ÷ 124 → **67.64%**.
- **Mr. Anand, Plan B (bonus)**: 10,000 + 12,500 (5:4) = 22,500 + 7,500 (1:3) = 30,000 + 7,500 (1:4) = **37,500** × 22.98 = 8,61,750 − STT 1,723.50 = 8,60,026.50 − STCG 10% on 7,500 × (22.93 − 19.95) = 22,350 → 2,235 → 8,57,791.50; gain 7,57,791.50 → **73.33%**.
- **Mr. Bacchan, Plan C (growth)**: 10,000 × 82.07 = 8,20,700 − STT 1,641.40 = 8,19,058.60; no STCG; gain 7,19,058.60 → **69.59%**.

ICAI: 53.64 and 22.93 are NAV net of 0.2% STT; alternatively take NAV without STT (Proviso 5 to s. 48: STT not deductible in computing capital gain); the cost of the last bonus units "can also be taken as zero".

Note (app-written): ICAI's answer dates the 40% dividend "30.10.2007"; the question table says 31.10.2007 (SM p. 8.30 vs 8.46). Figures unaffected.

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**TYK P12 (Mr. X, 10,000 units of ₹ 10 on 1.7.2007, dividends reinvested)**: 9-month yield = 153.33 × 9 ÷ 12 = 115% → value 31.3.2008 = 1,00,000 + 1,15,000 = 2,15,000; NAV = (2,15,000 − dividend 10,000) ÷ 10,000 = **₹ 20.50**; units = 10,000 + 10,000 ÷ 20.50 (487.80) = 10,487.80. Dividend 2009 = 10,487.80 × 10 × 20% = 20,975.60; 11,296.11 = 20,975.60 ÷ X + 10,487.80 → NAV 31.3.2009 = **₹ 25.95**. NAV 31.3.2010 = 1,00,000 (1 + 0.7352 × 33 ÷ 12) ÷ 11,296.11 = **₹ 26.75**.

**TYK P13 (same data as HPR)**: HPR 115% → NAV 31.3.2013 **₹ 20.50**; NAV 31.3.2014 **₹ 25.95**; NAV 31.3.2015 = 1,00,000 (1 + 2.0217) ÷ 11,296.11 = **₹ 26.75**.

## 6.2 Entry and Exit Load

Asked: Sep 2025 Q5(b) 7m | Asked: May 2026 Q3(b) 4m

- Loads = sales charges to compensate **distribution costs**; funds without them = **no-load funds**.
- **Entry load** — at purchase; **% added to NAV** → fewer units allotted. At present no scheme has an entry load.
- **Exit load** — at redemption or switch; **% deducted from NAV**; goes to the **scheme's pool**, not the AMC. Most schemes have none after a stipulated holding period.

10.5% of marks

| Formula | Terms / logic | SM figures |
|---|---|---|
| $r_2 = r_1 \div (1 - \text{initial exp}) + \text{recurring exp.}$ | (TYK PI answer writes the same as $R_2 = R_1 \div (1 - \text{Initial expenses \%}) + \text{Recurring expenses \%.}$) $r_1$ = return investor earns on his own; $r_2$ = return the fund must earn; initial expenses reduce the amount actually invested, so the fund must earn $r_1$ on the smaller base, plus cover annual recurring expenses. | SM Example (Mr. X): 0.10 ÷ (1 − 0.06) + 0.02 = **12.64%**. TYK PI (Mr. A): 16% ÷ (1 − 0.055) + 1.5% = **18.43%**. |
| Public Offer Price = NAV ÷ (1 − Front end Load); Redemption Price = NAV ÷ (1 + Back End Load) | ICAI's main method. | TYK P2 (ELSS NAV ₹ 10, POP ₹ 10.204, redemption ₹ 9.80): main — front-end 0.0199 say **2%**, back-end **2.04%**; alternative — front-end (10.204 − 10) ÷ 10 = **2.04%**, exit (10 − 9.80) ÷ 10 = **2.00%**. |
| Alternative: Front End Load = (Public Offer Price − NAV) ÷ NAV; Exit Load = (NAV − Redemption Price) ÷ NAV | Load as % of NAV (app-written general form of ICAI's alternative working). | |
| Issue price = Opening NAV + Entry Load + Dividend Equalization; Repurchase price = Opening NAV − Exit Load + Dividend Equalization | (app-written general form of ICAI's TYK P20 working.) Equalisation = income per unit accumulated up to that date, collected from new holders / paid to leavers so income per unit for others is unchanged. | TYK P20 below. |

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**TYK P20 (open-ended, 300 lakh units, NAV ₹ 18.75):** income per unit April 22.95 ÷ 300 = 0.0765; equalisation on 6 lakh issued 0.4590 → 23.4090 on 306; May 0.1125 (34.425) → 0.1890 / 57.8340; equalisation paid on 3 lakh repurchased (0.5670) → 57.2670 on 303; June 0.1500 (45.45) → **(i) available 102.7170 lakh = ₹ 0.3390/unit**; 70% paid 0.2373 = 71.9019; retained 30.8151. **(ii) Issue price** 18.750 + 0.375 + 0.0765 = **₹ 19.2015**. **(iii) Repurchase** 18.750 − 0.375 + 0.1890 = **₹ 18.564**. **(iv)** 5,625 + appreciation 425.47 + issue 6 × 19.2015 = 115.209 + income 102.825 = 6,268.504 − repurchase 3 × 18.564 = 55.692 − distributed 71.9019 = 6,140.9101 ÷ 303 lakh = **₹ 20.2670**.

Note (app-written): ICAI's repurchase working labels the 0.1890 line "Add: Dividend Equalization paid on Issue Price"; it is the equalisation on repurchase (SM p. 8.51). Figures unaffected.

## Exam steps

**Exam steps — NAV:** 1 each holding × market price (index-adjust shares bought at old index; impair unlisted) → 2 + cash (opening cash + income − expenses paid − distributions) + accrued income → 3 − accrued expenses and payables → 4 ÷ units (new money: units = amount ÷ NAV at that time).

**Exam steps — HPR payout vs reinvestment:** 1 cash: $[(NAV_1 - NAV_0) + D_1 + CG_1] \div NAV_0$ → 2 reinvest: $(D_1 + CG_1)$ × units ÷ reinvestment NAV = additional units → 3 (value of all units at $NAV_1$ − price paid) ÷ price paid → 4 pick higher %. With loads: buy at NAV + entry load, sell at NAV − exit load; then tax, then real return.

10.5% of marks

**Exam steps — Dividend vs Bonus plan:** 1 units = investment ÷ entry NAV → 2 dividend plan: units × ₹ 10 × % ÷ that day's NAV, running total → 3 bonus plan: units × ratio, running total → 4 final units × final NAV (− STT, − STCG on units under a year) → 5 gain ÷ investment × 12 ÷ months (or IRR).

**Exam steps — back-solve NAV (TYK P12/13):** 1 period HPR (annual × months ÷ 12) → 2 value = investment × (1 + HPR) → 3 NAV = (value − dividend) ÷ units → 4 dividend ÷ NAV = new units → 5 next NAV = dividend ÷ (final units − previous units).

**Exam steps — issue/repurchase with equalisation (TYK P20):** 1 income per unit month by month on units then outstanding → 2 issue = NAV + load + accumulated income/unit; repurchase = NAV − load + accumulated income/unit → 3 closing net assets = opening + appreciation + issue proceeds + income − repurchases − distribution → 4 ÷ closing units.

**Exam steps — expected NAV (TYK P21):** 1 split NAV into equity and cash → 2 $E(R) - R_f$ = Sharpe × σ, then β = $[E(R) - R_f]$ ÷ Treynor → 3 equity × (1 − β × market fall) → 4 cash − monthly cost → 5 add.

9.6% of marks

# ⑤ Ch 05 — Security Valuation

| Attempt | Q | M | Asked (MCQ / numerical / theory) |
|---|---|---|---|
| Sep 2025 | Q2(a) | 6 | CAPM required return plus constant-growth dividend model (existing vs revised inputs): intrinsic price and buy/hold/sell advice (Numerical) |
| Sep 2025 | Q3(a) | 6 | Bond refunding decision: initial outlay (call premium, issue cost, tax) and NPV of refunding 12% debentures with 10% debentures (Numerical) |
| Sep 2025 | Q6(a) | 7 | FCFF valuation with WACC from a target capital structure and value per share vs market price; FCFE needed to give the same equity value (constant growth) (Numerical) |
| Jan 2026 | MCQ CS-I 1-3 | 6 | Repo/reverse repo of a GOI bond: classify the two parties, accrued interest and dirty price, first-leg and second-leg proceeds (MCQ) |
| May 2026 | Q1(b) | 4 | H-model intrinsic value (long-run growth = terminal ROE x retention; H = half of transition period) and % overvaluation against market price (Numerical) |
| May 2026 | Q5(a) | 5 | Two-stage FCFE valuation: PV of high-growth FCFE, and implied terminal growth rate for a given market price (Numerical) |

## 4. Required Return on Equity

Asked: Sep 2025 Q2(a) 6m

→ Investors judge an asset by its contribution to **systematic risk** (not diversifiable). CAPM = relatively **objective** estimate; widely used.

→ Return on share A = $R_f$ + β × Market risk premium.

→ β from data (used in TYK P17): **β = Cov(security, market) ÷ Variance of market**.

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**SM Example (4):** $R_f$ 5%, β 1.5, MRP 4.5% → 0.05 + 1.5(0.045) = **11.75%**.

## 6–6.1 Valuation of Equity Shares; Dividend Based Models

Asked: Sep 2025 Q2(a) 6m

Asked: May 2026 Q1(b) 4m

→ Three approaches: (1) **Dividend** based, (2) **Earning** based, (3) **Cash flow** based.

→ Dividend model assumptions: (a) dividend paid **annually**; (b) first dividend at **end of year 1**; (c) sale at end of a year at **ex-dividend** price.

→ Value = PV of expected dividend stream at **$K_e$** (required return / opportunity cost).

| Formula | Terms / logic | SM figures |
|---|---|---|
| One-year: $P_0 = (D_1 + P_1) \div (1 + K_e)$ | Dividend and sale price both received at end of year 1. | $D_1$ 6, $P_1$ 36, $K_e$ 20% → **₹35** |
| Zero growth: $P_0 = D \div K_e$ | Same dividend forever = perpetuity. | — |
| Constant growth (Gordon): $P_0 = D_1 \div (K_e - g) = D_0(1+g) \div (K_e - g)$ | $D_0$ just paid, $D_1$ next; g forever; needs $K_e$ > g. Rearranged (as in TYK P6 answer): **$K_e = D_1/P_0 + g$**. | TYK P8, P9, P13, P15 |

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**TYK P8:** $K_e$ = 8 + 1.5(12 − 8) = **14%**; $P_0$ = 3(1.10)/(0.14 − 0.10) = 3.30/0.04 = **₹82.50**. **TYK P9:** $K_e$ 9 + 1.5(13 − 9) = 15% → 2.00/0.08 = **₹25.00** (alt. 2.00(1.07)/0.08 = ₹26.75); β 1.75 → $K_e$ 16% → **₹22.22** (alt. ₹23.78). **TYK P13:** k = 9 + 0.75(15 − 9) = 13.5%; P = 2.5 × 1.1/0.035 = **₹78.57**. **TYK P15:** $K_e$ 13.8% → 2.14/0.068 = **₹31.47**; all changes together: 9.18% + 1.3(13% − 9.18%) = 14.146%, g 10% → 2.20/0.04146 = **₹53.06**, alt. 11% + 1.3(15% − 11%) = 16.20% → **₹35.48**; separately: inflation +2% → $R_x$ 15.80% → **₹24.32**; g 10% → **₹57.89**; β 1.3 → 14.2% → **₹29.72**.

9.6% of marks

*Note (app-written): TYK P9 main answer divides $D_0$ (2.00) by $K_e - g$; the "alternative" with $D_1$ = 2.00(1.07) follows the SM formula. TYK P15(ii) "all together" gives two ICAI versions (₹53.06 and ₹35.48); the second ($R_f$ and $R_m$ both +2%) matches the separate inflation working ($R_x$ 15.80%).*

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***TYK P5 (holding period):*** *$D_0$ 14, g 9%: $D_1$ 15.26, $D_2$ 16.63, $D_3$ 18.13, $P_3$ 360, $K_e$ 13% → 13.50 + 13.02 + 12.56 + 249.48 =* ***₹288.56****; (ii) 9% forever → 15.26/0.04 =* ***₹381.50****; (iii) $P_3$ = 18.13 × 1.09/0.04 = 19.76/0.04 =* ***₹494****.* ***TYK P14 (bonus 1:5 in yr 4, 5% costs):*** *dividends 20 × 3 yrs, then 24; sale 900 × 1.2 × 0.95 = 1,026 → PV* ***₹563.68*** *vs cost 500 × 1.05 = 525 → gain ₹38.68, buy; max price 563.68 × 100/105 =* ***₹536.84****.*

***Variable growth*** *— many growth rates allowed, but* ***one rate must be for infinity****. Company stages: growth, transition, maturity.*

| Formula | Terms / logic |
|---|---|
| ***Two-stage:*** $P_0 = \Sigma_{t=1..n}\ D_0(1+g_1)^t \div (1+K_e)^t + P_n \div (1+K_e)^n$ | PV of super-growth dividends + PV of price at end of super growth. |
| $P_n = D_0(1+g_1)^n(1+g_2) \div (K_e - g_2)$ | Gordon at end of year n using $D_{n+1}$. $g_1$ finite/super growth; $g_2$ normal growth; $P_n$ = price at beginning of normal growth. Discount $P_n$ by n years (not n+1). |
| ***Three-stage:*** PV(high-growth dividends) + PV(transition dividends) + PV(price at start of stable growth) | High growth → advantage depleted by competition → gradual decline (transition) → stable growth. |
| ***H-Model:*** $P_0 = D_0(1+g_n) \div (r - g_n) + D_0 \times H \times (g_c - g_n) \div (r - g_n)$ | Value at normal growth + ***premium for abnormal growth***. $g_n$ long-run normal growth; $g_c$ current (initial short-term) growth; ***H = half of the transition period***; growth declines ***linearly*** over 2H. Also usable for FCFE. |

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***TYK P3:*** *$D_0$ 4; $D_1$ 4.80, $D_2$ 5.76, $D_3$ 6.336; TV = 6.336/(0.15 − 0.10) = 126.72 at end of yr 2 → 4.80 × 0.8696 + 5.76 × 0.7561 + 126.72 × 0.7561 =* ***₹104.34****.* ***TYK P17:*** *β = 30%/24% = 1.25; $K_e$ = 11 + 1.25(18 − 11) =* ***19.75%****; dividends 2.80, 3.92, 5.49, 7.68, 10.76 → PV 16.36; TV PV = 10.76(1.15)/(0.1975 − 0.15) × 0.406 = 105.77 →* ***₹122.13****.* ***TYK P6 ($K_e$ from price):*** *(i) 1.12/20 + 12% =* ***17.6%****; (ii) 20% for 5 yrs then 10%: at 18% PV ₹20.23, at 19% ₹17.89 → 18% + (0.23/2.34) × 1% =* ***18.10%****.* ***TYK P16:*** *now 2(1.06)/0.02 =* ***₹106****; with project (no dividend 3 yrs, $D_4$ 2.50, g 7%) 2.50/0.01 × 1/$(1.08)^3$ =* ***₹198.46****; prices ₹214.33 (yr 1), ₹231.48 (yr 2), ₹250 (yr 3); sell 10, 9, 8 shares → ₹2,143.30, ₹2,083.32, ₹2,000; 973 shares left → yr 4 income* ***₹2,432.50****.*

*Note (app-written): TYK P16 gives ₹198.46 and then ₹198.45 for the same figure (rounding).*

## 6.3–6.3.1 Cash Flow Based Models; FCFF

Asked: Sep 2025 Q6(a) 7m

→ *DDM ignores cash needed for* ***long-term capex*** *and* ***short-term working capital*** *→ not the true free cash flow. Free cash flow models discount cash left after meeting these.*

→ ***FCFF*** *discounted at* ***cost of capital ($K_0$)*** *→ value of the* ***firm****.* ***FCFE*** *discounted at* ***$K_e$*** *→ value of* ***equity*** *(like DDM).*

| FCFF from | Formula |
|---|---|
| (a) Net income | NI + Interest(1 − t) + Depreciation ∓ Capex ∓ ΔNon-cash NWC |
| (b) EBIT | EBIT(1 − t) + Depreciation ∓ Capex ∓ ΔNon-cash NWC |
| (c) EBITDA | EBITDA(1 − t) + Depreciation × t ∓ Capex ∓ ΔNon-cash NWC (Dep × t = tax shield on depreciation) |
| (d) FCFE | FCFE + Interest(1 − t) + Principal repaid − New debt issued + Preferred dividend (add back what went to/came from non-equity providers) |
| (e) Cash flows | CFO + Interest(1 − t) ∓ Capex (US GAAP; if interest already adjusted in CFO, do not add again) |

→ **Capex** (one year) = Purchase of fixed assets − Sale of fixed assets (investing activities).

→ **ΔNon-cash WC:** (1) WC = CA − CL; (2) Non-cash NWC = CA − Cash & bank − CL; (3) same for previous year; (4) change = current − previous; (5) **increase** = more WC needed → **deduct** from FCFF.

→ Value of firm: **one-stage** = PV of stable-period FCFF; **two-stage** = PV explicit (or transition) period FCFF + PV stable period FCFF; **three-stage** = PV explicit + PV transition + PV stable.

### 6.3.2 Free Cash Flow to Equity (FCFE)

Asked: Sep 2025 Q6(a) 7m

Asked: May 2026 Q5(a) 5m

→ Cash for equity holders after operating expenses, interest, **net debt** obligations and reinvestment (WC, capex).

→ FCFE = Net income ∓ Capex + Depreciation ∓ ΔNon-cash NWC + New debt issued − Debt repayments + Net issue of preference shares − Preference dividends.

→ Or FCFE = Net profit + Dep − ΔNWC − Capex + New debt − Debt repayment + Net pref issue − Pref dividend (Capex = addition to fixed assets to sustain the base).

→ FCFE can be valued by **multistage growth** (two-stage, three-stage, H-model) exactly like dividends.

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**TYK P10:** shares 1,300/40 = 32.5 cr; EPS 290/32.5 = **₹8.923**; FCFE = 8.923 − [(1 − 0.27)(47 − 39) + (1 − 0.27)(3.45)] = 8.923 − [5.84 + 2.5185] = **₹0.5645**; $K_e$ = 8.7 + 0.1(10.3 − 8.7) = 8.86%; $P_0$ = 0.5645(1.08)/(0.0886 − 0.08) = 0.60966/0.0086 = **₹70.89**.

TYK P10 formula (ICAI answer): FCFE = Net income − [(1 − b)(capex − dep) + (1 − b)(ΔWC)], where b here = debt ratio (27%). Logic (app-written): only the equity-funded part (1 − b) of net capex and ΔWC reduces FCFE; the debt-funded part comes from new borrowing.

### 8.16 Refunding of Bonds

Asked: Sep 2025 Q3(a) 6m

→ Issuer refunds before maturity when rates **fall**: new low-coupon bonds replace old high-coupon bonds; needs a **call feature**; cost = **call premium** (repay above face value).

→ Capital budgeting (**NPV**) decision: refund if PV of net cash savings > PV of outflows (call premium, interest in transition period etc.). New bonds usually issued before old are refunded → **overlap interest** on both.

| Step | Build-up |
|---|---|
| Initial outlay | (Face + Call premium) − (New issue − Issue cost) − Tax saving on (call premium + unamortised old issue cost written off) |
| Annual net saving | Interest saving − tax on [interest saving + (old amortisation lost − new amortisation)] |
| Discount rate | After-tax cost of new debt (TYK P25 uses PVIFA 7% = 10% × (1 − 0.30)) (app-written explanation) |
| NPV | PV of annual net saving − Initial outlay; positive → refund |

9.6% of marks

ICAI ·

**TYK P25:** cost of calling 300 + 12 = 312; net new proceeds 300 − 6 = 294; tax saving 0.30(12 + 9) = 6.3 → initial outlay **₹11.7 m**. Saving 300 × 2% = 6.00 less tax 0.30 × [6 + (9 − 6)/6] = 1.95 → **₹4.05 m** p.a.; × PVIFA(7%, 6) 4.766 = 19.30 → NPV **₹7.60 m** → refund.

## 8.17.6 Repo and Reverse Repo

Asked: Jan 2026 MCQ CS-1 1-3 6m

→ Raise funds by **selling securities** and agreeing to **repurchase** later at a specified price (includes interest at agreed rate). Also called **Ready Forward Contract** (sell spot/ready, buy back forward).

| Point | Repo | Reverse Repo |
|---|---|---|
| RBI rate | Rate at which **RBI lends** to banks against G-Secs | Rate at which **banks lend** to RBI |
| Perspective | **Seller of securities** (acquires funds) | **Supplier of funds**; name depends on who initiated |
| Purpose | Meet **deficiency** of funds | Reduce **excess liquidity** |
| Rate level | Comparatively **higher** | Lower |
| Aim | Contain **inflation** | Control **money supply** |

| Formula (TYK P31) | Terms / logic |
|---|---|
| Dirty Price = Clean Price + Interest Accrued (= 10/100 × 100 × 262/360 in TYK P31) | Clean price excludes accrued interest; buyer pays both. |
| First leg (start proceed) = Nominal value × Dirty price/100 × (100 − Initial margin)/100 | Cash lent against the bond, less margin (haircut). |
| Second leg (repayment) = Start proceed × (1 + Repo rate × days/360) | Repurchase amount = first leg + repo interest. |

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**TYK P31:** 10% GoI 2028, ₹8 cr, clean 99.42, 262 days accrued → dirty 99.42 + 10 × 262/360 = **106.70**; first leg 8,00,00,000 × 106.70/100 × 98/100 = **₹8,36,52,800**; second leg × (1 + 0.0565 × 14/360) = **₹8,38,36,604**.

### Exam steps

**Exam steps — dividend / CAPM valuation:** $K_e$ by CAPM (β = Cov/Var if needed) → identify $D_0$ vs $D_1$ → constant growth: $D_1/(K_e - g)$ → compare with market price → under/over-valued → buy/hold/sell.

**Exam steps — two-stage / three-stage:** grow $D_0$ year by year in high growth → terminal price at end of year $n = D_{n+1}/(K_e - g_2)$ → discount dividends and $P_n$ at $K_e$ ($P_n$ by $n$ years) → sum. For implied $K_e$ or $g$: trial two rates, interpolate.

**Exam steps — H-model:** $g_n$ (e.g. ROE × retention) → $H$ = transition years ÷ 2 → $P_0 = D_0[(1 + g_n) + H(g_c - g_n)] \div (r - g_n)$ (app-written: SM formula with $D_0$ taken common) → % over/under-valuation vs market price.

9.6% of marks

**Exam steps — FCFF / FCFE:** FCFF = EBIT(1 − t) + Dep − Capex − ΔNon-cash WC → discount at WACC (target weights) → firm value − debt = equity → ÷ shares vs price. FCFE = NI + Dep − Capex − ΔWC + net borrowing (or Net income − [(1 − b)(capex − dep) + (1 − b)(ΔWC)], b = debt ratio) → discount at $K_e$; multistage like DDM.

**Exam steps — bond value / YTM / duration:** V = I × PVIFA + F × PVIF (halve rate, double periods if half-yearly) → YTM by trial + interpolation (or approx. formula) → duration = Σ t × PV of CF ÷ $P_0$ → Volatility = Duration/(1 + yield) → change in price = price × Δy × Volatility/100 → immunize: duration = holding period, solve weights.

**Exam steps — bond refunding:** initial outlay = face + call premium − net new proceeds − tax on (call premium + unamortised old cost) → annual saving = interest saving − tax on [interest saving + old amortisation − new amortisation] → discount at after-tax cost of new debt → NPV > 0 → refund.

**Exam steps — money market / repo:** T-bill/CP/CD yield = (FV − P)/P × 365/days → price = FV ÷ (1 + interest rate × days/365) (TYK P30 alt.). Repo: borrower = repo, lender = reverse repo → dirty = clean + coupon × days/360 → first leg = nominal × dirty/100 × (1 − margin) → second leg = first leg × (1 + repo rate × days/360).

8.2% of marks

# 14 Ch 14 – Mergers, Acquisitions and Corporate Restructuring

| Attempt | Q | M | Asked (MCQ / numerical / theory) |
|---|---|---|---|
| Sep 2025 | MCQ CS-V 13-15 | 6 | Acquisition by share exchange: EPS of both firms from P/E, post-acquisition EPS and post-acquisition P/E of acquirer (MCQ) |
| Jan 2026 | MCQ CS-IV 10-12 | 6 | Bank merger: swap ratio using weights on book value, market price, CRAR and NPA; shares issued; post-merger EPS (MCQ) |
| Jan 2026 | Q1(b) | 4 | Complying with SEBI promoter-holding limit (84% to 75%) by bonus issue to minority holders at the same P/E: P/E, bonus ratio, post-bonus price and free-float market cap (Numerical) |
| May 2026 | MCQ CS-III 7-9 | 6 | Takeover defence and restructuring terms: poison pill, greenmail, carve-out (identify the tactic from a description) (MCQ) |
| May 2026 | Q6(a) | 7 | Acquisition with cash, hybrid equity and hybrid convertible-bond offers: synergies, P/E, fair value per share, shares offered, ranking of the offers (Numerical) |

## 4.1 Gains from Mergers or Synergy

Asked: May 2026 Q6(a) 7m

| Formula | Terms / logic | SM example figures |
|---|---|---|
| **Synergy = $V_{AB}$ – ($V_A$ + $V_B$)** | $V_A$ acquirer, $V_B$ stand-alone target, $V_{AB}$ combined. Gain exists only if combined > sum of parts. | 175 – (100 + 50) = **25** |
| **Net gain = Synergy – Premium** | Cost of acquisition = premium over market value + integration costs. | 25 – 10 = **15** |
| $V_A$ + $V_B$ + Synergy = Combined value | Exhibit "merger gains": combined value made of acquirer value, stand-alone value, synergy, less transaction cost. | — |

- → Value sources (Anslinger & Copeland exhibit): normal gain from the market, industry gain over the market, financial leverage, **operating improvements**, NPV of subsequent investment.
- → Acquisitions need not be synergy-driven: **operating improvements** and better post-merger integration give abnormal returns even in unrelated business. **Managerial talent** is the single most important instrument; tie executive pay to post-merger performance; equity stakes make executives think like shareholders.

Numericals on synergy, true cost and gains (TYK P2, P16, P24, P25, P31) are worked under section 13, after the exchange ratio is defined.

## 4.4 Arranging Finance for Acquisition

Asked: May 2026 Q6(a) 7m

- → After the Definitive Agreement: secretarial/legal work by both companies; CFO moves to "financing the acquisition".
- → Key decision: **cash or stock or part of each** (part of the Definitive Agreement). All-equity deal → no financing need. Pay by shares even with funds available: keep a **war chest** for further acquisitions; acquirer's shares are **overpriced**.
- → **LBO** financing is challenging; strong companies shore up long-term funds after the takeover, immediate funding by **bridge financing**.

8.2% of marks

## 5.2 Defensive Tactics

Asked: May 2026 MCQ CS-III 7-9 6m

| Tactic | What the target does |
|---|---|
| **Divestiture** | Divests / spins off some businesses as an independent subsidiary → less attractive. |
| **Crown jewels** | Divestiture of the prized assets; not allowed in some countries (UK) once the deal is known. |
| **Poison pill** | Makes itself unattractive, e.g. issues substantial convertible debentures to existing holders, convertible on a takeover threat → voting shares rise sharply. |
| **Poison put** | Issues bonds that encourage holders to cash in at higher prices → cash drain. |
| **Greenmail** | Incentive to the bidder not to pursue — e.g. buys bidder's shares at a price above market. |
| **White knight** | Offers to be acquired by a friendly company to escape the hostile one (management retained). |
| **White squire** | Sells shares to a company not interested in a takeover → management retains control. |
| **Golden parachutes** | Hefty compensation to managers ousted by takeover → acquirer's interest falls. |
| **Pac-man defence** | Counter-bid for the acquirer → acquirer defends itself, may call off. |

Hostile takeovers are harder to consummate; friendly takeover is the better course.

Note (app-written): the SM defines poison pill as "tactics used by the acquiring company to make itself unattractive"; from the context (defensive tactic, issue to its existing shareholders) it is the **target** company that does this (SM p.12).

## 7.2 Different Forms

Asked: May 2026 MCQ CS-III 7-9 6m (carve-out)

- → **7.2.1 Sell-off / partial sell-off:** sale of an asset, factory, division, product line or subsidiary for cash or securities; partial sell-off = unit not fitting core strategy. Market may undervalue the combination (no synergy); sell-off raises cash to repay debt. 1980s raiders bought with debt and sold subsidiaries — works only if sum of parts > whole.
- → **7.2.2 Spin-off:** part separated into a new firm; existing shareholders get **proportionate ownership** (no change in ownership); management parted; **no fresh cash**. Reasons: separate identity; avoid takeover by spinning off a valuable division; separate regulated and unregulated businesses. Example: Future Group spun off Ezone.
- → **7.2.3 Split-up:** entire firm broken into a series of spin-offs; parent ceases to exist (4 divisions → 4 new companies, original wound up). Smaller units easier to manage → spin-off and split-up likely to enhance value.
- → **7.2.4 Equity carve-out:** like spin-off but some shares of the new company sold by public offer (IPO) → **brings cash**; partial sell-off; parent keeps a **controlling stake**. Used when a subsidiary grows faster / has higher valuation; unlocks value. Separate board, some parent directors shared. Carving out a burden (debt-laden, no track record) won't succeed.
- → **7.2.5 Demerger / division of family-managed business:** ~80% of Indian private companies are family-managed; pressure to cede to professionals; hive off unprofitable divisions to meet succession problems. Steps: identify **core and non-core** operations → reduce interest burden by **debt restructuring** and sale of surplus assets → use proceeds for acquisitions and rejuvenation. Acquisitions must improve scale, lower cost and generate synergies; also improve productivity of existing operations.

| Point | Spin-off | Equity carve-out |
|---|---|---|
| New entity's shares | Given to existing shareholders in proportion | Some sold to the public by IPO |
| Cash to parent | No fresh cash | Brings cash |
| Ownership | Same shareholders, same proportion | Parent keeps controlling stake; public holds the rest |

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**TYK T3** (equity carve-out vs spin-off) → 7.2.

8.2% of marks

| Demerger formula | Terms / logic | SM figures |
|---|---|---|
| Net worth of resulting co. = assets transferred − outside liabilities transferred (book value) | Remaining co. = total − demerged part. | TYK P22: Pharma 25,100 − 4,100 = **21,000**; FMCG 44,900 − 20,900 = **24,000** (total 70,000 − 25,000 = 45,000) ₹ lakh |
| EPS of new co. = expected MPS ÷ sector P/E; shares = profit ÷ EPS | Shares needed so that sector P/E gives the expected price. | 24.50 ÷ 25 = **0.98**; 1,470 ÷ 0.98 = **1,500 lakh** shares |
| Ratio = new shares ÷ existing shares | Issued free to existing holders. | 1,500 : 3,000 = **1 share of Fortune Pharma for 2** of Fortune India (0.50 : 1) |

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**TYK P22 (Fortune India demerger):** FMCG EPS 11,400 ÷ 3,000 = 3.8 × P/E 42 = **₹ 159.60**; book value per share Pharma 21,000 ÷ 1,500 = **₹ 14**, FMCG 24,000 ÷ 3,000 = **₹ 8**.

## 13 Acquisition through Shares

Asked: Sep 2025 MCQ CS-V 13-15 6m · Asked: Jan 2026 MCQ CS-IV 10-12 6m · Asked: Jan 2026 Q1(b) 4m · Asked: May 2026 Q6(a) 7m

- → **Steps:** estimate value of acquirer's own equity → value of target's equity → **maximum number of shares** that can be exchanged → analyse pessimistic and optimistic scenarios.
- → **Exchange ratio (ER)** = number of acquirer's shares given for **each** target share. SM example: B's 1,00,000 shares at ₹ 230 = cost ₹ 2,30,00,000; A valued at ₹ 200 → ER 230 ÷ 200 = **1.15** → **1,15,000** shares. Compare cash vs shares on EPS, capital structure etc.
- → How the number of shares is set: (1) acquirer compares its **value per share with and without** the merger; (2) target compares its value with the **value of shares it would receive**; (3) final terms negotiated in light of (1) and (2) — reflect relative bargaining power.
- → **Fewer shares issued → acquirer's holders better off, target's worse off**; both must expect some benefit, subject to synergy risk.
- → **Impact of P/E:** P/E is the reciprocal of cost of equity and reflects perceived risk; combined P/E can differ from both. Value = **P/E × E**, so combined value = combined earnings × combined P/E. A lower combined P/E can offset synergy; a higher one can raise value even without synergy → take the likely combined P/E into account when setting the ER.

### (a) Swap ratio and post-merger EPS

| Formula | Terms / logic | SM figures |
|---|---|---|
| EPS = PAT ÷ N; MPS = EPS × P/E; EPS = MPS ÷ P/E; Earnings = Market cap ÷ P/E | N shares. Use whichever two are given to get the third. | TYK P1: C 15,00,000 ÷ 10 = 1,50,000; B 60,00,000 ÷ 17 = 3,52,941 |
| ER (market price basis) = MPS of target ÷ MPS of acquirer | Target holders get the same market value. | Illus 3: 60 ÷ 75 = 0.8 (4 for 5) |
| ER (EPS basis) = EPS of target ÷ EPS of acquirer | Target holders keep the same earnings; with no synergy, acquirer's EPS also unchanged. | Illus 2: 2.25 ÷ 4 |
| ER (book value basis) = BVPS of target ÷ BVPS of acquirer | BVPS = (share capital + reserves) ÷ N (no. of shares). | TYK P17: 32 ÷ 40 = 0.8 |
| Shares to be issued = target's shares × ER | Total shares after merger = acquirer's shares + shares issued. | TYK P6: 1,80,000 × 0.5 = 90,000 |
| **EPS after merger = Total Earnings ÷ Total No. of Shares after merger** | SM form; total earnings include any synergy. | TYK P6: 21,60,000 ÷ 6,90,000 = 3.13 |
| **Equivalent EPS = EPS of merged entity after merger × Share Exchange ratio** | Earnings per **old** target share; compare with target's pre-merger EPS. | TYK P6: 3.13 × 0.5 = 1.57 |

8.2% of marks

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**Illus 2 (X and Y):** earnings 12,00,000 + 4,50,000 = 16,50,000. (a) EPS-ratio: shares 3,00,000 + 2,00,000 × 2.25/4 = 1,12,500 → 4,12,500; EPS ₹ **4**; Y's equivalent 4 × 2.25/4 = ₹ **2.25** (both unchanged). (b) 0.5 : 1 → 4,00,000 shares; EPS ₹ **4.125**; X gains **0.125**; Y equivalent 2.0625 vs 2.25 → loses **0.1875**.

**Illus 3 (A and B):** MPS basis 50,000 × 60/75 = 40,000 shares → 2,40,000; EPS 13,00,000 ÷ 2,40,000 = ₹ **5.42**. For B's EPS not to fall: EPS B 6, A 5 → **6 for 5**, 60,000 shares (3,00,000 ÷ 5); EPS 13,00,000 ÷ 2,60,000 = 5 → B holders get 60,000 × 5 = 3,00,000 as before.

**TYK P3 (MK/NN):** 160 : 200 = 4 : 5 → 2,40,000 shares, total 14,40,000; EPS 78,00,000 ÷ 14,40,000 = ₹ **5.42**. No loss: EPS 6 ÷ 5 → **6 for 5**, 3,60,000 shares, EPS ₹ **5.00**, NN earnings 3,60,000 × 5 = 18,00,000. **TYK P12 (M Co/N Co):** 4,00,000 × 160/200 = 3,20,000 → 19,20,000; EPS 1,04,00,000 ÷ 19,20,000 = ₹ **5.42**; EPS basis 6/5 = **1.20** → 4,80,000 shares, EPS ₹ **5**, N gets 24,00,000.

**TYK P4 (ABC/XYZ):** EPS 5 and 3; MPS basis 28/42 × 6,00,000 = 4,00,000 → 14,00,000; EPS ₹ **4.86**; same earnings: 3/5 × 6,00,000 = 3,60,000 → EPS ₹ **5**; ER **0.6 : 1**.

**TYK P6 (A/T, 0.5 : 1):** 90,000 shares; EPS ₹ **3.13**; equivalent EPS of T ₹ **1.57**; MPS at P/E 10 = ₹ **31.30**; merged value 6,90,000 × 31.30 = ₹ **2,15,97,000**.

**TYK P11 (Rama acquires Krishna, 1 : 1):** 2,00,000 shares → 6,00,000; EPS 17,00,000 ÷ 6,00,000 = ₹ **2.83**; Rama +0.33, Krishna −0.67; at P/E 14 MPS ₹ **39.62**; value 2,37,72,000 vs 2,10,00,000 → gain **27,72,000**: Rama 18,48,000, Krishna 9,24,000 (2 : 1).

**TYK P1 (B buys lower-P/E C):** shares 15,00,000 ÷ 12 = 1,25,000 → 6,25,000; EPS (3,52,941 + 1,50,000) ÷ 6,25,000 = ₹ **0.80** vs ₹ **0.71** — buying a lower-P/E firm by share exchange raises reported EPS.

*Note (app-written): TYK P12's recommendation says the 6-for-5 ratio is "based on market shares"; it is the EPS-based ratio (6/5 = 1.20) worked just above (SM p.64).*

### (b) Post-merger MPS, market value and gains to each group

| Formula | Terms / logic | SM figures |
|---|---|---|
| **Expected Market Price = EPS after merger × P/E** (P/E × E = P) | P/E given: acquirer's unchanged, adjusted (e.g. −10%), or combined. | TYK P10: 15 × (10 × 0.9) = 135 |
| Market value of merged firm = Market price × total shares (= P/E × combined earnings) | Gain from merger = post-merger market value − pre-merger market values. | TYK P7: 240.02 − 220 = 20.02 crore |
| **Gain to shareholders = Post-merger value (shares held × market price) − Pre-merger market value** | Fraction owned = shares held ÷ total shares; value owned = fraction × merged value. | TYK P14: 67.57% / 32.43% |
| Target's price per old share = market price after merger × ER (TYK P14: 6.47 × 4/5) | Compare with target's pre-merger MPS. | TYK P14: 6.47 × 4/5 = 5.18 |

8.2% of marks

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**Illus 4 (Simpson/Wilson, combined earnings 3,50,000, P/E 10, value 35,00,000):** shares issued 20,000 → ER 2, total 70,000, Wilson owns 2/7 = **10,00,000**, Simpson 5/7 = **25,00,000**; 25,000 → 2.5, 75,000, 1/3 = **11,66,667**, 2/3 = **23,33,333**; 30,000 → 3, 80,000, 3/8 = **13,12,500**, 5/8 = **21,87,500**.

**TYK P7 (Mark/Mask):** EPS 10 and 4; MPS 100 and 20; swap **0.2** (1 for 5) → 20 lakh shares; EPS 2,400 ÷ 220 = **₹ 10.91**; MPS **₹ 109.10**; value **240.02 crore**; gain 20.02 crore: Mark 218.20 − 200 = **18.20**, Mask 21.82 − 20 = **1.82** crore.

**TYK P10 (Mani/Ratnam):** EPS 10 and 4; MPS 100 and 20; swap **0.20**; EPS 6,000 ÷ 400 = **₹ 15**; P/E 9 → MPS **₹ 135**; value **₹ 540 crore**; each group 200 → 270 crore, gain 70 crore = **₹ 35** per Mani share, **₹ 7** per Ratnam share.

**TYK P14 (P/R, 4 for 5, tax 30%):** PAT 10.50 and 9.45 crore; EPS 0.42 and 0.63; MPS 5.04 and 5.67; values 126 + 85.05 = **211.05 crore**; R gets 12 crore shares (37 total) → P 67.57% = **142.61**, R 32.43% = **68.44** (alt. ₹ 5.7040 5 per share → 142.60 / 68.45). At P/E 12: EPS 19.95 ÷ 37 = 0.539, MPS **₹ 6.47**, value 239.39 crore; P +1.43 (**+28.4%**); R 6.47 × 4/5 = 5.18 vs 5.67 → −0.49 (**−8.64%**).

**TYK P24 (RIL/SIL, 1 for 4):** values 2,00,00,000 and 50,00,000; shares 12,50,000; EPS **2.4**; MPS **₹ 24**; value 3,00,00,000 → gain 50,00,000: RIL 40,00,000, SIL 10,00,000 (both better off). Synergy +20%: earnings 36,00,000, EPS **₹ 2.88**, MPS **₹ 28.80**.

### 13 (c) Maximum / minimum exchange ratio

| Formula | Terms / logic | SM figures |
|---|---|---|
| **Maximum number of shares after merger = Combined PAT ÷ EPS of acquirer; maximum shares to be exchanged = that − existing shares; max ER = shares to be exchanged : target's shares** | SM steps (TYK P13). Combined earnings ÷ acquirer's EPS = maximum total shares; less existing shares = maximum shares to issue. | TYK P13: 200 ÷ 8 = 25 lakh − 15 = 10 lakh → 10 : 16 = **5 : 8** |
| **Maximum shares after merger = Combined Market Capitalisation ÷ Current Market Price of acquirer; less existing shares = shares to be exchanged; ratio to target's shares** | SM steps (TYK P13, P8). Combined value = combined market cap, or combined earnings × expected P/E. | TYK P8: 512 × 6.25 = 3,200 lakh ÷ 250 = 12.8 lakh − 10 = 2.8 lakh → **0.70** |
| Target's condition (app-written): ER × MPS after merger ≥ target's MPS before merger (or ER × EPS after merger ≥ target's EPS) | SM step (2): target compares value received with its own. | TYK P8: 1 : 1 gives ABC 228.56 vs 160 |
| Range: Market Price Based = $MPS_S/MPS_R$ (lower limit); Intrinsic Value Based = intrinsic value of target ÷ MPS of acquirer (max. limit) — S = target, R = acquirer as in TYK P27 | Negotiated ratio nearer the lower limit when acquirer has higher EPS, P/E, ROE, growth. | TYK P27: 0.40 to 0.50 |
| EPS = EAT ÷ N; P/E = MPS ÷ EPS; BVPS = Equity Value ÷ N; ROE = EAT ÷ EF; **Growth Rate = ROE × Retention Ratio**, Retention Ratio = 1 − D/P Ratio | N = no. of shares, EF = equity funds, D/P = dividend payout. Decompose MPS into EPS × P/E and EPS into ROE × BVPS. | TYK P27: R 22.21% × 0.80 = 17.77% |

8.2% of marks

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**TYK P13 (Longitude/Latitude):** MPS 8 × 15 = ₹ **120**, S × 10 = ₹ **50**; EPS max **5 : 8**; MPS: 1,800 + 800 = 2,600 lakh ÷ 120 = 21.67 lakh − 15 = 6.67 lakh → 6.67 : 16 = **0.4169 : 1**.

**TYK P8 (XYZ/ABC, 0.7 : 1):** profits 400 and 112 lakh; P/E 6.25 and 5.71; 2,80,000 shares → 12,80,000; EPS ₹ **40**; ABC 40 × 0.7 = 28 → no change for either. At 1 : 1: 14,00,000 shares, EPS ₹ **36.57**, MPS ₹ **228.56**; ABC gain ₹ **68.56** (vs 160); XYZ loss ₹ **21.44** (vs 250). Max ER: 3,199.98 (or 3,200) − 2,500 = 699.98 (700) lakh ÷ 250 = 2.80 lakh → **0.70**.

**TYK P9 (XYZ/ABC):** EPS 2 and 1, P/E 10 and 10. ABC at P/E 6.4 → ₹ **6.40**; ER 6.40 ÷ 20 = **0.32** (20 ÷ 6.40 = 3.125 ABC shares per XYZ share); post EPS 6,25,000 ÷ 2,90,000 = ₹ **2.16**. Same EPS: 6,25,000 ÷ 2 = 3,12,500 − 2,50,000 = 62,500 → **0.50**.

**TYK P5 (EPS-maximising CEO):** acquirer 840 lakh ÷ 42 = 20 lakh shares, EPS ₹ 4; target 157.50 ÷ 105 = 1.50 lakh, EPS ₹ 10.50 → max ER **4 : 10.50 = 1 : 2.625** (2.625 shares per target share).

**TYK P27 (R/S):** EPS 2.665 / 1.56; P/E 18.76 / 12.82; BVPS 12 / 10; ROE 22.21% / 15.60%; g 17.77% / 10.92%; ratio 20/50 = **0.40** (lower) to 25/50 = **0.50** (max); expect near the lower limit.

**TYK P28 (BA/DA):** EPS 2.10 / 1.2375; P/E 19.05 / 12.12; BVPS 12 / 10; ROE 17.50% / 12.37%; g 10.50% / 4.95%; range 15/40 = **0.375** to 20/40 = **0.5**, nearer lower. At 0.4 : 1: 1,32,000 shares, EPS ₹ **2.341** → BA +0.241, DA 2.34 × 0.4 = 0.936 → **−0.3015**; MPS 2.341 × 19.05 = ₹ **44.6** → BA +4.6, DA 44.60 × 0.4 = 17.84 − 15 = **+2.84**.

Note (app-written): TYK P13 — 6.6667 ÷ 16 = 0.4167; ICAI's 0.4169 comes from rounding to 6.67 first (SM p.64). TYK P8 — 36.5714 × 6.25 = 228.57; ICAI prints 228.56 in (ii)–(iii) and 228.57 in (iv) (SM p.59–60). TYK P27/P28 printed "6,80,00", "3,00,00", "1,98,00" are typos for 6,80,000, 3,00,000, 1,98,000 (SM p.49–50).

### (d) Synergy, true cost, value of acquisition, max/min price

| Formula | Terms / logic | SM figures |
|---|---|---|
| **True Cost of Merger = Value of acquirer after merger × target holders' % − market value of target** | SM form (TYK P2); value after merger = VA + VB + PV of gains; % = new shares ÷ total shares. | TYK P2: 20% × 1,450 − 250 = 40 lakh |
| Value of Acquisition = Value of Merged Entity − Value of Yes Ltd.; Gain to shareholders = Share of Yes Ltd. in merged entity − Value of Yes Ltd. before merger | SM form (TYK P16), values by DCF; $TV_5 = CF_5(1 + g) \div (K_e - g)$. | TYK P16: share 1/1.5 |
| **Maximum Price = (Market Value + Synergy Gain + Saving of Overpayment) ÷ No. of shares** | SM form (TYK P31); alt.: MP = PE × EPS × ER. | TYK P31: 3,50,00,000 ÷ 10,00,000 = 35 |
| **Minimum Price = (Value of Management Holding + PV of loss of remuneration) ÷ No. of shares held** | SM form (TYK P31); promoter-managers lose perks/salary on sale. | TYK P31: 1,26,00,000 ÷ 4,00,000 = 31.50 |
| Cost of Capital Ke = Dividend ÷ Price + Growth %; Price = Dividend ÷ (Ke − g) | Higher growth under the acquirer raises target value. | TYK P25: 0.60/20 + 0.07 = 0.10; 0.60/(0.10 − 0.08) = 30 |

8.2% of marks

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**TYK P2 (Elrond/Doom, ER 0.5):** Doom gets 5 lakh shares = 5 ÷ 25 = **20%**; value after merger 1,000 + 250 + 200 = **1,450 lakh**; true cost 290 − 250 = **₹ 40 lakh**.

**TYK P16 (Yes/No):** PV of CF 1–5: Yes **882.44**, merged **1,679.27**; TV Yes 367.50 ÷ 0.10 = 3,675 × 0.497 = 1,826.475; merged 657.20 ÷ 0.09 = 7,302.22 × 0.497 = 3,629.20. Value of Yes **2,708.915** before, **5,308.47** after; value of acquisition **2,599.555**; Yes holders' share 5,308.47 × 1/1.5 = 3,538.98 → gain **830.065 lakh**.

**TYK P31 (PQR/XYZ):** 2,40,00,000 + 80,00,000 + 30,00,000 = 3,50,00,000 → max **₹ 35** (alt. ER 0.875 × EPS 4 × P/E 10 = 35); promoters 40% × 10,00,000 × 24 = 96,00,000 + 30,00,000 = 1,26,00,000 ÷ 4,00,000 → min **₹ 31.50**.

**TYK P25 (AFC/BCD):** (i) Ke 10%, price at 8% growth **₹ 30**, increase (30 − 20) × 5,00,000 = **₹ 50,00,000**. (ii) 1 for 4: immediate gain to BCD 100 − 4 × 20 = ₹ 20; merged 1,000 + 150 = 1,150 lakh ÷ 11.25 lakh = **₹ 102.22** → BCD gain 102.22 − 80 = **₹ 22.22**, AFC gain **₹ 2.22**.

*Note (app-written): in TYK P25(ii) the BCD gains (₹ 20, ₹ 22.22) are per AFC share received, i.e. per 4 BCD shares (compared with 4 × ₹ 20), not per BCD share (SM p.78).*

### 13 (e) Cash acquisitions

| Formula | Terms / logic | SM figures |
|---|---|---|
| **EPS-neutral cash (borrowed) : [Net profit of acquirer + Net profit of target − x × 15% × (1 − 0.30)] ÷ No. of shares of acquirer = present EPS** | SM form (TYK P5), x = cash paid (₹ lakhs); after-tax interest on borrowing reduces earnings; no new shares. | TYK P5: (95.75 − 0.105x) ÷ 20 = 4 → x = 150 lakh |
| Gain per acquirer share (own cash) = (Earnings of target − Loss of earning in cash) ÷ No. of shares × P/E ratio | SM labels (TYK P25(iii)); cash paid loses its 10% earning; gain to target = price paid − market price. | TYK P25(iii): 1,50,000 ÷ 10,00,000 = 0.15 × 12.5 |
| Net cost = shares × issue price + liabilities taken/settled + new debentures − realisation of assets − cash taken over; Net Present Value = PV of Cash Inflow + PV of Demerger − Cash Outflow = Cash inflow × PVAF(16%, 6) + disposal × PVF(16%, 6) − outflow | Acquisition as a capital budgeting decision. | TYK P30: NPV 8,49,000 |

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**TYK P5(ii):** cash at 15% borrowing, tax 30% → **₹ 150 lakh** keeps EPS at ₹ 4.

**TYK P25(iii) (AFC pays ₹ 22 cash, cash earns 10%):** 12,50,000 − 11,00,000 = 1,50,000 → EPS +0.15; P/E 100/8 = 12.50 → AFC gain **₹ 1.88** per share (alt. 81,50,000 ÷ 10,00,000 = 8.15 × 12.50 = 101.875 − 100); BCD gain 22 − 20 = **₹ 2**.

**TYK P30 (Tiger/Leopard):** 35,000 × 15 = 5,25,000 + external liabilities 5,00,000 + 13% debentures 3,00,000 = 13,25,000 − debtors/inventories 2,00,000 − cash 50,000 = **10,75,000**; NPV 5,00,000 × 3.684 + 2,00,000 × 0.410 − 10,75,000 = 18,42,000 + 82,000 − 10,75,000 = **₹ 8,49,000** → acquire.

### 13 (f) Valuation-based exchange ratio (links to 4.2)

8.2% of marks

| Formula | Terms / logic | SM figures |
|---|---|---|
| NAV per share = (net assets − contingent liability expected) ÷ N | Net assets from balance sheet. | TYK P23: H (1,300 − 300) ÷ 3.50 = 285.71 |
| Earnings capitalisation value = (maintainable profit ÷ capitalisation rate) ÷ N | Profit capitalised at the given rate. | TYK P23: H 300/0.08 ÷ 3.50 = 1,071.43 |
| Fair value = weighted average of methods; Exchange ratio = fair value of target ÷ fair value of acquirer | Weights given (1 : 3; or simple average). | TYK P23: 156.3475 ÷ 875 = 0.1787 |
| Business value ÷ issue price = shares; allocate on **fully-paid equivalents** | Partly-paid share counted at paid-up proportion. | TYK P26: 25 lakh equivalents |
| Enterprise multiple: price = multiple × EBIDAT − debt | Consideration for equity; discharged in shares at market price. | TYK P21: 7 × 115.71 − 240 = 569.97 |

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**TYK P23 (H/B):** NAV H ₹ **285.71**, B 31.50 ÷ 0.65 = ₹ **48.46**; earnings cap. H ₹ **1,071.43**, B 10/0.08 ÷ 0.65 = ₹ **192.31**; fair value (1 × NAV + 3 × EC) ÷ 4: H ₹ **875**, B ₹ **156.3475**; ER **0.1787** H share per B share (contingent liability assumed to materialise in full; alternatively deduct it from EC value too).

**TYK P26 (AB/XY):** AB price (570 + 430) ÷ 2 = ₹ **500**; DCF 105 × 0.93 + 120 × 0.86 + 125 × 0.79 + 120 × 0.74 + 300 × 0.68 = **592.40**; net assets 350 − 100 = **250**; average **421.20 lakh** → 4,21,20,000 ÷ 500 = **84,240** shares; fully-paid equivalents 20 + 5 = 25 lakh → fully paid 84,240 × 20/25 = **67,392**, partly paid **16,848**.

**TYK P21 (T/E):** T's offer: consideration **569.97 lakh**, shares 569.97 ÷ 220 = **2,59,000**; NPAT (516.57 − interest 58 + 30) × 0.70 = **300**; EPS 300 ÷ 14.59 = ₹ **20.56**; pre-merger T EAT 240 ÷ 12 = EPS 20 → P/E 220/20 = 11 → MPS ₹ **226.16**. E's plan: 6 lakh × 110 = **660** ÷ 220 = **3 lakh** shares; EPS 300 ÷ 15 = ₹ **20**; MPS ₹ **220**. Advantages (same industry): synergy, cost reduction, operating efficiency; better market share; avoidance of competition.

*Note (app-written): TYK P26 — ICAI prints "(120x0.74)x(300x0.68)"; the operator is "+", and 300 = year-5 earnings 100 + terminal value 200; the total 592.40 is right (SM p.79).*

### 13 (g) Risk: expected value of debt and equity

→ Equity = total value − debt in each state; **limited liability → equity cannot be negative**: take equity 0 and debt = total value in that state; then expected value = Σ probability × value.

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**TYK P15 (Simple/Dimple):** Simple equity 360, 90, −50 → 0; debt 460, 460, 410 → expected debt **450**, equity **126** (total 576). Dimple equity 985, 760, 525 → **758**; debt **65** (total 823). Merged: equity **884**, debt **515** (ICAI adds the two stand-alone expected values).

### 13 (h) Weighted swap ratio, promoter holding, free float, bonus/split, bank merger

8.2% of marks

| Formula | Terms / logic | SM figures |
|---|---|---|
| **Swap ratio = Σ (ratio of target to acquirer for each parameter × weight)** | SM layout: EPS 1 : 4 i.e. 4.0 × 40%, etc.; weights by the Board. | TYK P17: 4 × 40% + 0.8 × 25% + 2 × 35% = 2.5 |
| Bank: for "lower is better" items (Gross NPA) use acquirer ÷ target | App-written logic: ratio must show how good the weak bank is relative to the strong one. | TYK P29: NPA 5 : 40; CAR 5 : 16 |
| CAR / CRWAR = Total Capital ÷ Risky Weighted Assets; GNPA Ratio = Gross NPA ÷ Gross Advances × 100 | SM forms; Risky Weighted Assets = Total Capital ÷ CAR (derived). | TYK P29: 6,180 ÷ 41,100 = 15.04% |
| **Promoter % after = (promoter shares of A + promoter shares of T × ER) ÷ total shares** | Target promoters' shares converted at the ER. | TYK P17: (4.75 + 5 × 2.5) ÷ 28.75 = 60% |
| **Free float market cap = MPS × non-promoter shares** | Public's holding only. | TYK P17: 69.56 × 28.75 × 40% = 799.94 |
| Total market cap = free float cap ÷ non-promoter % | Back out MPS = total cap ÷ N; EPS = MPS ÷ P/E. | TYK P18: 500 ÷ 50% = 1,000 lacs |
| Post BVPS = (capital + reserves of both) ÷ total shares; equivalent per target share = × ER | Assets taken at book value; total net worth unchanged by the swap. | TYK P18: 1,800 ÷ 3.48825 = 516.02 |
| Bonus 1 : 2 → capital × 1.5 (from reserves); split ₹ 100 → ₹ 5 → shares × 20 | Earnings and net worth unchanged; spread over more shares. | TYK P20: 105 lakh shares |

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**TYK P17 (Efficient/Healthy):** MPS 50 / 100; EPS 5 / 20; profit 50 / 150 lakh; BVPS 400/10 = 40, 240/7.5 = 32; ER **2.5** → 18.75 lakh shares; promoters 17.25 ÷ 28.75 = **60%**; EPS 200 ÷ 28.75 = **₹ 6.956**; MPS **₹ 69.56**; market cap **₹ 1,999.85 lakh**; free float **₹ 799.94 lakh**.

**TYK P19 (E/H, same data doubled):** MPS 50 / 100, EPS 5 / 20, BVPS 40 / 32; ER **2.5** → 37.50 lakh; promoters 34.50 ÷ 57.50 = **60%**; EPS 400 ÷ 57.50 = **₹ 6.956**; MPS **₹ 69.56**; cap **₹ 3,999.70 lakh** (≈ 4,000); free float **₹ 1,599.88 lakh**.

**TYK P18 (Abhiman/Swabhiman):** BVPS 550 / 70; total cap 1,000 / 390 lacs; MPS 500 / 39; EPS 50 / 9.75; ER 0.1273 × 25% + 0.195 × 50% + 0.078 × 25% = **0.148825** → 1,48,825 shares; BVPS **₹ 516.02** (per Swabhiman share 76.80); EPS (100 + 97.50) ÷ 3.48825 = **₹ 56.62** (8.43); MPS × 10 = **₹ 566.20** (84.26).

**TYK P20 (Abhiman/Abhishek):** BVPS 500 / 60; cap 800 / 320; MPS 400 / 32; EPS 40 / 8; profits 80 / 80 lakh; ER 0.03 + 0.10 + 0.02 = **0.15** → 1.50 lakh shares; BVPS 1,600 ÷ 3.5 = **₹ 457.14**; EPS 160 ÷ 3.5 = **₹ 45.71**; MPS **₹ 457.10**; promoters 1.00 + 0.90 = 1.90 ÷ 3.50 = **54.29%**; free float (3.5 − 1.9) × 457.10 = **₹ 731.36 lakh**. Bonus 1 : 2 + split to ₹ 5: capital 350 + 175 = 525 lakh → 5.25 lakh × 20 = **105 lakh** shares; EPS 160 ÷ 105 = **1.523**; BV (525 + 1,075) ÷ 105 = **₹ 15.238**.

**TYK P29 (Weak/Strong Bank):** NPA 5/40 × 30% = 0.0375; CAR 5/16 × 28% = 0.0875; MPS 12/96 × 32% = 0.0400; BVPS 12/120 × 10% = 0.0100 → **0.1750**. BVPS: W (230 − preliminary 50) = 180 ÷ 15 = 12; S 6,000 ÷ 50 = 120. Shares 15 lakh × 0.175 = **2.625 lakh** (₹ 26.25 lac); capital reserve 180 − 26.25 = **153.75**; merged balance sheet **₹ 57,570 lac** (capital 526.25, reserves 5,500, deposits 48,000, other liabilities 3,390; cash 2,900, other banks 2,000, investments 20,100, advances 30,500, other assets 2,070). RWA = capital ÷ CAR: 180/5% = 3,600, 6,000/16% = 37,500 → CAR 6,180 ÷ 41,100 = **15.04%**. Gross NPA 40% × 3,500 = 1,400 + 5% × 27,000 = 1,350 = **₹ 2,750 lac**.

Note (app-written): ICAI stops at Gross NPA ₹ 2,750 lac; as a ratio on merged advances 2,750 ÷ 30,500 = 9.02% (SM p.83).

Jan 2026 Q1(b) (SEBI promoter limit 84% → 75% by bonus to minority at the same P/E) uses the promoter-%, bonus and free-float logic above: bonus shares go to non-promoters only, so promoter shares ÷ new total = 75%.

8.2% of marks

## Exam steps

**Exam steps — swap ratio and post-merger EPS/MPS:** EPS, MPS, P/E for both (derive missing one: MPS = EPS × P/E) → ER on the stated basis (target ÷ acquirer) → shares issued = target's shares × ER → post EPS = total earnings (+ synergy) ÷ total shares → target's equivalent EPS = post EPS × ER → post MPS = post EPS × given P/E → compare each group with pre-merger figures.

**Exam steps — maximum exchange ratio:** EPS-based: combined PAT ÷ acquirer's EPS − existing shares = max shares to exchange → ratio to target's shares. MPS-based: combined market capitalisation (or earnings × P/E) ÷ acquirer's MPS − existing shares → ratio to target's shares. Range question: lower = market price ratio, upper = intrinsic value ÷ acquirer's MPS; say which end and why (EPS, P/E, ROE, growth).

**Exam steps — gains and true cost:** $V_A$, $V_B$ → $V_{AB}$ (sum + synergy, or P/E × combined earnings, or DCF) → gain = $V_{AB} - (V_A + V_B)$ → each group: % holding × $V_{AB}$ − own pre-merger value → true cost = target's share − $V_B$ (share deal). Max price = ($V_B$ + synergy + savings) ÷ target's shares; min price = (holding + lost benefits) ÷ shares.

**Exam steps — weighted swap / promoter / free float:** per-share MPS, EPS, BVPS for both (from market cap, P/E, capital + reserves; from free float ÷ public %) → each ratio T ÷ A (bank NPA: A ÷ T) × weight → sum = ER → shares issued → promoter % = (A's promoters + T's promoters × ER) ÷ total → post EPS, MPS (A's P/E), market cap → free float = MPS × public shares → bonus/split if asked.

**Exam steps — cash acquisition:** EPS-neutral: (net profits of both − x × interest rate × (1 − tax rate)) ÷ acquirer's shares = acquirer's present EPS → x. Own cash: lost return on cash vs target earnings → EPS change × P/E. Project view: net consideration (shares + liabilities + debentures − realisations − cash taken) → NPV of division cash flows + disposal value.

**Exam steps — demerger:** split assets and outside liabilities at book value → net worth of each → new co. EPS = expected MPS ÷ sector P/E → shares = profit ÷ EPS → ratio to old shares → remaining co. MPS = EPS × its P/E → BVPS = net worth ÷ shares.

**Exam steps — financial restructuring:** total sacrifices (capital reduction, waivers) + revaluation gains = amount available → write off P&L debit, fictitious assets, provisions, asset write-downs (must tally) → new capital (old reduced + conversions + fresh issue) → cash as balancing figure → new balance sheet.

7.1% of marks

# 11 Ch 11 — International Financial Management

| Attempt | Q | M | Asked (MCQ / numerical / theory) |
|---|---|---|---|
| Sep 2025 | Q5(a) | 7 | Foreign (Nepal) import project: NPV with NC cash flows converted at forward rates from the inflation differential and discounted at the Indian rate (Numerical) |
| Jan 2026 | Q5(b) | 6 | GDR issue: number of GDRs for a dollar requirement and cost of GDR to the company (discount, flotation cost, dividend growth) (Numerical) |
| May 2026 | Q1(a) | 6 | Foreign project with a currency swap at spot for the initial and terminal principal; forward rates for technical fees and the excess resale proceeds; PV of facilitation fee; NPV in rupees and accept/reject (Numerical) |
| May 2026 | Q3(a) | 6 | Project in Europe with 50% debt: forward rates by interest rate parity, cheapest borrowing option, then APV (unlevered NPV at 12% plus PV of tax shield at 5%); ICAI gives a main and an alternative answer (Numerical) |

## 1.5 Adjusted Present Value (APV)

Asked: May 2026 Q3(a) 6m

→ **Value-additive** approach: each cash flow discounted separately at a rate matching **its own risk**.

→ Test **basic viability** first; if feasible, stop. If not, add the other complexities (tax shields, subsidies).

| Formula | Terms / logic |
|---|---|
| $APV = -I_0 + \Sigma\, X_t \div (1 + k^*)^t + \Sigma\, T_t \div (1 + i_d)^t + \Sigma\, S_t \div (1 + i_d)^t$ | $I_0$ = PV of investment outlay (net of usable blocked funds); $X_t$ = operating CF, discounted at **$k^*$ = unlevered cost of capital**; $T_t$ = tax saving from financing mix (interest tax shield); $S_t$ = before-tax value of interest subsidies from project-specific financing; last two at **$i_d$ = before-tax cost of debt** (they are as certain as debt). |

→ **Blocked funds:** balances abroad that cannot be remitted. **Direct** = exchange-control bar. **Indirect** = heavy local/withholding tax on remittance. If released for the new project, they **reduce the investment outlay**.

## 1.6 Scenarios · 1.6.1 A foreign company investing in India

Asked: Sep 2025 Q5(a) 7m; May 2026 Q1(a) 6m

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**Illus 1 (Perfect Inc., US, JV in Baddi):** 55% revenue share + royalty $0.01/bottle; outlay ₹ 200 cr, scrap ₹ 5 cr; depreciation 20% of 195 = 39.00; WC 50 + 5 p.a.; 24 cr bottles at ₹ 7.50 (+₹ 1 each year, plus that year's inflation); production cost 40%. Tax India 30%, US 35% with credit, paid next year. Spot ₹ 57; inflation India 6% (falls 0.50% p.a.), US 5%. Only **50% remittable** in the year, rest next year. Rate 15%.

PPP rates yr 0–6: 57, 57.54, 57.82, 57.82, 57.54, 56.99, 56.18. Inflated prices 7.95, 8.97, 9.98, 10.97, 11.96 → 55% share 104.94 … 157.87. Royalty ₹ 13.81 … 13.68 cr. PBT 37.77 … 69.40; Indian tax 11.33 … 20.82. FCF: −250.00, 71.77, 68.59, 74.15, 79.51, 164.89, −20.82. Remittance (50% + previous 50%): −250.00, 35.88, 70.17, 71.37, 76.84, 122.20, 61.62 → in $ mn −43.86, 6.24, 12.14, 12.34, 13.35, 21.44, 10.97. US tax 35% (next year) less Indian tax credit → net tax; net CF −43.86, 6.24, 11.92, 10.47, 11.84, 20.02, 7.18, −3.84. **NPV @15% = −$4.15 mn → do not invest.**

Logic: compute project CF in host currency → apply remittance restriction → convert at PPP rates → home tax less foreign tax credit (with the one-year lag) → discount at the parent's required rate.

7.1% of marks

## 2.3 Global Depository Receipts (GDRs)

Asked: Jan 2026 Q5(b) 6m

- → **Negotiable certificate**, denominated in a currency not native to the issuer, representing its publicly traded local-currency equity shares. Mostly USD, a few in Euro/Pound. Issued in US = ADR; outside US = GDR. Can represent debt in theory, rarely in practice.
- → Created when local shares are delivered to the depository's **local custodian bank**; depository bank (e.g. Bank of New York) issues DRs in USD; traded on a foreign exchange, OTC, or among **QIBs**. Indian issues are GDRs as they are marketed globally.
- → Why Indian companies used them: equity (no principal repayment, lower servicing cost); no other form of term forex funding was available; fetched **higher prices** than a domestic issue.

**Impact of GDRs on Indian capital market** (TYK T2):

- → Market shifting from **Bombay to Luxemburg**.
- → **Arbitrage** possibility in GDR issues.
- → Market **no longer independent** of the world — investors must track worldwide events.
- → **Retail investors sidelined** — GDR/FII placements + free pricing end easy money on discounted rights/public issues.
- → Result: considerable foreign investment flowed into India.

**Markets of GDRs:**

- → Sold primarily to **institutional investors**; demand dominated by **emerging market funds**.
- → FIIs likely to **switch** from ordinary shares into GDRs.
- → Major demand: **UK, USA (QIBs), South East Asia** (Hong Kong, Singapore), some continental Europe (France, Switzerland).

**Mechanism:** company issues ordinary shares → kept with custodian/depository bank → GDRs issued to foreign investors.

**Characteristics:**

- → Economic benefits of shareholders but **no voting rights**.
- → Settled through **CEDEL & Euro-clear**; listed on **Luxemburg** stock exchange.
- → Traded **OTC** between professional market makers; **freely traded**; marketed globally in more than one currency.
- → Dividends paid in issuer currency, converted into dollars by depository → **exchange risk with investor**.
- → **Cancellation** any time after a cooling period of **45 days**: depository asks domestic custodian to release underlying shares; BSE/NSE price on date of redemption advice = cost of acquisition.

| Formula | Terms / logic | SM example figures |
|---|---|---|
| Gross issue = Net requirement ÷ (1 − flotation %) | Funds needed are net of issue cost. | Illus 4: $10 mn ÷ 0.98 = $10.204 mn |
| Price per GDR = MP × shares per GDR × (1 − discount) ÷ exchange rate | Converted to $ at expected rate. | 250 × 2 × 90% = ₹ 450 ÷ 60 = $7.50 |
| No. of GDRs = Gross issue ÷ price per GDR | | 10.204 ÷ 7.50 = **1.3605 mn** |
| Cost of GDR $K_e$ = $D_1$ ÷ Net proceeds per GDR + g | $D_1$ and net proceeds both per GDR (in ₹). | 40 ÷ 441 + 0.12 = **21.07%** |

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**Illus 4 (X Ltd., plant in US):** need $10 mn net, flotation 2%; MP ₹ 250 (FV ₹ 100); 2 shares per GDR at 10% discount; ₹ 60/$; dividend 20% (₹ 20 per share → $D_1$ ₹ 40 per GDR), growth 12%. Net proceeds per GDR ₹ 450 × 0.98 = ₹ 441.00. (a) **1.3605 million GDRs**; (b) **cost 21.07%**. Also see Illus 3: 2.75 ÷ 198 + 0.10 = 11.39%.

7.1% of marks

## Exam steps

**Exam steps — foreign project NPV (home currency approach):** 1 forward rates by PPP (inflation) or IRP (interest), watch direct vs indirect quote → 2 foreign CF (inflate if real; tax; depreciation; WC; terminal) → 3 apply remittance restriction / withholding tax / home tax less credit → 4 convert each year → 5 add home-currency CF → 6 discount at home rate → 7 accept if NPV > 0.

**Exam steps — foreign currency approach:** 1 $1 + RP = (1 + k_{home}) \div (1 + i_{home})$ → 2 $k_f = (1 + RP)(1 + i_f) - 1$ → 3 discount foreign CF → 4 foreign NPV × spot.

**Exam steps — APV:** 1 forward rates (IRP) if foreign flows → 2 base NPV: operating CF at unlevered $k^*$ → 3 PV of interest tax shields (and subsidies) at before-tax cost of debt → 4 APV = base NPV + PV shields (− issue costs / + usable blocked funds).

**Exam steps — GDR:** 1 gross issue = net ÷ (1 − flotation) → 2 price per GDR = MP × shares × (1 − discount) ÷ rate → 3 number = gross ÷ price → 4 $K_e = D_1$ per GDR ÷ net proceeds per GDR + g.

**Exam steps — netting:** 1 convert all to common currency → 2 matrix: receipts (row) vs payments (column) → 3 net position per affiliate → 4 net transfers vs gross, % saving.

5.9% of marks

# 12 Ch 12 – Interest Rate Risk Management

| Attempt | Q | M | Asked (MCQ / numerical / theory) |
|---|---|---|---|
| Sep 2025 | Q2(b) | 4 | FRA final settlement amount for a borrower, for a higher and a lower actual rate (formula with days/year basis) (Numerical) |
| Jan 2026 | Q4(a) | 6 | Interest rate options: strategy for borrower (buy cap) and depositor (buy floor), premium from lump-sum premium rate, net gain/loss from hedging (Numerical) |
| May 2026 | Q4(c) [or] | 4 | Zero-cost collar on a floating-rate loan: value of cap = value of floor (intrinsic value, PV of payoffs); solve the floor rate (Numerical) |
| May 2026 | Q5(b) | 7 | Re-pricing gap (RSA - RSL): negative gap, favourable/unfavourable scenario, Earnings at Risk = gap x rate change x 1/4 versus 8% of NII limit, hedging decision (Numerical) |

## 3.1.1 Gap Exposure

Asked: May 2026 Q5(b) 7m

→ Gap (mismatch) risk: assets, liabilities and off-balance sheet items with different **principal amounts, maturity dates or re-pricing dates**. Most important for banks.

| Formula | Terms / logic | Effect on NII |
|---|---|---|
| **Gap = RSA – RSL** | RSA = interest rate sensitive assets; RSL = interest rate sensitive liabilities (re-priced in the time band). Gap = measure of rate sensitivity. | **Positive** (asset sensitive): rate **rise** → NII up. **Negative** (liability sensitive, RSL > RSA): rate **fall** → NII up; rise → NII down. |
| **EaR = Gap × assumed change in interest rate** | Earnings at Risk: how much earnings may be hit by an adverse rate move. Rate change estimated from past trends / forecasts. | Bank fixes an EaR limit (based on last/current year income) and a **trigger point** at which on- or off-balance sheet **hedging** starts. |

Gap figures can be augmented with the average coupon in each time band to estimate NII; periodic gap analysis shows exposure over distinct maturities and the portfolio change needed to alter the risk profile. (In May 2026 Q5(b) the gap was for a quarter, so EaR used × 1/4, compared with a limit of 8% of NII.)

**Limitations of Gap report:**

→ Measures only the **time difference** in re-pricing; ignores **basis** and **embedded option** risks.

→ Assumes all items in a time band mature/re-price **simultaneously**; ignores effect on **economic/market value** of assets, liabilities and off-balance sheet positions.

→ Ignores differences in **timing of payments** due to changed rate environment.

→ Assumes **parallel shift** in yield curve, which seldom happens.

→ Ignores variability in **non-interest revenue and expenses**.

## 4.1.2 Forward Rate Agreements (FRAs)

Asked: Sep 2025 Q2(b) 4m

**Features (TYK TI — short note):**

→ Agreement by which a **borrower/lender** protects itself from unfavourable rate changes; **OTC**, not exchange-traded (unlike futures).

→ Used by banks to fix interest cost on anticipated deposits or revenue on variable-rate loans indexed to a benchmark (SOFR, €STER, MIBOR).

→ **Off-balance sheet**; **no transfer of principal** — principal is **notional**.

→ **Cash-settled** at maturity for the profit/loss. **Seller** pays buyer the increased interest cost if RR > FR; **buyer** pays seller if RR < FR. So a **borrower buys** an FRA (protection against a rise).

→ Differential is **discounted at the actual (post-change) rate** because it is settled at the **beginning** of the period, not at the end.

5.9% of marks

| Formula | Terms / logic | SM example figures |
|---|---|---|
| **Settlement = N × (RR − FR) × (dtm/DY) ÷ [1 + RR × (dtm/DY)]** | **N** = notional principal. **RR** = reference rate for the contract maturity on the settlement date (SOFR/€STER/MIBOR). **FR** = agreed forward rate. **dtm** = FRA period in days. **DY** = day-count basis, 360 or 365. Numerator = extra (or saved) interest for the FRA period; denominator = discount at RR for the same period (paid upfront). Positive → seller pays buyer; negative → buyer pays seller. | $1 m, FR 5%, 3-month SOFR 6%: 1 bp on $1 m for 90 days = **$25** → $25 × 100 bp = **$2,500** → buyer receives $2,500 ÷ [1 + 0.06(90/360)] = **$2,463.05**. |
| **a × b FRA** | Settlement after **a** months; deposit/loan matures **b** months from contract date → FRA period = b − a months. Use RR for that (b − a) period. | **1 × 4**: deposit placed after 1 month, maturing 4 months from contract (3-month period). |

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**TYK P1 (Parker & Co.):** borrow ₹ 60 crore for 3 months in 6 months; FRA 9.30%. (i) Actual 9.60%: 60 cr × (0.096 − 0.093)(3/12) ÷ [1 + 0.096(3/12)] = 60 cr × 0.00075 ÷ 1.024 = **₹ 4,39,453.13** → banker pays Parker. (ii) Actual 8.80%: 60 cr × (−0.00125) ÷ 1.022 = **− ₹ 7,33,855.19** → Parker pays banker. ICAI: 90/360, 90/365, 91/360 or 91/365 also accepted.

**TYK P2 (TM Fincorp, bought 6 × 9, ₹ 100 crore at 5.25%):** 6 × 9 = 3-month rate in 6 months → use 3-month MIBOR **5.50%** (not 6 or 9-month). (a) Profit **25 basis points**. (b) 100 cr × (5.50% − 5.25%)(90/360) ÷ [1 + 0.055(90/360)] = **₹ 6,16,523** profit.

## 4.2.2 Interest Rate Options (IRO)

Asked: Jan 2026 Q4(a) 6m

→ Also called **Interest Rate Guarantee (IRG)**; a **right, not an obligation**; acts as **insurance** — protection against adverse moves while keeping benefit of favourable moves. Cost = **premium**.

→ An IRO is basically a **series of FRAs** exercisable at pre-set benchmarked rates for each period (3 months, 6 months ...).

## 4.2.2.1–4.2.2.2 Cap Option and Floor Option

Asked: Jan 2026 Q4(a) 6m

→ **Cap** (call option): buyer pays premium for the right to receive the excess interest whenever the index rises **above the cap rate**; no obligation if rates fall. Bought by a **borrower** on floating-rate debt. Evolved from guarantees on floating-rate loans; OTC caps from **1985** (banks stripped guarantees from FRNs); spurred by the **leveraged buyout** boom of the 1980s (lenders required LBO borrowers to buy caps); activity declined as LBOs fell.

→ **Cap is characterised by: notional principal**; **benchmark** (SOFR, MIBOR, PLR) of a specified maturity; **cap rate** (= strike); **period** incl. payment and reset dates. Payments follow swap conventions; reset intervals usually equal the benchmark's term.

→ **First period is excluded** — its rate is known on the agreement date.

→ **Floor** (OTC): protects the buyer against a **fall** in rates — bought by a **lender/depositor**. Payoff when rate < floor rate = payoff from **selling an FRA**; above floor → nothing, premium lost. Floor = **European put on an FRA**; multi-period floor = bundle of European puts on a sequence of FRAs.

5.9% of marks

| Formula | Terms / logic | SM example figures |
|---|---|---|
| **Cap payoff = N × max(0, r − $r_c$) × (dt / days in year)** | r = actual spot rate on reset date; $r_c$ = cap rate (decimal); **dt** = days from reset date to payment date. Paid on the **next payment date** (in arrears). | $1 m, 6-month SOFR cap 5%, 15 Jan–15 Jan, reset 15 July; SOFR 5.5% → on next 15 Jan (184 days): $1,000,000 × (0.055 − 0.050)(184/360) = **$2,555.56**. |
| **Floor payoff = N × max(0, $r_f$ − r) × (dt / days in year)** | $r_f$ = floor (strike) rate; dt = days from last reset to payment date. | — |
| **Premium per reset period = (Lump-sum premium rate × A) ÷ PVAF(i, t)**, i.e. P = rp × A ÷ [(1/i) × (1 − 1/(1+i)$^t$)] | Upfront lump-sum premium spread into equal amounts per reset period. **A** = principal; **rp** = premium rate; **i** = fixed rate per period; **t** = number of periods. | TYK P3 below. |

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**TYK P3 (XYZ Ltd):** £ 15 m, 6-month SONIA, 24 months; cap 8%; lump sum premium 1% (= £ 150,000); fixed rate 7% p.a. → 3.5% per half-year, 4 periods. PVAF(3.5%, 4) = 0.966 + 0.933 + 0.901 + 0.871 = **3.671** → premium **£ 40,861** per period (formula route: **£ 40,595**). SONIA 9%, 9.5%, 10% → amount from bank = (SONIA − 8%) × 15 m × 6/12 = **£ 75,000; £ 112,500; £ 150,000** = **£ 337,500**. Less premium 3 × 40,861 = £ 122,583 → net **£ 214,917** (with £ 40,595: **£ 215,715**). Risk of £ 337,500 reduced by £ 214,917.

**TYK P7 (textile manufacturer):** floating loan ₹ 40,00,000; to peg cost at 8.5%, **buy caps**: notional ₹ 40,00,000, strike 8.5%, settlement 31 March each year till 31 March 2016. Premium 0.75% paid once on 1 Oct 2012 = **₹ 30,000**. Compensation: 2013 (10.2%) 1.70% → **₹ 68,000**; 2014 (11.5%) 3.00% → **₹ 1,20,000**; 2015 (9.25%) 0.75% → **₹ 30,000**; 2016 (8.25% < 8.5%) → **nil**. Cost never exceeds 8.5%.

Note (app-written): TYK P3 states the loan as "SONIA + 10.00%"; the spread does not affect the cap payoff (only SONIA vs 8% matters). ICAI spreads the premium over 4 half-years (PVAF 3.5%, 4) but deducts it for only the 3 reset periods shown (first period excluded), so the premium deducted totals only £ 122,583 (three of four instalments). The exact PVAF(3.5%, 4) is 3.6731, giving about £ 40,838; ICAI's £ 40,861 uses the rounded 3.671 and its alternative £ 40,595 does not reconcile exactly. Keep ICAI's figures.

## 4.2.2.3 Interest Rate Collars

Asked: May 2026 Q4(c) 4m

- → Collar = **buy a Cap + sell a Floor** → floating rate **bounded** on both high and low side.
- → Limits benefit from a big fall in rates, but **cheaper than a cap alone**: floor premium received offsets cap premium.
- → **Zero-cost collar:** premium earned on the floor **exactly offsets** the cap premium (value of cap = value of floor; solve for the unknown floor or cap rate).

| Formula | Terms / logic |
|---|---|
| **Collar payment = N × [max(0, r − $r_c$) − max(0, $r_f$ − r)] × (dt / days in year)** | First bracket = cap receipt (r above cap); second = floor payment owed (r below floor); between $r_f$ and $r_c$ → nil. r = actual spot rate on reset date; dt = term of the index in days. |

## Exam steps

**Exam steps — FRA:** identify buyer (borrower) / seller → FRA period from a × b (b − a months) and pick RR for that period → (RR − FR) × N × dtm/DY → ÷ [1 + RR × dtm/DY] → sign: + seller pays buyer, − buyer pays seller → state who pays whom.

5.9% of marks

**Exam steps — Cap / Floor:** borrower buys cap, depositor buys floor → premium = rp × N, spread per period ÷ PVAF(fixed rate per period, n) → each reset: payoff = N × max(0, r − $r_c$) [or max(0, $r_f$ − r)] × period → net = payoff − premium per period → total net gain / risk reduced.

**Exam steps — Collar:** buy cap + sell floor → payoff = cap receipt − floor payment each period → zero-cost: PV of cap = PV of floor → solve unknown rate → effective rate lies between floor and cap.

**Exam steps — Plain vanilla / OIS:** fixed leg N × AIC × 180/360 → floating leg N × R × actual days/360 (OIS: compound daily, holidays carry previous rate) → net = difference → solve fixed rate from net if given.

**Exam steps — Comparative advantage swap:** table of each party's rates → each borrows where relatively cheaper → list receipts/payments under swap → net cost vs cost without swap → gain to each.

**Exam steps — Gap / EaR:** RSA − RSL per time band → sign (positive gains on rise, negative gains on fall) → EaR = gap × rate change × fraction of year → compare with EaR limit → hedge if beyond trigger.

4.0% of marks

# ② Ch 02 — Risk Management

| Attempt | Q | M | Asked (MCQ / numerical / theory) |
|---|---|---|---|
| Sep 2025 | Q4(c) [or] | 4 | Parameters used to identify currency risk (SM Ch 2 section on currency risk) (Theory) |
| Jan 2026 | Q3(c) | 4 | Political risk: host-government actions that signal it, and techniques to mitigate it (SM Ch 2 section on political risk) (Theory) |
| May 2026 | MCQ CS-IV 10-12 | 6 | Value at Risk of a two-stock portfolio (annual VaR at 99%), diversification benefit at 90%, effect of change in correlation (MCQ) |

## 3.3 Example (VAR computation)

Asked: May 2026 MCQ CS-IV 10-12 6m

| Formula | Terms / logic | SM example figures |
|---|---|---|
| σ in ₹ = σ% × Value of holding | Daily SD of price converted to rupee volatility. | 2% × ₹ 2 crore = ₹ 4 lakh |
| 1-day VAR = 2 × $\sigma_{₹}$ | 2 = number of SDs for the confidence level (normal table, cumulative area). 99% → **2 = 2.33**. | ₹ 4 lakh × 2.33 = **₹ 9.32 lakh** |
| n-day VAR = 1-day VAR × √n | SD scales with square root of time (n trading days). | √10 × ₹ 9.32 lakh = **₹ 29.47 lakh** |
| Portfolio variance (₹): $V = \sigma_A^2 + \sigma_B^2 + 2 \times r \times \sigma_A \times \sigma_B$ | $\sigma_A$, $\sigma_B$ = rupee daily SD of each holding; r = correlation. Portfolio σ = √V. | TYK PI: $2^2 + 2^2 + 2 \times 0.3 \times 2 \times 2 = 10.4$ → σ = ₹ 3.22 lakh |
| Alternative (%): $\sigma_p^2 = \sigma_A^2 w_A^2 + \sigma_B^2 w_B^2 + 2\sigma_A\sigma_B r\, w_A w_B$ | w = weight of each holding in portfolio value; then $\sigma_{₹} = \sigma_p\%$ × total value. | 0.25 + 0.25 + 0.15 = 0.65 → $\sigma_p$ = 0.80623%; ₹ 400 lakh × 0.80623% = ₹ 3.22 lakh |

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**SM Example 3.3 (X Ltd):** holding ₹ 2 crore, daily SD 2%, 252 trading days, 99% → Z 2.33; σ ₹ 4 lakh; 1-day max loss **₹ 9.32 lakh**; 10-day **₹ 29.47 lakh**.

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**TYK PI (XYZ + ABC):** ₹ 200,00,000 in each, daily SD 1% each, r 0.3, 10-day 99% VAR. Daily SD of each ₹ 2 lakh; V = 10.4; σ = **₹ 3.22 lakh** (alt.: 0.65 → 0.80623% × ₹ 400 lakh); 10-day σ = ₹ 3.22 lakh × √10 = **₹ 10.18 lakh**; Z for 1% = 2.33 (1% of a normal distribution lies more than 2.33 SDs below the mean); 10-day 99% VAR = 2.33 × ₹ 10.18 lakh = **₹ 23.72 lakh**.

*Note (app-written): ICAI's alternative working prints the variance as "0.65%"; in % units it is 0.65 (%²), whose root is 0.80623%. The answer ₹ 3.22 lakh is unaffected.*

4.0% of marks

## 4.2 Political Risk

**Asked: Jan 2026 Q3(c) 4m**

| Identified from host-govt actions | Assess country risk | Mitigation techniques |
|---|---|---|
| → *Insistence on* ***resident investors or labour***. | → ***Political ranking*** *published by business magazines.* | → ***Local sourcing*** *of raw materials and labour.* |
| → ***Restriction on conversion*** *of currency.* | → *Evaluating country's* ***macro-economic conditions***. | → ***Joint ventures***. |
| → ***Expropriation*** *of foreign assets by local govt.* | → *Analysing* ***popularity of current government*** *and its stability.* | → ***Local financing***. |
| → ***Price fixation*** *of products.* | → ***Advice from embassies*** *of home country in host country.* | → ***Prior negotiations***. |

## 4.4 Currency Risk

**Asked: Sep 2025 Q4(c) [or] 4m**

*Depends on Government action and economic development. Parameters to identify:*

- → ***Government action***: *visual impact on currency (e.g.* ***Brexit*** *took the pound to its lowest since the 1980s).*
- → ***Nominal interest rate***: *per* ***interest rate parity (IRP)***, *exchange rate depends on the country's nominal interest rate.*
- → ***Inflation rate***: ***purchasing power parity*** *theory — inflation impacts currency value.*
- → ***Natural calamities***: *negative impact.*
- → ***War, coup, rebellion*** *etc.: far-reaching impact on exchange rates.*
- → ***Change of government*** *and its attitude towards foreign investment.*

*Management: covered in the later foreign exchange risk chapter.*

*Note (app-written): the diversification-benefit line in the sticky above follows from the SM formula; the SM itself does not define the term (it was tested in May 2026).*

### Exam steps

**Exam steps (single-asset VAR):** $\sigma_{₹}$ = σ% × value → 1-day VAR = 2 × $\sigma_{₹}$ (99% → 2.33) → n-day VAR = 1-day VAR × √n.

**Exam steps (two-asset portfolio VAR):** rupee σ of each holding → $V = \sigma_A^2 + \sigma_B^2 + 2r\sigma_A\sigma_B$ → $\sigma_p = \sqrt{V}$ → × √n for n days → × 2. Diversification benefit = sum of individual VARs − portfolio VAR; lower r → lower portfolio VAR.

3.4% of marks

# ① Ch 01 – Financial Policy and Corporate Strategy

| Attempt | Q | M | Asked (MCQ / numerical / theory) |
|---|---|---|---|
| Sep 2025 | Q2(c) | 4 | Key points that make an organisation financially sustainable (Theory) |
| Jan 2026 | Q2(b) | 4 | External funding requirement (AFN) next year at 65% and 95% fixed-asset capacity usage with a given sales growth and payout ratio (Numerical) |
| May 2026 | Q1(c) | 4 | Sustainable growth rate versus target growth: concerns on an additional loan request and two alternative courses of action (Theory) |

## 6. Balancing Financial Goals vis-à-vis Sustainable Growth

Asked: Sep 2025 Q2(c) 4m

TYK T2 (answer: para 6).

- Sustainable growth forces managers to consider the **financial consequences of sales increases** and set sales growth goals **consistent with operating and financial policies**; conflict if growth objectives exceed sustainable growth.
- Rightness for **future stakeholders** too: fuel industry's **save fuel campaign = demarketing** → example of **stable growth strategy** under resource constraints. Variants of stable growth: **incremental growth, profit, pause** strategies.
- **Too fast or too slow** growth both harm the enterprise; finance should keep growth speed **close to the sustainable growth ratio**.

**What makes an organisation financially sustainable?** It must:

- have **more than one source** of income;
- have **more than one way** of generating income;
- do **strategic, action and financial planning** regularly;
- have **adequate financial systems**;
- have a **good public image**;
- be clear about its values (**value clarity**);
- have **financial autonomy**.

### 6.1 Sustainable Growth Rate

Asked: May 2026 Q1(c) 4m

- **SGR** (Robert C. Higgins) = **maximum rate of growth in sales** achievable given the firm's **profitability, asset utilisation, desired dividend payout and debt (financial leverage) ratios**.
- Measures how much a firm can grow **without borrowing more money**; beyond it, the firm must borrow from another source.
- Variables: **net profit margin** on new and existing revenues; **asset turnover** (sales ÷ total assets); **assets to equity** ratio; **retention rate** (fraction of earnings retained).

| Formula | Terms / logic |
|---|---|
| **SGR = ROE × (1 − Dividend payout ratio)** | ROE = return on equity; (1 − d) = retention rate. Logic: only new equity is retained earnings, and the asset to beginning-of-period equity ratio is constant (debt grows in step) → sales and assets grow no faster than retained earnings plus the debt they support. |

Note (app-written): ROE itself is built from the SM's other three variables — net profit margin × asset turnover × assets to equity — so SGR moves up with any of the four.

Assumptions — the business wants to:

- maintain a **target capital structure** without issuing new equity;
- maintain a **target dividend payout** ratio;
- increase sales **as rapidly as market conditions allow**.

3.4% of marks

→ Consistent with evidence that corporations are **reluctant to issue new equity**; if willing to issue equity, there is in principle **no financial constraint** on growth. Formula is directly predicated on ROE.

→ **Twin cornerstones: growth strategy** and **growth capability** — strategy without infrastructure to execute (or the reverse) → long-term growth impossible.

→ **Concepts of sustainability: very weak** — overall stock of capital assets remains constant; **weak** — preserve critical resources to support all over a long horizon; **strong** — preserve resources under primacy of **ecosystem functioning**.

**What makes an organisation sustainable?** It must:

→ have a **clear strategic direction**;

→ be able to **scan its environment** to identify opportunities;

→ be able to **attract, manage and retain competent staff**;

→ have adequate **administrative and financial infrastructure**;

→ demonstrate **effectiveness and impact** to leverage further resources;

→ get **community support** and involvement.

**Use of SGR model:**

→ **Borrower requests additional financing** (actual growth > SGR): risk of **too much debt, too little equity** → either **raise additional equity**, or **reduce the rate of expansion** to a level sustainable without more financial leverage.

→ **Mature firms** (actual growth < SGR): find productive uses of excess cash — **increase dividends / buy back stock**, **reduce debt**, or hold more **lower-earning liquid assets** (these lower SGR); or **acquire rapidly growing companies** to raise actual growth.

→ **Inflation:** growth = volume + inflation; inflationary asset increase must be financed like real growth → more external financing and higher historical-cost D/E → if creditors fix historical D/E, **inflation lowers SGR**.

**Mitsubishi Corporation (SM case):** dropped fixed 3-year "midterm management plan" for a long-term circa-2020 growth vision ("New Strategic Direction"); aim — stable earnings through business cycles by a portfolio diversified by business model, industry, market and geography; "Three Corporate Principles". Summary:

→ **Future pull** approach eyeing 2020 — double the business with a diversified but focussed portfolio;

→ **Clear portfolio strategy** — select winning businesses by proactive reshaping;

→ Grow and deliver returns while keeping **financial discipline**.

## 6.2 External Funding Requirement (EFR)

Asked: Jan 2026 Q2(b) 4m

→ Growth needs both current and non-current assets to rise. When management also has concerns on **Debt-to-Equity** or **Current Ratio**, compute EFR and **its types** (short-term, long-term, equity).

### Exam steps

**Exam steps — EFR (both approaches):** 1 → $S_1 = S \times (1 + g)$, $\Delta S = S_1 - S$ → 2 → retained profit = $m \times S_1 \times (1 - d)$ → 3 → asset increase = (TA − spontaneous liabilities) × $g$ (or (TA ÷ $S$ − P ÷ $S$) × $\Delta S$) → 4 → EFR = 3 − 2 → 5 → short-term from ratio condition, long-term from FA ÷ LTL, balance = equity → 6 → check D/E (and current ratio if given).

*App-written: if the question says fixed assets have spare capacity (as in Jan 2026), grow only the assets that must grow with sales before step 3.*

3.4% of marks

**Exam steps — SGR theory:** 1 → SGR = ROE × (1 − d), growth without new equity or more leverage → 2 → target growth > SGR + loan request = too much debt, too little equity → 3 → two alternatives: raise additional equity, or cut expansion to SGR level (mature firm below SGR: dividends/buyback, repay debt, liquid assets, acquisitions).

3.4% of marks

# 15 Ch 15 – Startup Finance

| Attempt | Q | M | Asked (MCQ / numerical / theory) |
|---|---|---|---|
| Sep 2025 | Q1(c) | 4 | Steps in a business succession strategy (evaluate key positions, map competencies, identify workforce competencies, bridge leader) (Theory) |
| Jan 2026 | Q6(c) [either] | 4 | Role of government in creating a sustainable environment for start-ups in India (Theory) |
| May 2026 | Q2(c) | 4 | True/False with justification on start-up finance: valuation of early-stage firms, venture capital staging, bootstrapping, angel investors (Theory) |

## S.1.1 Bootstrapping

Asked: May 2026 Q2(c) 4m

**Bootstrapping** = start and build a company from **personal finances** or **operating revenues** of the new company. Logic: more money at inception → complacency and wasteful spending (marketing, offices, equipment); own savings → cautious approach, curbs waste, keeps promoter on toes. Methods:

- **(a) Trade credit**: suppliers reluctant to new businesses (want cash/credit card). Way out: **well-crafted financial plan**, visit supplier (owner if small, **CFO** if big firm), communication skills, reference of someone known to the supplier. May get **half on credit**, balance on delivery. Trick: get goods shipped and **sell before paying**. Reduces working capital needed, especially in **retail**.
- **(b) Factoring**: receivables **sold to a commercial finance company**; factor collects and does the paperwork; can be on a **non-notification basis** (customers not told). Merits: cuts receivable-maintenance costs (bookkeeping, collections, credit verification) — often cheaper than the factor's fee; **frees money tied in receivables**, esp. sales to businesses/government with long payment delays.
- **(c) Leasing**: lease equipment instead of buying → lower capital cost, lessee claims tax benefit. SM figures: photocopier at **₹ 5,000 per month** vs ₹ 1,00,000 or more to buy; automobile/van to avoid ₹ 5,00,000 or more. **Lessor**: depreciation tax benefit, possible capital appreciation, profit on lease. **Lessee**: smaller payments, ability to **walk away** at end of lease term, tax benefit on lease rentals.

## S.1.2 Angel Investors

Asked: May 2026 Q2(c) 4m

- Context: India ranks low on entrepreneurship, innovation and ease of doing business — outdated draconian rules, red tapism, time-consuming procedures, lack of general support; changing in recent times.
- Invest in small startups or entrepreneurs; often among the entrepreneur's **family and friends**; **one-time** investment or **ongoing injection** through difficult early stages.
- **More favourable terms** than other lenders — invest in the **entrepreneur** rather than the viability of the business; focus on helping first steps, not possible profit → different from VCs.
- Also called **informal investors, angel funders, private investors, seed investors, business angels**: affluent individuals; capital in exchange for **ownership equity or convertible debt**; some via crowdfunding platforms or angel networks.
- Use **their own money** (VCs manage **pooled money** in a strategically managed fund). Vehicle may be an LLC, business, trust or investment fund.
- Startup fails early → angel **loses investment completely**; so professional angels look for a defined **exit strategy**, acquisitions or IPOs.

3.4% of marks

### 5.1.3.4 Stages of funding for VC

Asked: May 2026 Q2(c) 4m

1. **Seed Money**: low level financing to prove a new idea.
2. **Start-up**: early-stage firms needing funds for marketing and product development.
3. **First-Round**: early sales and manufacturing funds.
4. **Second-Round**: working capital for early stage companies selling product but not yet in profit.
5. **Third Round (Mezzanine financing)**: expansion money for a newly profitable company.
6. **Fourth-Round (Bridge financing)**: finances the "going public" process.

Indicative risk matrix (SM) — risk falls as the stage advances:

| Stage | Funds locked (years) | Risk perception | Activity financed |
|---|---|---|---|
| Seed Money | 7–10 | Extreme | Concept/idea or R&D for product development; low level of financing |
| Start Up | 5–9 | Very High | Initialising prototype operations or developing products and marketing |
| First Stage | 3–7 | High | Started commercial production and marketing |
| Second Stage | 3–5 | Sufficiently high | Expanding market, growing WC needs, not earning profit |
| Third Stage | 1–3 | Medium | Market expansion, acquisition, product development for a profit-making company (Mezzanine) |
| Fourth Stage | 1–3 | Low | Facilitating public issue i.e. going public (Bridge) |

### 7. Startup India Initiative

Asked: Jan 2026 Q6(c) 4m

Launched by GoI on **16 January 2016**. Per **GSR Notification 127(E) dated 19 Feb 2019**, an entity is a Startup if:

→ (i) Up to **10 years** from incorporation/registration, as a **private limited company** (Companies Act, 2013), **registered partnership firm** (s. 59, Partnership Act, 1932) or **LLP** (LLP Act, 2008) in India.

→ (ii) **Turnover** for any financial year since incorporation/registration has not exceeded **₹ 100 crore**.

→ (iii) Working towards **innovation, development or improvement** of products/processes/services, or a **scalable business model** with high potential of employment generation or wealth creation.

→ Proviso: entity formed by **splitting up or reconstruction** of an existing business is **not** a Startup.

SM box (startupindia.gov.in) words the third test as innovation, development, deployment or commercialisation of new products, processes or services driven by technology or intellectual property.

**Government support (SM):**

→ **Startup India and StandUp India** (2016): widespread awareness, boost to entrepreneurial mindset.

→ SIDBI-run **Electronic Development Fund (EDF)**: Government became a **Limited Partner** in a fund for the first time.

→ Easy finance: **Mudra Scheme**; tax benefits: **100% tax holiday u/s 80-IAC**; **exemption from angel taxation**.

→ **Startup India Seed Fund Scheme (SISFS)**, DPIIT, Jan 2021: outlay **₹ 945 crore** for proof of concept, prototype, product trials, market entry, commercialisation; **~3,600 entrepreneurs through 300 incubators** in next 4 years. Eligibility: DPIIT-recognised, incorporated **not more than 2 years** ago at application, idea with market fit, viable commercialisation and scope of scaling. Seed fund up to **₹ 50 lakh**. Priority sectors: social impact, waste management, water management, financial inclusion, education, agriculture, food processing, biotechnology, healthcare, energy, mobility, defence, space, railways, oil and gas, textiles.

**Other reasons India is a sustainable environment for start-ups:**

3.4% of marks

→ **(i) Pool of talent**: millions graduating from colleges and B-schools start own ventures (earlier went only to big companies).

→ **(ii) Cost effective workforce**: young country, over **10 million** join the workforce every year → lower cost of setting up and running a business.

→ **(iii) Increasing use of internet**: affordable telecom; reached rural areas; **second-largest internet user base** after China.

→ **(iv) Technology**: quick, simple, efficient processes; easy data storage; startups in **AI and blockchain**.

→ **(v) Variety of funding options**: angel investors, VCs, seed funding investors; easing of **FDI norms**, most sectors on **100% automatic route**.

## 8.3 Business succession strategy

Asked: Sep 2025 Q1(c) 4m

1. **Evaluate key leadership positions**: risk or impact assessment to find critical roles — those that transform the business or set strategic direction.
2. **Map competencies required** for those positions: qualifications, behavioural and technical competencies.
3. **Identify competencies of current workforce**: internal options that can deliver Step 2, training and development needs, and whether to **search outside**.
4. **Bridge leader**: in family businesses, an **outsider** appointed to develop the business and prepare **young family members** for leadership.

## Exam steps

**Exam steps (4m theory):** 1 → name the SM heading (e.g. "Characteristics of VC financing") → 2 → write the SM list items in SM order, key term first → 3 → one line of SM logic per item; for true/false, state the SM fact that decides it (e.g. seed stage = extreme risk, 7–10 years; angels use own money; bootstrapping = personal finances/operating revenues).

**Lists to have ready:** 9 innovative sources · 9 pitch points · 4 unicorn features · 3 bootstrapping methods · 4 VC "generally" · 4 VC characteristics · 7 VC advantages · 6 VC stages · 6 investment-process steps · 3 fund structures · 3 startup tests + proviso · 5 sustainability reasons · 5 needs · 4 strategy steps · 3 challenges.

3.4% of marks

# ③ Ch 03 — Advanced Capital Budgeting Decisions

| Attempt | Q | M | Asked (MCQ / numerical / theory) |
|---|---|---|---|
| Sep 2025 | MCQ CS-I 1-3 | 6 | Sensitivity analysis of NPV: % change in NPV when initial cost, annual cash inflow, or cost of capital is varied adversely by 10% (MCQ) |
| Jan 2026 | MCQ CS-V 13-15 | 6 | Choice between a sure product and an uncertain one: profit from contribution minus fixed cost, and expected value of profit using a probability distribution (MCQ) |

## 4.1 Statistical techniques — 4.1.1 Probability

Asked: Jan 2026 MCQ CS-V 13-15 6m

→ Probability = chance an event occurs; certain event = **1**, impossible = **0**; probabilities of all outcomes add to 1.

| Formula | Terms / logic | SM example figures |
|---|---|---|
| $E(R)/ENCF = \Sigma_{i=1..n} NCF_i \times P_i$ | $NCF_i$ net cash flows, $P_i$ probability. Expected (mean) cash flow: each possible cash flow × its probability, added. | Example: 3,00,000 × 0.3 + 2,00,000 × 0.6 + 1,20,000 × 0.1 = 90,000 + 1,20,000 + 12,000 = **2,22,000**. |
| $ENPV = \Sigma_{t=1..n} ENCF \div (1 + k)^t$ | t period, k discount rate. Find ENCF year by year, then discount at k. ENCF includes inflows and outflows. | Illus 3, 4 below. |

*Note (app-written): the SM text after the first table says chances of "₹ 1,00,000" are 10%; the table and the working use ₹ 1,20,000 (SM p.13).*

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**Illus 3 (single period):** outlay 10,000, k 10%. ENCF A 12,000, B 16,000. NPV A = 0.909 × 12,000 − 10,000 = **₹ 908**; B = 0.909 × 16,000 − 10,000 = **₹ 4,544** → B.

**Illus 4 (multi-period):** ENCF yr 1 6,000, yr 2 4,800, yr 3 4,200; PV = 6,000 × 0.909 + 4,800 × 0.826 + 4,200 × 0.751 = 12,573 − 10,000 = ENPV **₹ 2,573**.

**TYK P21 (Jumble — expected utility):** same idea with utilities: Σ utility × probability. A = −10 − 12 + 16 + 6 + 2 = **2**; B = −6 − 0.45 + 16 + 5 + 3 = **17.55** → select **B** (higher expected utility).

## 4.3 Other techniques — 4.3.1 Sensitivity analysis

Asked: Sep 2025 MCQ CS-I 1-3 6m

→ **CIMA:** modelling and risk-assessment procedure in which changes are made to significant variables to see the effect on the planned outcome; attention then on variables of special significance.

→ Change **only one variable at a time** (others constant). The more sensitive the NPV (or IRR), the **more critical** the variable.

→ **Steps:** 1. find variables that influence NPV/IRR; 2. establish mathematical relationship among them; 3. analyse effect of change in each on NPV/IRR.

| Formula | Terms / logic | SM example figures |
|---|---|---|
| Change in NPV = (NPV − NPV (Revised)) ÷ NPV | As ICAI TYK P10 working. Method 1: change one variable by a given % adversely (any %, e.g. 10%), recompute NPV. Highest % change = most sensitive. | TYK P10: inflow −10% → (22,605 − 8,345) ÷ 22,605 = 63.08%. |
| % change in a variable to make NPV = 0 = NPV ÷ PV of that variable's cash flows × 100 | App-written summary of ICAI's TYK P9/P11 working. Method 2 (break-even): how far the variable can move before NPV becomes zero. **Smallest %** = most sensitive. | TYK P11: savings 4,914 ÷ 22,792 = 21.56%; plant 4,914 ÷ 10,000 = 49.14%. |
| (IRR − X) ÷ X = cost of capital sensitivity | X = cost of capital; % rise in X before NPV = 0. ICAI TYK P12 writes it with figures; general form app-written. | (16 − X) ÷ X = 60% → X = 10%. |

3.4% of marks

→ **Advantages:** (1) **Critical issues** — identifies critical factors for success/failure; (2) **Simplicity**.

→ **Disadvantages:** (1) **Assumption of independence** of variables — unlikely in real life; (2) **ignores probability** of changes.

→ **TYK T2:** measures how sensitive estimated parameters (cash flows, discount rate, life) are to estimation errors; studies responsiveness of NPV/IRR to selling price, quantity, returns; answers "what if" (e.g. flows ₹ 50,000 instead of ₹ 80,000; life 3 years not 5). Not a method to remove risk — only to analyse and measure it. Cash flows may be worst (pessimistic), most likely, most optimistic. Three steps: identify variables; define quantitative relationship; analyse impact on NPV.

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**Illus 9 (X Ltd):** cost 400 cr, 5 cr units, SP 100, VC 50, FC 50 cr, 6%, 3 yrs (PVAF 2.673). Net inflow 200 cr → NPV **134.60**. 2.5% adverse: cost 410 → 124.60 (−7.43%); SP 97.5 → 101.19 (**−24.82%**); VC 51.25 → 117.89 (−12.41%); FC 51.25 → 131.26 (−2.48%); units 4.875 → 117.89 (−12.41%). **Selling price most sensitive.**

**TYK P10:** cost 1,20,000, inflow 45,000, 4 yrs, 10% (3.169; 11% 3.103). NPV **22,605**. Cost +10% → 10,605 (**53.08%**); inflow −10% → 8,345 (**63.08%**); k 11% → 19,635 (**13.14%**) → most sensitive to **annual cash inflow**.

**TYK P9 (XYZ):** cost 10,00,000, margin 20, volumes 20,000/30,000/30,000, 10%. NPV 13,10,293 − 10,00,000 = **3,10,293**. Break-even: (a) SP: 5 − 40 = 10,00,000 ÷ 65,514 = 15.26 → 5 55.26, fall **7.9%**; (b) unit cost 44.74, rise **11.85%**; (c) volume fall 3,10,293 ÷ 13,10,293 = **23.68%**; (d) outlay rise 3,10,293 ÷ 10,00,000 = **31.03%**; (e) life: 2 yrs NPV −1,40,800; yr 3 PV 4,50,600 ÷ 360 = 1,252 per day → 112 days → 2 years 112 days, fall **22.97%**. Alt (10% adverse): SP → NPV −82,796 (126.68%); cost → 48,234 (84.46%); volume → 1,79,263 (42.22%); outlay → 2,10,293 (32.22%).

**TYK P11 (Red Ltd):** NPV 14,914 − 10,000 = **4,914**. NPV-zero: plant **+49.14%**; running cost PV 7,878 → **+62.38%**; savings PV 22,792 → **−21.56%** → **savings most sensitive**. Alt 10%: 20.35%, 16.04%, 46.38%.

**TYK P12 (Easygoing):** (i) outlay = 3.274 × 57,500 = **1,88,255** (at IRR NPV = 0); (ii) k 10% → NPV 57,500 × 3.791 − 1,88,255 = **29,727.50**; (iii) 29,727.50 ÷ 3.791 = 7,841.60 = 7.8416% of FC → FC **1,00,000**; (iv) SP = 60 ÷ 30% = 200; sales = (57,500 + 1,00,000) ÷ 0.70 = 2,25,000 → **1,125 units**; (v) BEP = 1,00,000 ÷ 140 = **714.285 units**.

**TYK P13 (Unnat):** NPV = [2,00,000 × 13.50 − 10,00,000] × 3.605 − 50,00,000 = **11,28,500**. Break-even: SP 28.43 (**5.23%**); volume 1,76,812 (**11.59%**); VC 18.07 (**9.51%**). (d) Expected volume 1,95,000 → ENPV **8,85,163**; worst (1,75,000) −88,188; best 23,45,188. 30% chance of negative NPV > acceptable 20% → **reject**.

*Note (app-written): TYK P13(c) says variable cost 18.07 "represents a fall of 9.51%" — it is a rise from 16.50 (SM p.74).*

## Exam steps

**Exam steps — inflation NPV:** check if k is nominal or real and if CFs are nominal or real → match them (Nominal Cash Flow = (1 + Inflation Rate)$^t$ × Real Cash Flow, or Real Discount Rate = (1 + Nominal Discount Rate)/(1 + Inflation Rate) − 1) → if revenue and cost inflation differ, compound each separately → (R − C)(1 − T) + DT (shield not inflated) → discount → NPV.

**Exam steps — ENPV, SD, CV:** ENCF = $\Sigma$ NCF$_i$ × P$_i$ (each year) → discount → ENPV → $\sigma^2$ = $\Sigma$ (NCF$_j$ − ENCF)$^2$ P$_j$ → $\sigma$ (multi-year: Hillier $\sigma^2 = \Sigma (1+r)^{-2i} \sigma_i^2$) → CV = $\sigma$ ÷ mean → same mean: lower $\sigma$; different mean: lower CV.

3.4% of marks

**Exam steps — RADR / CE:** RADR = Risk free rate + Risk premium (or $IR_f + (k_o - IR_f)$ Risk Factor; or table by CV) → discount unadjusted CF. CE: $NCF_t \times \alpha_t$ → discount at **risk-free rate** only → NPV; lower $\Sigma\alpha$ = riskier = higher RADR.

**Exam steps — sensitivity:** base NPV → change one variable (given % adverse) → revised NPV → % change = (base − revised) ÷ base → highest % = most sensitive. Or NPV-zero method: % = NPV ÷ PV of that variable → lowest % = most sensitive. k: (IRR − X) ÷ X.

**Exam steps — scenario / best-worst:** NPV for worst, most likely, best → expected NPV with probabilities → σ, CV → worst-case probability: dependent = year-1 probability, independent = that probability$^n$ → compare with acceptable risk level.

**Exam steps — decision tree:** draw left to right (□ decision, ○ chance) → path NPVs → joint probability = product along path → roll back right to left: chance node = Σ value × p; decision node = highest → decide at first node.

**Exam steps — replacement:** tax on sale of old = (BV − MV) × t → net outflow = cost of new − (MV + tax saving) → Δ CF = [(ΔS ± ΔC) − ΔD](1 − t) + ΔD → PV of Δ CF + PV of new salvage → NPV → accept if > 0.

**Exam steps — optimum replacement cycle:** for each life n: PV of cost + running costs − PV resale at n → EAC = PV ÷ PVAF(k, n) → lowest EAC = optimum cycle. Replace now vs later: compare PV of each timing with new asset's NPV included.

3.4% of marks

# ④ Ch 04 — Security Analysis

| Attempt | Q | M | Asked (MCQ / numerical / theory) |
|---|---|---|---|
| Sep 2025 | Q4(c) [either] | 4 | Technical analysis: favourable and unfavourable arguments (Theory) |
| Jan 2026 | Q4(c) | 4 | Technical analysis: identify chart patterns (uptrend, continuation/flag, head and shoulders, triangle/wedge) from descriptions, any four (Theory) |
| May 2026 | Q4(c) [either] | 4 | Elliott Wave Theory of technical analysis (Theory) |

## 2.4.2 Elliot Wave Theory

Asked: May 2026 Q4(c) 4m

TYK T3

- → **Ralph Elliot, 1934**; inspired by Dow Theory and nature; based on **75 years** of price movements and charts.
- → **Wave** = movement of market price from one change in direction to the next change in the same direction; caused by buying/selling impulses from demand-supply pressures.
- → **Impulsive patterns (basic waves)**: **3 or 5 waves** in the direction of the basic movement (bull or bear phase).
- → **Corrective patterns (reaction waves)**: **3 waves** against the basic direction; correct the earlier rise (bull) or fall (bear).
- → Waves 1, 3, 5 = directional; waves 2, 4 = corrective.
- → **Complete cycle**: five-wave impulse + three-wave correction (a, b, c) = **8 waves**; two phases, bullish and bearish; then a fresh cycle starts.

## 2.8 Interpreting Price Patterns

Asked: Jan 2026 Q4(c) 4m

| Pattern | Shape | Signal |
|---|---|---|
| (a) Channel | Series of uniformly changing tops and bottoms | Upward slope → rising prices; downward → declining |
| (b) Wedge | Tops and bottoms change in **opposite** directions, or same direction at **different rates** | — |
| (c) Head and shoulders | Large hump (head) between two smaller humps (shoulders); **neckline** joins where head meets shoulders. Price after 2nd shoulder is crucial. | Most important **reversal** pattern. Below neckline → drop ≈ distance from top of head to neckline. |
| (i) H&S top | Left shoulder, head, right shoulder | **Bearish**; fall below neckline → **sell** |
| (ii) Inverse H&S | Inverse of the top formation | **Bullish**; rise above neckline → **buy** |
| (d) Triangle / coil | — | **Uncertainty**; hard to predict breakout direction |
| (e) Flags and pennants | — | Previous trend likely to **continue** |
| (f) Double top | — | **Bearish**, price expected to fall |
| (g) Double bottom | — | **Bullish**, price expected to rise |
| (h) Gap | Opening price today − previous close | Wider gap → stronger signal of **continuation**; up-gap in rising market = eager buyers; lower gap in falling market = extreme selling pressure |

3.4% of marks

## 2.10 Evaluation of Technical Analysis

Asked: Sep 2025 Q4(c) 4m

| Arguments in favour | Arguments against |
|---|---|
| → **Crowd psychology** makes trends persist; tools spot them early. | → No **convincing explanation** for tools used. |
| → Shifts in demand and supply are **gradual**, not instantaneous; detected early. | → Empirical evidence for **random walk** casts a shadow. |
| → Fundamental information is **assimilated over time**; price moves in the same direction till fully assimilated. | → By the time a trend is signalled it may **already have taken place**. |
| | → **Self-defeating**: more users → less value. |

Nutshell: may not work in a rational, efficient market; helpful in real-world imperfect markets; best used **with fundamental analysis**.

### Exam steps

**Exam steps — EMA:** $a = 2 \div (n+1)$ (or as given) → skip non-trading days → diff = close − previous EMA → × $a$ → add to previous EMA → carry forward → rising EMA = bullish, take long position (TYK PI).

**Exam steps — Run test:** sign each change (+/−) → count runs $r$, $n_1$, $n_2$ → $\mu_r = 2n_1n_2/(n_1+n_2) + 1$ → $\sigma_r$ → limits $\mu \pm t\,\sigma$ ($df$ = changes − 1) → $r$ inside = weak-form efficient.

**Exam steps — Moving average:** sum of $n$ most recent closes → (SM) two-item centred total of adjacent sums → $\div 2n$ (SM: ÷ 10 for 5-day) → compare price line with MA line using buy/sell signals.

**Exam steps — Intrinsic value:** $D_1 = D_0(1+g)$ or $bE_1$ → check $g < k$ → $P_0 = D_1/(k-g)$ → price < value buy, > value sell.

3.4% of marks

# 7 Ch 07 — Securitization

| Attempt | Q | M | Asked (MCQ / numerical / theory) |
|---|---|---|---|
| Sep 2025 | Q3(c) | 4 | Risks associated with the use of blockchain technology (SM section on risks of blockchain) (Theory) |
| Jan 2026 | Q6(c) [or] | 4 | Securitisation instruments: name and characteristics of pass-through, pay-through (own debt securities) and strips structures (Theory) |
| May 2026 | Q3(c) | 4 | Securitisation of NPAs: benefit to balance sheet versus regulatory and valuation challenges (Theory) |

## 7. Securitization Instruments

Asked: Jan 2026 Q6(c) 4m

Divided by **maturity characteristics** into three:

### 7.1 Pass Through Certificates (PTCs)

→ Originator transfers the **entire cash receipt** (interest + principal) from assets sold → **direct claim** of investors on all securitized assets via SPV.

→ Investors hold **proportional beneficial interest** in assets held in trust by SPV.

→ **Prepayment** of principal distributed proportionately to all holders.

→ On final payment, **all securities terminate simultaneously** (single maturity).

→ **Skewness of cash flows** in early stage if principal repaid before schedule.

**7.2 Pay Through Security (PTS)** — overcomes PTC's limitation of all cash flows being passed on and of **single maturity**.

→ SPV issues **debt securities backed by the assets** → can restructure **tranches** from varying maturities of receivables.

→ Permits **desynchronisation** of servicing of securities from asset cash flows.

→ SPV may **reinvest surplus** funds short term (e.g. on early retirement of receivables, for short-term yield).

→ Freedom to issue several debt tranches with **varying maturities**.

| Point | PTC (Pass Through) | PTS (Pay Through) |
|---|---|---|
| Cash flows | All passed on immediately to investors | Desynchronised from asset cash flows |
| Investor's claim | Direct, proportional beneficial interest in assets | SPV's debt securities backed by the assets |
| Maturity | Single maturity; all terminate together | Several tranches with varying maturities |
| Prepayment / surplus | Prepayment passed proportionately; early-stage skewness | Surplus can be reinvested short term |

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**TYK TI:** Differentiate between PTS and PTC → ICAI: refer para 7 (table above).

**7.3 Stripped Securities** — cash flows of underlying securities divided into two or more new securities:

→ **Interest Only (IO)** — holder gets only interest. **Principal Only (PO)** — holder gets only principal.

→ **Highly volatile** → less preferred by investors.

→ **Rates rise** → PO price falls (borrowers postpone paying cheaper loans); IO value rises (more interest calculated).

→ **Rates fall** → borrowers prepay to borrow fresh at lower rates; IO tends to fall due to prepayments of principal.

→ Prices of IOs and POs mainly determined by **perception of investors**.

3.4% of marks

*Note (app-written): SM p.10 sentence "However, when interest rate due to prepayments of principals, IO's tends to fall" is incomplete; read as: when interest rates fall, prepayments of principal rise and IO value falls (this matches the SM's own prepayment logic on p.9).*

## 10.2 Risks associated with Blockchain

Asked: Sep 2025 Q3(c) 4m

- → **Wider risk perspective** — members have different **risk appetites** → conflict in designing monitoring controls; who manages risk if no one is in charge; **accountability**.
- → **Reliability depends on technology** — if the **consensus mechanism** is tampered with, ledger information becomes inaccurate and unreliable.
- → **No central authority** to administer/enforce protocol amendments → difficult process control; users of public blockchains struggle to understand **general IT controls** and their effectiveness.
- → **Information overload** — huge, frequently updated data challenges monitoring; hard to find **competent people** for monitoring controls.

## 11. Securitization in India

Asked: May 2026 Q3(c) 4m (NPAs)

- → **Citi Bank** pioneered it in India by bundling **auto loans**; later sales tax deferrals, aircraft receivables etc.
- → **SARFAESI Act, 2002** (Securitization and Reconstruction of Financial Assets and Enforcement of Security Interest) — to tackle **NPAs without approaching the Court**.
- → Now an important funding source for **micro finance companies and NBFCs**; **commercial MBS** emerging.
- → Market **dominated by a few players** — ICICI Bank, HDFC Bank, NHB.
- → **CRISIL**: volumes may reach near pre-pandemic highs of **₹ 1.9 trillion** (FY19 & FY20) once macro-situation and rates stabilise; key funding for non-banks, investment avenue for banks growing retail assets.
- → **SEBI** allows **FPIs** to invest in securitized debt of **unlisted companies** up to a limit.

*Note (app-written): the SM has no separate section on securitizing NPAs. For a May 2026-type question, build the answer from SM points only: balance-sheet benefit → 3.1 (off-balance sheet financing, capital release, Capital-to-Weighted Asset Ratio); challenges → 6.2 taxation, 6.3 accounting (esp. without recourse), 6.6 foreclosure laws, 9.2 legal risk, 8.2 pricing (discounting best estimate of expected cash flows); India context → SARFAESI Act, 2002.*

### Exam steps

**Exam steps (4m theory):** 1 define (repackaging illiquid assets into marketable securities / RBI definition) → 2 list the SM points with the bold key term → 3 one line each in SM wording → 4 for "differentiate", use the PTC vs PTS table.

**Lists to know cold:** features 6 · originator benefits 4 · investor benefits 3 · primary 3 + secondary 6 participants · mechanism 7 steps · problems 6 · instruments 3 · risks 3 (+3 market) · blockchain applications 5 · blockchain risks 4 · tokenization similarities 4.

2.8% of marks

# 13 Ch 13 – Business Valuation

| Attempt | Q | M | Asked (MCQ / numerical / theory) |
|---|---|---|---|
| Sep 2025 | Q1(a) | 6 | Cost of equity by CAPM, EPS from a profit statement, equity value from an EBITDA multiple, and enterprise value (market cap + debt - cash) (Numerical) |
| Jan 2026 | Q4(b) | 4 | EVA: operating income, EVA with cost of equity from CAPM, and dividend per share that can be paid before value declines (Numerical) |

## 4. Measuring Cost of Equity — 4.1 CAPM, 4.2 APT

Asked: Sep 2025 Q1(a) 6m

| Formula | Terms / logic | SM use |
|---|---|---|
| 4.1 $R = rf + \beta (rm - rf)$ | R = expected rate of return (cost of equity); rf = risk free rate of return; β = Beta value of the stock; rm = market rate of return. Returns are a reward for assuming risk; two buckets: risk-free + market-relative risk. | Illus 1, 2; TYK P7, P10, P16 |
| 4.2 $Rf + \beta1(RP1) + \beta2(RP2) + ....\beta j(RPn)$ | Each macro-economic factor (e.g. interest rate, sector growth) has its own beta; RP = risk premium of the factor. E(R) = **linear function** of factors. | — |

→ CAPM's weakness: **oversimplification of risk**. APT (Stephen Ross, 1970s) uses **multiple factors**; APT itself does not name the factors — analysts must develop them for their economy.

## 5. Relative Valuation

Asked: Sep 2025 Q1(a) 6m

→ Asset, earnings and cash-flow approaches give **intrinsic value**. Relative valuation gives a **'relative' value** by **comparative analysis with peers** (similar asset and risk dispositions); increasingly used **with** the intrinsic approaches to validate them. Also called **'valuation by multiples'**. (TYK T2 → this para.)

→ Steps: (1) find the **drivers** for the multiple; (2) determine results through **financial ratios**; (3) find **comparable firms** and compare; (4) **iterate** to smoothen deviations.

| Multiple | Best suited (SM) |
|---|---|
| EV/EBITDA | Popular; gives critical inputs. |
| EV/Invested Capital | **Capital-intensive** enterprises; misfit for asset-light firms or those with significant investment properties. |
| EV/Sales | **Cash-rich**, huge order book, organic growth through own capital. |
| P/E (equity based) | "Celebrated status"; may not reflect forex risk of an exporter (Brexit example). |
| PEG = PE Ratio ÷ Growth Rate | **High growth or sunrise** industries. |

→ **Step 2 (ratio):** a factor-based approach helps choose the right ratio. **Step 3 (comparables)** — most challenging: same size, capital structure, head count can still differ in revenue model (X: **FTE pricing** vs Y: **Unit Transfer Pricing**) → "take benchmarks with a pinch of salt". Niche market → look at similar companies in other industries; or take closely relevant firms and run a **regression** (last resort).

→ **Assumptions:** (i) market is **efficient**; (ii) function between fundamentals and multiples is **linear**; (iii) comparables similar in **structure, risk and growth**.

→ EV two ways: (a) take entity value and **adjust for debt**; (b) **balance sheet** approach.

| Formula | Terms / logic | SM figures |
|---|---|---|
| EV = EBITDA × EBITDA multiple | Value of the whole business (debt + equity holders). | Illus 1, 2 |
| Equity value = EV – Debt + Surplus funds (cash) | Debt belongs to lenders; surplus cash belongs to shareholders. | Illus 2: 3,000 – 300 + 500 = 3,200 |
| EV = Market capitalisation + Debt – Cash & cash equivalents | Market cap = shares × CMP (balance sheet approach). | Illus 3: 8,40,000 + 2,00,000 – 5,00,000 |

2.8% of marks

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**Illus 2 (A Ltd):** ke = 4.5% + 0.9(12% − 4.5%) = **11.25%** (capitalisation rate). EBITDA = GP 1,000 − indirect exp. 400 = **600** (₹ '000); earning value 600/0.1125 = 5,333.33 ÷ 1,00,000 shares = **₹ 53.33** per share. (b) 600 × 5 = 3,000 − debt 300 + surplus 500 = equity value **₹ 3,200 thousand**.

**Illus 3 (H K Ltd):** 70,000 shares × CMP ₹ 12 = market cap ₹ 8,40,000 + long-term debt ₹ 2,00,000 − cash ₹ 5,00,000 = EV **₹ 5,40,000**.

## 6.3 Economic Value Added (EVA)

Asked: Jan 2026 Q4(b) 4m

→ Holistic measure of financial performance and economic contribution. A company creates value only if **returns exceed its cost of capital**. EVA **separates operations from financing**. Negative EVA → not generating value; positive → producing value.

→ **Why EVA:** ROI, ROCE, asset/earnings/FCF methods are "number-crunching"; EVA makes **management accountable** for decisions (e.g. choosing only the earlier break-even project although both had positive IRR) and shows whether returns **cover the cost of capital**.

| Formula | Terms / logic | SM use |
|---|---|---|
| **EVA = NOPAT − (Invested Capital × WACC)** = NOPAT − Capital Charge | Excess of returns over the weighted average cost of invested capital. | All EVA items |
| NOPAT = EBIT − tax (= EBIT(1 − t)) + non-cash expenses | Operating cash return after tax; depreciation left untouched (operating expense); add back book entries such as **provision for bad and doubtful debts**. | Illus 5 |
| Invested Capital = Total assets − non-interest-bearing current liabilities (creditors, billing in advance) + non-cash adjustments | Also = **equity + long-term debt**, generally at the **start of the year**. Adjust non-cash items in both NOPAT and capital. | Illus 5, TYK P14 |
| Capital charge = Invested Capital × WACC; WACC (ko) = ke × E/(E+D) + kd × D/(E+D), kd post-tax = rate × (1 − t) | Hurdle the operations must clear. | TYK P9–P16 |
| Taxable income = Net income ÷ (1 − t); EBIT = Taxable income + Interest | Back out operating income from PAT (used in TYK answers). | TYK P10, P11, P13 |
| Financial leverage = PBIT ÷ PBT = PBIT ÷ (PBIT − Interest) | Gives PBIT when only leverage and interest are known. | TYK P9 |
| EVA dividend per share = EVA ÷ No. of shares | Maximum DPS before the value of the entity starts declining. | TYK P11, P13 |

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**Illus 5 (A Ltd, ₹ lac):** NOPAT = EBIT 410 − tax 123 + bad-debt provision 20 = **307**. Invested capital = 1,300 − 400 = 900 + 20 = **920** (current liabilities include 100 tax liability). WACC = 800/900 × 8.45% + 100/900 × 12%(1 − 0.30) = **8.44%**. Capital charge 920 × 8.44% = **77.65** → EVA = 307 − 77.65 = **229.35**.

2.8% of marks

ICAI ·

**TYK P9 (Jatayu):** interest 10% × 400 = 40; 1.5 = PBIT/(PBIT − 40) → PBIT **₹ 120 lakhs**; NOPAT 120 × 0.7 = **84**; WACC = 14% × 300/700 + 7% × 400/700 = **10%**; EVA = 84 − 0.10 × 700 = **₹ 14 lakhs**.

**TYK P10 (RST):** ke = 8.5% + 1.36 × 9% = **20.74%**; kd = 11% × 0.7 = 7.70%; WACC = 20.74 × 125/165 = 7.70 × 40/165 = 15.71 + 1.87 = **17.58%**. Taxable 25,00,000/0.7 = 35,71,429 + interest 4,40,000 = EBIT **40,11,429**; EVA = 28,08,000 − 29,00,700 = **−₹ 92,700**.

**TYK P11 (Tender):** 15 lac/0.7 = 21,42,857 + interest 10,00,000 = operating income **₹ 31,42,857**; EVA = 22,00,000 − 13% × 95,00,000 (12,35,000) = **₹ 9,65,000**; ÷ 6,00,000 shares = **₹ 1.6083** per share.

**TYK P12 (Orange / Grape / Apple):** kd post-tax 10.40% / 8.45% / 9.75%; WACC **13.52% / 15.225% / 17.95%**; EVA = 25,000 × 0.65 − WACC × 1,00,000 = **2,730 / 1,025 / −1,700** → **Orange** best (highest EVA, lowest WACC). Interest 12,800 / 6,500 / 3,000 → NI 7,930 / 12,025 / 14,300 → EPS 1.30 / 1.45 / 1.43 × PE 11 = **₹ 14.30 / 15.95 / 15.73**; market cap **87,230 / 1,32,385 / 1,57,300**. PE should be adjusted for different financial risk.

**TYK P13 (Delta):** 15,00,000/0.6 = 25,00,000 + interest 15,00,000 = EBIT **₹ 40,00,000**; EVA = 24,00,000 − 0.126 × 1,00,00,000 = **₹ 11,40,000**; ÷ 2,50,000 = **₹ 4.56**/share. No dividend → value expected to **increase** (higher growth, higher EBIT).

**TYK P14 (XYZ Inc, $ mn):** NOPAT 180 − 63 = 117 − 12% × capital employed 605 (100 + 325 + 180) = 72.60 → EVA **44.40**. MVA = market value of equity 10 mn × $50 = 500 − equity fund 425 = **75**.

**TYK P15 (Herbal Gyan):** investment = WC 20 + PPE 80 + patent (offer value) 40 = 140; EVA = 12 − 0.15 × 140 = **−₹ 9 lakh**.

**TYK P16 (Constant Engineering):** β = 1 (moves with market) → ke = 10% + 1(12% − 10%) = 12%; investment = NW 200 + LTD 400 + patent 100 = 700; WACC = 12 × 300/700 + 10 × 400/700 = 5.14% + 5.71% = **10.85%**; EVA = 84 − 10.85% × 700 = **₹ 8.05 crore**.

*Note (app-written): TYK P15–P16 include the unrecorded patent at market/offer value in invested capital. In TYK P16 ICAI takes cost of debt = 10% (the government bond rate) with no tax adjustment, as no debt rate or tax rate is given (SM p.67).*

## Exam steps

**Exam steps — Relative valuation / EV (Sep 2025 type):** ke = rf + β (rm − rf) → adjust EBITDA for extraordinary / pending items → EV = EBITDA × multiple → Equity value = EV − debt + surplus cash (÷ shares) → or EV = shares × CMP + debt − cash.

**Exam steps — EVA (Jan 2026 type):** PAT ÷ (1 − t) = PBT → + interest = EBIT → NOPAT = EBIT(1 − t) (+ non-cash items) → invested capital (equity + debt, or assets − non-interest CL, + unrecorded patents) → WACC (ke by CAPM, kd(1 − t), weights) → capital charge → EVA → EVA ÷ shares = max DPS. MVA = market value − invested capital.

**Exam steps — DCF / FCFF:** high-growth WACC and stable WACC (CAPM + kd(1 − t)) → FCFF = EBIT(1 − t) − (capex − dep) − ΔWC each year → PV at high-growth WACC → TV = terminal-year FCF ÷ (stable cost of capital − growth rate) → discount TV by PVF of the last explicit year → firm value (− debt + non-core assets = equity, as in SVA).

2.8% of marks

**Exam steps — Unlisted firm / new project:** proxy $\beta_e$ → de-gear $\beta_a = \beta_e \times E/[E + D(1 - t)]$ → average → re-gear at own D/E → CAPM ke → WACC → discount FCFs.

**Exam steps — Several methods / exchange ratio:** NAV per share (deduct contingent liability if it will materialise) → earnings capitalisation (maintainable PAT ÷ rate) → PE × EPS → DCF → weighted / average fair value → range (seller's minimum to buyer's maximum) → exchange ratio = target value ÷ acquirer value; allocate shares on fully-paid equivalents.

**Exam steps — Chop-shop / distressed:** segment data × pure-play ratios (sales, assets, operating income) → average = break-up value vs market. Distressed: DCF value of equity (1 − Probability of distress) + Distress sale value of equity (Probability of distress); APV = Unlevered Firm Value + Tax Benefits of Debt − (Unlevered firm value − Distress Sale Value) × Probability of Distress.